

National Technical University of Athens
School of Electrical and Computer Engineering
Division of Computer Science

Energy Management and Profit Allocation Mechanisms for Prosumer Coalitions

Diploma Thesis

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Supervisor: Dimitris Fotakis
Professor, NTUA

Athens, July 2026



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Athens, July 2026

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Περίληψη

Η αυξανόμενη διείσδυση της οικιακής φωτοβολταϊκής παραγωγής, των μπαταριών και των έξυπνων μετρητών δημιουργεί νέες δυνατότητες συντονισμού στις ενεργειακές κοινότητες. Ωστόσο, η συνεργασία όλων των μελών μπορεί να είναι υπολογιστικά απαιτητική και προϋποθέτει διαφανείς κανόνες κατανομής κερδών. Η παρούσα εργασία αναπτύσσει ένα υπολογιστικό πλαίσιο που αξιοποιεί τη συμπληρωματικότητα μεταξύ ζήτησης, παραγωγής και αποθήκευσης, χωρίς να απαιτεί συντονισμό ολόκληρης της κοινότητας.

Με βάση δεδομένα κατανάλωσης και παραγωγής ανά 15 λεπτά από το ιρλανδικό σύνολο έξυπνων μετρητών, διατυπώνεται μοντέλο μικτού ακέραιου γραμμικού προγραμματισμού για την παθητική λειτουργία, τον ατομικό προγραμματισμό μπαταριών και τη συνεργατική λειτουργία συνασπισμών με εσωτερική ανταλλαγή ενέργειας. Το κέρδος συνεργασίας ορίζεται ως η πρόσθετη εξοικονόμηση έναντι της βελτιστοποιημένης αυτόνομης λειτουργίας και κατανέμεται βάσει των θετικών οριακών συνεισφορών αποχώρησης των μελών.

Για τον σχηματισμό υποσυνασπισμών, εκτιμώνται ανά ζεύγος βαθμολογίες συντονισμού σε αντιπροσωπευτικές ημέρες, επιλεγμένες μέσω k-medoids και προσθήκης ακραίων ημερών. Οι ομάδες σχηματίζονται με ακριβή επίλυση MILP σε μικρότερα προβλήματα και με ευρετική τοπικής αναζήτησης σε μεγαλύτερα. Στη συνέχεια, αξιολογούνται στο πλήρες ετήσιο σύνολο δεδομένων και ελέγχονται βάσει κριτηρίου ευστάθειας αποδεκτής μετακίνησης.

Τα αποτελέσματα δείχνουν ότι κατάλληλα διαμορφωμένοι υποσυνασπισμοί διατηρούν μεγάλο μέρος του οφέλους του μεγάλου συνασπισμού, ιδιαίτερα όταν υπάρχει ισορροπία μεταξύ παραγωγών και καταναλωτών. Η επιλογή των μελών είναι σημαντικότερη στις μικρότερες ομάδες, όπου περιορίζεται η αντιστοίχιση τοπικού πλεονάσματος και ζήτησης. Δεν εντοπίζονται αποδεκτές επικερδείς μετακινήσεις μεταξύ των εξεταζόμενων ομάδων. Επιπλέον, η απόκριση ζήτησης μπορεί να συμπληρώσει την αποθήκευση, αλλά προσφέρει περιορισμένο πρόσθετο όφελος στην κλίμακα της κοινότητας.

Συνολικά, η εργασία παρέχει ένα επεκτάσιμο πλαίσιο για συνεργατική ενεργειακή διαχείριση, κατανομή κερδών και σχηματισμό υποσυνασπισμών, με σημείο αναφοράς τον μεγάλο συνασπισμό.

Λέξεις-κλειδιά: Ανανεώσιμες Ενεργειακές Κοινότητες, Συνεργατική Θεωρία Παιγνίων, Ενεργειακή Βελτιστοποίηση, Κατανομή Κερδών, Σχηματισμός Συνασπισμών, Απόκριση Ζήτησης, Ενεργειακή Ευελιξία

Abstract

The increasing penetration of residential photovoltaic generation, battery storage and smart metering creates new opportunities for coordination in energy communities. However, full-community cooperation can be computationally demanding, while practical deployment requires transparent profit-sharing rules. This thesis develops a computational framework that exploits complementarity between demand, photovoltaic generation and battery flexibility without requiring coordination of the entire community.

Using 15-minute consumption and generation profiles from the Irish smart-meter dataset, a mixed-integer linear optimization model is formulated for passive operation, individual battery scheduling and cooperative operation with internal energy exchange. Cooperation gain is defined as the additional saving beyond the optimized stand-alone operation of the users. The resulting surplus is allocated through a rule based on positive leave-one-out marginal contributions, providing an interpretable and computationally scalable approach.

The thesis also studies subcoalition formation. Instead of exhaustively evaluating all coalitions, pairwise coordination scores are estimated on representative days selected through k-medoids clustering and the explicit inclusion of extreme days. Balanced subcoalitions are formed through exact MILP for smaller instances and a local-search heuristic for larger ones. The selected groups are evaluated on the full annual dataset and tested under an accepted-migration stability criterion after profit allocation.

The results distinguish the value of individual battery scheduling from the additional value of cooperation. Well-formed subcoalitions preserve a large share of the grand-coalition benefit, particularly when prosumers and consumers are well balanced. The score-based method is especially valuable for smaller groups, where member selection strongly affects the matching of local surplus and demand. No accepted profitable migrations are found for the examined subcoalitions. A targeted demand-response extension shows that load shifting can complement battery storage, although its additional value remains limited at the residential-community scale.

Overall, the thesis provides a scalable framework for cooperative energy management, profit allocation and the formation of smaller subcoalitions benchmarked against the grand coalition.

Keywords: Renewable Energy Communities, Cooperative Game Theory, Energy Optimization, Profit Allocation, Coalition Formation, Demand Response, Energy Flexibility

Acknowledgements

I would like to first express my deep gratitude to my parents, Fotis and Maria, and to my sister, Katerina, for their unwavering love and continuous support throughout these years. Thank you for believing in me, standing by me in every difficulty, and never ceasing to encourage me to pursue my dreams. Your presence and support have been a constant point of reference throughout this journey.

I would also like to offer my sincere thanks to my supervisor, Professor Dimitris Fotakis, whom I have had the privilege to have as a teacher and mentor during my studies. Through our collaboration, he taught me, above all, a more meaningful way of thinking. I am truly grateful for his trust, patience, and ongoing guidance, not only during the preparation of this diploma thesis, but at every important step of my path. With his advice, honesty, and steady interest, he encouraged me to continuously improve and strive to become the best version of myself.

I would like to thank all the professors and fellow students who contributed, each in their own way, to making this journey so special. The knowledge, experiences, and moments we shared will stay with me.

Finally, a special thank you goes to my friends, Nikos, Andreas, Giorgos, Panagiotis, and Vasilis. They shared with me both the easy and the difficult moments and were my companions throughout this entire route. Thank you for your support, advice, countless discussions, laughter, and all the moments that made these years truly unforgettable.

Athens, July 2026
Kyriakos Katsiadramis

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Chapter 1

Εκτενής Ελληνική Περίληψη

1.1 Αντικείμενο της εργασίας και βασική ιδέα

Η παρούσα διπλωματική εργασία μελετά τη συνεργατική λειτουργία μιας οικιακής ενεργειακής κοινότητας με φωτοβολταϊκή παραγωγή, μπαταρίες και δυνατότητα εσωτερικής ανταλλαγής ενέργειας μεταξύ των συμμετεχόντων. Το κεντρικό ερώτημα είναι αν μια ομάδα ετερογενών χρηστών μπορεί να οργανωθεί με τρόπο που να μειώνει το συνολικό ενεργειακό κόστος, να παραμένει υπολογιστικά εφικτός για μεγαλύτερες κοινότητες και ταυτόχρονα να δημιουργεί αποδεκτά κίνητρα συμμετοχής για τους ίδιους τους χρήστες.

Η αφετηρία της εργασίας είναι ότι οι τελικοί χρήστες του ηλεκτρικού συστήματος δεν είναι πλέον μόνο παθητικοί καταναλωτές. Ένας οικιακός χρήστης μπορεί να καταναλώνει ενέργεια, να παράγει ενέργεια μέσω φωτοβολταϊκών, να αποθηκεύει μέρος της ενέργειας σε μπαταρία και να προσαρμόζει μέρος της ζήτησής του. Οι χρήστες που έχουν και κατανάλωση και παραγωγή αντιμετωπίζονται ως prosumers. Η οικονομική αξία μιας ενεργειακής κοινότητας δεν εξαρτάται μόνο από το πόση παραγωγή ή αποθήκευση υπάρχει συνολικά, αλλά και από το πότε εμφανίζονται η ζήτηση, η παραγωγή και η ευελιξία κάθε χρήστη. Αν ένας χρήστης έχει πλεόνασμα φωτοβολταϊκής παραγωγής την ίδια στιγμή που ένας άλλος χρήστης εισάγει ενέργεια από το δίκτυο, η τοπική συνεργασία μπορεί να αξιοποιήσει μέρος αυτής της ενέργειας εσωτερικά, αντί να την εξάγει σε χαμηλότερη τιμή και να αγοράζει άλλη ενέργεια σε υψηλότερη τιμή.

Για τον λόγο αυτό, η εργασία δεν αντιμετωπίζει την κοινότητα ως απλό άθροισμα ανεξάρτητων νοικοκυριών. Αντίθετα, αναπτύσσεται ένα ενιαίο υπολογιστικό πλαίσιο που συνδέει τέσσερα επίπεδα αποφάσεων. Το πρώτο επίπεδο είναι η ατομική λειτουργία κάθε χρήστη, χωρίς συνεργασία. Το δεύτερο είναι η συνεργατική βελτιστοποίηση μιας ομάδας χρηστών, όπου επιτρέπεται εσωτερική ανταλλαγή ενέργειας και κοινός προγραμματισμός μπαταριών. Το τρίτο είναι ο διαμοιρασμός του πρόσθετου κέρδους που δημιουργείται από τη συνεργασία. Το τέταρτο είναι ο σχηματισμός μικρότερων υποσυνασπισμών, ώστε να διατηρείται μεγάλο μέρος της αξίας του πλήρους συντονισμού χωρίς να απαιτείται πάντα λειτουργία ολόκληρης της κοινότητας ως ένας μεγάλος συ-

νασπισμός.

Η εμπειρική βάση της εργασίας είναι το Irish smart-meter dataset. Χρησιμοποιούνται προφίλ κατανάλωσης και φωτοβολταϊκής παραγωγής σε χρονική ανάλυση 15 λεπτών για το έτος 2020. Από το διαθέσιμο ημερολογιακό έτος αξιολογούνται οι 343 ημέρες για τις οποίες υπάρχουν πλήρη προφίλ χρηστών και τιμών με 96 χρονικά διαστήματα ανά ημέρα. Οι αρχικές ετήσιες αξιολογήσεις γίνονται σε 20 μετρημένους χρήστες του dataset. Για μεγαλύτερα πειράματα, όπου εξετάζεται η κλιμάκωση της μεθόδου σε 30 και 50 χρήστες, οι μετρημένοι χρήστες συμπληρώνονται με συνθετικά οικιακά προφίλ που έχουν μετατραπεί στην ίδια δεκαπεντάλεπτη μορφή. Τα συνθετικά προφίλ δεν παρουσιάζονται ως επιπλέον πραγματικά μετρημένα νοικοκυριά, αλλά ως ρεαλιστικά πρόσθετα προφίλ για να μελετηθεί η κλιμάκωση.

Η εργασία οργανώνεται γύρω από τρία κύρια ερευνητικά ερωτήματα. Πρώτον, πόση αξία δημιουργείται όταν περνάμε από παθητική λειτουργία σε ατομικό προγραμματισμό μπαταρίας και στη συνέχεια σε πλήρη συνεργασία; Δεύτερον, πώς μπορεί το συνεργατικό όφελος να κατανεμηθεί με τρόπο ερμηνεύσιμο και πολύ φθηνότερο υπολογιστικά από κλασικές λύσεις της θεωρίας συνεργατικών παιγνίων, όπως ο Shapley value; Τρίτον, μπορεί μια μεγαλύτερη κοινότητα να χωριστεί σε μικρότερους υποσυνασπισμούς που ανακτούν το μεγαλύτερο μέρος του οφέλους του μεγάλου συνασπισμού και παραμένουν σταθεροί ως προς απλές ατομικές μετακινήσεις;

1.2 Βιβλιογραφικό πλαίσιο και θέση της εργασίας

Το εισαγωγικό μέρος της διπλωματικής τοποθετεί το πρόβλημα στη βιβλιογραφία των ενεργειακών κοινοτήτων, του peer-to-peer energy trading, της συνεργατικής θεωρίας παιγνίων, της επιλογής αντιπροσωπευτικών ημερών και της απόκρισης ζήτησης. Η βιβλιογραφία για τις ενεργειακές κοινότητες δείχνει ότι η απόδοσή τους επηρεάζεται από τεχνικούς, οικονομικούς και οργανωτικούς παράγοντες. Δεν αρκεί να υπάρχουν φωτοβολταϊκά και μπαταρίες σε μια περιοχή. Σημασία έχει η σύνθεση της κοινότητας, το ποιοι χρήστες έχουν παραγωγή, ποιοι έχουν ζήτηση σε διαφορετικές ώρες, πόση αποθήκευση υπάρχει και με ποιους κανόνες μοιράζεται το όφελος.

Η βιβλιογραφία του peer-to-peer trading προσφέρει ένα φυσικό υπόβαθρο για τοπικές ανταλλαγές ενέργειας μεταξύ prosumers. Προηγούμενες εργασίες έχουν δείξει ότι ο συνδυασμός τοπικής αγοράς ενέργειας και αποθήκευσης μπορεί να βελτιώσει τα αποτελέσματα των συμμετεχόντων σε σχέση με την ανεξάρτητη λειτουργία. Παράλληλα, η βιβλιογραφία των συνασπισμών δείχνει ότι δεν έχουν όλες οι ομάδες χρηστών την ίδια αξία. Η συνεργασία είναι πιο χρήσιμη όταν τα προφίλ είναι συμπληρωματικά: για παράδειγμα, όταν οι χρονικές στιγμές πλεονάσματος ενός prosumer συμπίπτουν με τη ζήτηση

άλλων χρηστών ή όταν η μπαταρία ενός μέλους μπορεί να βοηθήσει το συνολικό προφίλ της ομάδας.

Ένα δεύτερο σώμα βιβλιογραφίας αφορά τον δίκαιο και σταθερό διαμοιρασμό του οφέλους. Η θεωρία συνεργατικών παιγνίων παρέχει κλασικές έννοιες όπως ο Shapley value, ο πυρήνας και ο nucleolus. Ο Shapley value είναι ελκυστικός επειδή κατανέμει την αξία με βάση τη μέση οριακή συνεισφορά κάθε παίκτη σε όλες τις δυνατές σειρές έναξης. Όμως απαιτεί την αξιολόγηση όλων των δυνατών συνασπισμών, δηλαδή $2^{|\mathcal{N}|}$ τιμές συνασπισμών για $|\mathcal{N}|$ παίκτες. Αυτό γίνεται γρήγορα απαγορευτικό όταν κάθε τιμή συνασπισμού προκύπτει από ένα πρόβλημα λειτουργικής βελτιστοποίησης με μπαταρίες και εσωτερικές ανταλλαγές. Ο nucleolus μπορεί να δώσει ισχυρότερη οπτική σταθερότητας, αλλά επίσης απαιτεί σύνθετους υπολογισμούς πάνω στο σύνολο των συνασπισμών.

Η εργασία τοποθετείται ακριβώς σε αυτό το κενό. Δεν θεωρεί τις τιμές των συνασπισμών ως αφηρημένες ποσότητες. Τις υπολογίζει από πραγματικά προβλήματα ημερήσιου ενεργειακού προγραμματισμού. Στη συνέχεια, αντί να υπολογίζει εξαντλητικά όλους τους συνασπισμούς, προτείνει έναν κανόνα κατανομής βασισμένο στη θετική leave-one-out οριακή συνεισφορά του κάθε χρήστη στον πραγματοποιημένο συνασπισμό. Έτσι διατηρείται η ιδέα της συνεισφοράς, αλλά με γραμμικό αριθμό επιλύσεων γύρω από τον σχηματισμένο συνασπισμό.

Ένα τρίτο σχετικό ρεύμα είναι οι μέθοδοι επιλογής αντιπροσωπευτικών ημερών. Σε ενεργειακά προβλήματα μεγάλης χρονικής ανάλυσης, η αξιολόγηση όλων των ημερών και όλων των πιθανών ομάδων χρηστών είναι πολύ δαπανηρή. Η βιβλιογραφία δείχνει ότι η χρονική συμπίεση πρέπει να γίνεται με προσοχή, γιατί αν χαθούν ακραίες ή λειτουργικά κρίσιμες ημέρες, τα αποτελέσματα μπορεί να είναι παραπλανητικά. Στην παρούσα εργασία, οι αντιπροσωπευτικές ημέρες δεν αντικαθιστούν την τελική ετήσια αξιολόγηση. Χρησιμοποιούνται μόνο ως υπολογιστική γέφυρα για να υπολογιστούν pairwise scores μεταξύ χρηστών. Οι υποσυνασπισμοί που επιλέγονται με βάση αυτά τα scores αξιολογούνται τελικά σε ολόκληρο το διαθέσιμο ετήσιο σύνολο των 343 ημερών.

Τέλος, η απόκριση ζήτησης, δηλαδή η μετακίνηση φορτίου από ένα ακριβό σε ένα φθηνότερο χρονικό διάστημα, εξετάζεται ως δευτερεύουσα επέκταση. Η σχετική βιβλιογραφία αντιμετωπίζει την ευέλικτη ζήτηση ως οικονομικό πόρο που μπορεί να ενεργοποιηθεί με κατάλληλα κίνητρα και μηχανισμούς. Η παρούσα εργασία δεν σχεδιάζει πλήρη αγορά demand response. Εξετάζει ένα στοχευμένο σχήμα μετατόπισης φορτίου, ώστε να εκτιμηθεί πόσο πρόσθετο όφελος μπορεί να προκύψει όταν ένα μικρό μέρος της ζήτησης μεταφέρεται από ακριβές περιόδους σε ευνοϊκότερες περιόδους, αφού έχει ήδη ληφθεί υπόψη η αξία του battery scheduling.

1.3 Μοντέλο λειτουργίας και ορισμός συνεργατικού οφέλους

Το κεφάλαιο του μοντέλου βελτιστοποίησης παρουσιάζει τρεις διαδοχικούς τρόπους λειτουργίας: παθητική λειτουργία βάσης, ατομικό προγραμματισμό μπαταρίας και συνεργατική λειτουργία συνασπισμού. Θεωρείται σύνολο χρηστών $\mathcal{N} = \{1, 2, \ldots, n\}$ και ημερήσιος χρονικός ορίζοντας $\mathcal{T}$ με $T = 96$ περιόδους των 15 λεπτών, δηλαδή $\Delta t = 0.25\text{h}$. Για κάθε χρήστη i και περίοδο t είναι γνωστά η ζήτηση $d_{i,t}$, η τοπική παραγωγή $g_{i,t}$, η τιμή αγοράς από το δίκτυο π_t^{buy} και η τιμή πώλησης προς το δίκτυο π_t^{sell}. Υποτίθεται ότι $\pi_t^{\text{buy}} \geq \pi_t^{\text{sell}}$, κάτι που αντιστοιχεί στη συνηθισμένη περίπτωση όπου η αγορά ενέργειας είναι ακριβότερη από την αποζημίωση εξαγωγής.

Στην παθητική λειτουργία βάσης δεν λύνεται κάποιο πρόβλημα βελτιστοποίησης. Η τοπική παραγωγή χρησιμοποιείται πρώτα για να καλύψει την τοπική ζήτηση. Αν υπάρχει έλλειμμα, εισάγεται ενέργεια από το δίκτυο. Αν υπάρχει πλεόνασμα, εξάγεται ενέργεια προς το δίκτυο. Έτσι, για κάθε χρήστη ορίζονται η εισαγωγή βάσης $p_{i,t}^{\text{imp,base}} = \max\{0, d_{i,t} - g_{i,t}\}$ και η εξαγωγή βάσης $p_{i,t}^{\text{exp,base}} = \max\{0, g_{i,t} - d_{i,t}\}$. Το κόστος βάσης C_i^{base} προκύπτει από το κόστος εισαγωγής μείον τα έσοδα εξαγωγής.

Στο δεύτερο επίπεδο, κάθε χρήστης προγραμματίζει ανεξάρτητα τη δική του μπαταρία. Οι μεταβλητές απόφασης περιλαμβάνουν ισχύ φόρτισης $p_{i,t}^{\text{ch}}$, ισχύ εκφόρτισης $p_{i,t}^{\text{dis}}$, κατάσταση φόρτισης $E_{i,t}$, εισαγωγή από το δίκτυο και εξαγωγή προς το δίκτυο. Η μπαταρία περιορίζεται από τη χρησιμοποιήσιμη χωρητικότητα, τα μέγιστα όρια φόρτισης και εκφόρτισης, την απόδοση, την αρχική κατάσταση φόρτισης και την απαίτηση η τελική κατάσταση φόρτισης να ισούται με την αρχική. Χρησιμοποιείται δυαδική μεταβλητή ώστε η μπαταρία να μη φορτίζει και να εκφορτίζει ταυτόχρονα. Το αντικείμενο είναι η ελαχιστοποίηση του ατομικού κόστους, το οποίο περιλαμβάνει κόστος εισαγωγής, έσοδο εξαγωγής και έναν όρο κόστους υποβάθμισης μπαταρίας. Το βέλτιστο ατομικό κόστος συμβολίζεται C_i^{solo}.

Στο τρίτο επίπεδο, ένας συνασπισμός $S \subseteq \mathcal{N}$ λειτουργεί συνεργατικά. Τα μέλη του συνασπισμού μπορούν να ανταλλάσσουν ενέργεια εσωτερικά πριν αλληλεπιδράσουν με το εξωτερικό δίκτυο. Για κάθε διατεταγμένο ζεύγος χρηστών $i, j \in S$ εισάγεται μεταβλητή $q_{i,j,t} \geq 0$, που δηλώνει την ισχύ που μεταφέρεται από τον χρήστη i στον χρήστη j τη χρονική περίοδο t. Η ισορροπία ισχύος κάθε χρήστη περιλαμβάνει πλέον όχι μόνο την τοπική παραγωγή, την μπαταρία και το δίκτυο, αλλά και τη συνολική ενέργεια που λαμβάνει από άλλα μέλη και στέλνει προς άλλα μέλη. Οι περιορισμοί μπαταρίας παραμένουν ίδιοι, αλλά η αντικειμενική συνάρτηση ελαχιστοποιεί το συνολικό κόστος όλων των μελών του συνασπισμού. Προστίθεται επίσης ένας πολύ μικρός όρος regularization στις εσωτερικές μεταφορές, ώστε να αποφεύγονται περιττές κυκλικές ανταλλαγές χωρίς

ουσιαστική οικονομική σημασία.

Το συνεργατικό όφελος ενός συνασπισμού ορίζεται σε σχέση με την ήδη βελτιστοποιημένη ατομική λειτουργία των μελών του:

$$v(S) = \sum_{i \in S} C_i^{\text{solo}} - C^{\text{coal}}(S). \tag{1.1}$$

Ο ορισμός αυτός είναι σημαντικός γιατί αποδίδει στη συνεργασία μόνο το πρόσθετο όφελος που δεν θα μπορούσαν να πετύχουν τα μέλη αν προγραμμάτιζαν τις μπαταρίες τους μόνα τους. Το όφελος από το ατομικό battery scheduling μετριέται ξεχωριστά. Για τον μεγάλο συνασπισμό $\mathcal{N}$, η εργασία διακρίνει τρεις ποσότητες: το όφελος ατομικού scheduling σε σχέση με τη βάση, το πρόσθετο όφελος συνεργασίας σε σχέση με το solo scheduling και το συνολικό όφελος σε σχέση με την παθητική βάση.

Τα προβλήματα υλοποιούνται ως μικτά ακέραια γραμμικά προβλήματα, επειδή υπάρχουν συνεχείς μεταβλητές ισχύος και ενέργειας αλλά και δυαδικές μεταβλητές για αμοιβαία αποκλειόμενες λειτουργίες. Η επίλυση γίνεται με τον SCIP solver μέσω του OR-Tools. Στα βασικά πειράματα οι μπαταρίες έχουν ονομαστική χωρητικότητα 10 kWh, μέγιστη ισχύ φόρτισης και εκφόρτισης 3.3 kW, round-trip efficiency 0.95, χρησιμοποιήσιμη χωρητικότητα $0.85 \cdot 10 = 8.5$ kWh και αρχική κατάσταση φόρτισης ίση με το 50% της χρησιμοποιήσιμης χωρητικότητας.

Η ετήσια αξιολόγηση του μοντέλου για τους 20 χρήστες δείχνει καθαρά τη συνεισφορά κάθε επιπέδου. Σε 343 έγκυρες ημέρες του 2020, το συνολικό κόστος της παθητικής λειτουργίας είναι 7259.68 EUR. Με ατομικό battery scheduling μειώνεται σε 6670.89 EUR, δηλαδή προκύπτει εξοικονόμηση 588.79 EUR ή 8.11% σε σχέση με τη βάση. Όταν οι ίδιοι 20 χρήστες λειτουργούν ως μεγάλος συνασπισμός, το κόστος μειώνεται περαιτέρω σε 6373.42 EUR. Το πρόσθετο συνεργατικό όφελος σε σχέση με το solo scheduling είναι 297.47 EUR, δηλαδή 4.46% του solo κόστους. Συνολικά, ο μεγάλος συνασπισμός πετυχαίνει εξοικονόμηση 886.26 EUR ή 12.21% σε σχέση με την παθητική λειτουργία.

Τα αποτελέσματα αυτά δείχνουν ότι η ατομική μπαταρία έχει ήδη σημαντική αξία, επειδή μεταφέρει ενέργεια μέσα στη μέρα και μειώνει τις ακριβές εισαγωγές. Η συνεργασία όμως προσθέτει διαφορετικό είδος αξίας: επιτρέπει σε χρήστες με διαφορετικά προφίλ ζήτησης και παραγωγής να αξιοποιούν εσωτερικά την ενέργεια πριν απευθυνθούν στο δίκτυο.

1.4 Κατανομή του συνεργατικού κέρδους

Αφού υπολογιστεί το συνεργατικό όφελος, πρέπει να αποφασιστεί πώς θα μοιραστεί μεταξύ των συμμετεχόντων. Το κεφάλαιο της κατανομής κέρδους ξεκινά από τη

διάκριση μεταξύ δύο πηγών αξίας. Η πρώτη είναι η αξία που μπορεί να πετύχει κάθε χρήστης μόνος του, μέσω ατομικού battery scheduling. Η δεύτερη είναι το πρόσθετο πλεόνασμα που εμφανίζεται μόνο όταν οι χρήστες συνεργάζονται μέσα σε έναν συνασπισμό. Η εργασία επιλέγει να εγγυάται πρώτα σε κάθε χρήστη το ατομικό του όφελος και στη συνέχεια να μοιράζει μόνο το συνεργατικό πλεόνασμα.

Αν C_i^{base} είναι το κόστος βάσης και C_i^{solo} το βέλτιστο ατομικό κόστος, το ατομικό όφελος του χρήστη είναι

$$s_i = C_i^{\text{base}} - C_i^{\text{solo}}. \tag{1.2}$$

Για έναν συνασπισμό S, το συνολικό όφελος σε σχέση με τη βάση είναι $C^{\text{base}}(S) - C^{\text{coal}}(S)$. Το καθαρά συνεργατικό πλεόνασμα ορίζεται αφαιρώντας τα ατομικά οφέλη:

$$g(S) = \sum_{i \in S} C_i^{\text{solo}} - C^{\text{coal}}(S). \tag{1.3}$$

Επομένως, $g(S)$ είναι η επιπλέον εξοικονόμηση που δημιουργείται επειδή τα μέλη του S λειτουργούν από κοινού και όχι απλώς επειδή το καθένα έχει μπαταρία.

Ο προτεινόμενος κανόνας κατανομής βασίζεται στη θετική leave-one-out οριακή συνεισφορά. Για κάθε χρήστη i στον πραγματοποιημένο συνασπισμό, υπολογίζεται πόσο μειώνεται το συνεργατικό πλεόνασμα αν ο χρήστης αφαιρεθεί:

$$m_i = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\}). \tag{1.4}$$

Η ποσότητα αυτή ισοδύναμα γράφεται ως

$$m_i = C_i^{\text{solo}} + C^{\text{coal}}(\mathcal{N} \setminus \{i\}) - C^{\text{coal}}(\mathcal{N}). \tag{1.5}$$

Στο ακριβές μαθηματικό μοντέλο, λόγω της δυνατότητας του μεγάλου συνασπισμού να αναπαράγει τη λειτουργία μικρότερων συνασπισμών, οι συνεισφορές αυτές είναι μη αρνητικές. Στην αριθμητική υλοποίηση, πολύ μικρές αρνητικές τιμές μπορούν να κοπούν στο μηδέν ως προστασία από ανοχές solver.

Αν $G(\mathcal{N}) = g(\mathcal{N})$ είναι το συνολικό συνεργατικό πλεόνασμα και $w_i = m_i$ το βάρος συνεισφοράς, το συνεργατικό bonus του χρήστη i δίνεται από

$$b_i = G(\mathcal{N}) \frac{w_i}{\sum_{j \in \mathcal{N}} w_j}, \tag{1.6}$$

όταν το άθροισμα των βαρών είναι θετικό. Αν προκύψει εκφυλισμένη περίπτωση με θετικό πλεόνασμα αλλά μηδενικό συνολικό βάρος, το πλεόνασμα μοιράζεται ισόποσα. Το

τελικό όφελος που αποδίδεται στον χρήστη είναι

$$x_i = s_i + b_i, \tag{1.7}$$

ενώ η ισοδύναμη τελική κατανομή κόστους είναι $\alpha_i = C_i^{\text{solo}} - b_i$.

Ο κανόνας έχει τρεις βασικές πρακτικές ιδιότητες. Πρώτον, μοιράζει όλο το διαθέσιμο συνεργατικό πλεόνασμα και οι τελικές κατανομές κόστους αθροίζουν στο βέλτιστο κόστος του συνασπισμού. Δεύτερον, είναι ατομικά ορθολογικός ως προς το stand-alone optimized outcome: κάθε χρήστης κρατά το ατομικό του όφελος και λαμβάνει μη αρνητικό συνεργατικό bonus, άρα δεν χειροτερεύει σε σχέση με το να λειτουργούσε μόνος του. Τρίτον, είναι υπολογιστικά πολύ ελαφρύτερος από τον Shapley value. Για n χρήστες απαιτεί n ατομικά προβλήματα, ένα πρόβλημα μεγάλου συνασπισμού και n leave-one-out προβλήματα, δηλαδή $2n + 1$ επιλύσεις. Ο ακριβής Shapley value απαιτεί τιμές για όλους τους 2^n συνασπισμούς.

Η εργασία είναι επίσης προσεκτική ως προς τα όρια του κανόνα. Δεν ισχυρίζεται ότι ο κανόνας είναι πλήρης αξιωματική λύση συνεργατικών παιγνίων. Σε αντίθεση με τον Shapley value, δεν ικανοποιεί γενικά την προσθετικότητα και δεν χρησιμοποιεί όλες τις οριακές συνεισφορές σε όλα τα πιθανά υποσύνολα. Σε αντίθεση με τον nucleolus, δεν εγγυάται ότι η κατανομή βρίσκεται στον πυρήνα όταν αυτός είναι μη κενός. Ο στόχος του είναι πιο πρακτικός: να είναι ερμηνεύσιμος, αποδοτικός, ατομικά ορθολογικός και εφαρμόσιμος σε προβλήματα όπου οι τιμές των συνασπισμών είναι ακριβές βελτιστοποιήσεις.

Για να ελεγχθεί εμπειρικά η συμπεριφορά του, ο κανόνας συγκρίνεται με τον ακριβή Shapley value σε ένα μικρό πείραμα 8 παικτών για τον Ιούνιο του 2020. Το μηνιαίο κόστος βάσης είναι 159.46 EUR, το solo scheduling κόστος είναι 152.26 EUR και το κόστος του μεγάλου συνασπισμού είναι 138.36 EUR. Άρα το μηνιαίο συνεργατικό πλεόνασμα είναι 13.90 EUR. Και οι δύο μέθοδοι μοιράζουν ακριβώς τα ίδια 13.90 EUR, όπως απαιτεί η αποδοτικότητα.

Η διαφορά εμφανίζεται στο ποιοι χρήστες παίρνουν μεγαλύτερο μερίδιο. Ο προτεινόμενος κανόνας δίνει στους τέσσερις prosumers το 79.28% του συνεργατικού πλεονάσματος, ενώ ο Shapley value τους δίνει 69.71%. Αντίστοιχα, οι τέσσερις consumers λαμβάνουν 20.72% με τον προτεινόμενο κανόνα και 30.29% με τον Shapley value. Σε απόλυτους όρους, ο προτεινόμενος κανόνας μετακινεί 1.33 EUR του μηνιαίου πλεονάσματος προς την ομάδα των prosumers σε σχέση με τον Shapley. Η συνολική απόλυτη διαφορά των δύο κατανομών είναι 2.66 EUR και η μέση απόλυτη διαφορά είναι 0.33 EUR ανά παίκτη.

Η συμπεριφορά αυτή είναι συνεπής με τον σχεδιασμό του κανόνα. Στο συγκεκριμένο πείραμα, οι prosumers είναι οι κύριοι καθαροί πάροχοι εσωτερικής ενέργειας. Η αφαίρεσή τους από τον μεγάλο συνασπισμό μειώνει περισσότερο το συνεργατικό πλεόνασμα,

οπότε ο leave-one-out κανόνας τους δίνει μεγαλύτερο bonus. Αυτό δεν σημαίνει ότι οι consumers αδικούνται, γιατί και αυτοί κρατούν το stand-alone optimized αποτέλεσμα και λαμβάνουν μη αρνητικό συνεργατικό bonus. Σημαίνει ότι ο προτεινόμενος κανόνας δίνει μεγαλύτερη έμφαση στην οριακή σημασία του χρήστη μέσα στον πραγματοποιημένο συνασπισμό, ενώ ο Shapley value υπολογίζει μέσο όρο σε όλα τα πιθανά υποσύνολα.

1.5 Σχηματισμός υποσυνασπισμών

Το επόμενο βασικό κεφάλαιο εξετάζει πώς μπορεί μια κοινότητα να χωριστεί σε μικρότερους υποσυνασπισμούς. Θεωρητικά, ο μεγάλος συνασπισμός είναι το καλύτερο σχήμα ως προς το συνολικό συνεργατικό όφελος, επειδή μπορεί να αναπαράγει τη λειτουργία οποιασδήποτε συλλογής μικρότερων συνασπισμών. Πρακτικά όμως, ένας μεγάλος συνασπισμός μπορεί να είναι δύσκολος λόγω υπολογιστικού κόστους, ανάγκης ανταλλαγής πληροφοριών και οργανωτικής πολυπλοκότητας. Επομένως, το ζητούμενο δεν είναι οι υποσυνασπισμοί να ξεπεράσουν τον μεγάλο συνασπισμό, αλλά να ανακτήσουν όσο το δυνατόν μεγαλύτερο ποσοστό του οφέλους του με μικρότερη πολυπλοκότητα.

Μια partition $\mathcal{P} = \{S_1, S_2, \ldots, S_K\}$ χωρίζει το σύνολο χρηστών N σε ξένες ομάδες που καλύπτουν όλους τους χρήστες. Στην εργασία όλοι οι υποσυνασπισμοί έχουν ίσο μέγεθος m. Για μια ημέρα, το κέρδος του μεγάλου συνασπισμού είναι

$$G_N = \sum_{i \in N} C_i^{\text{solo}} - C_N,$$

ενώ το κέρδος μιας partition είναι το άθροισμα των συνεργατικών κερδών των ομάδων της:

$$G_{\mathcal{P}} = \sum_{k=1}^{K} \left(\sum_{i \in S_k} C_i^{\text{solo}} - C_{S_k} \right).$$

Η βασική μετρική αξιολόγησης είναι το ποσοστό ανάκτησης

$$R_{\mathcal{P}} = 100 \cdot \frac{G_{\mathcal{P}}}{G_N}.$$

Τιμή κοντά στο 100% σημαίνει ότι η partition διατηρεί σχεδόν όλο το όφελος που θα έδινε η πλήρης συνεργασία.

Η απευθείας αξιολόγηση όλων των υποψήφιων υποσυνασπισμών είναι υπολογιστικά μη εφικτή. Γι' αυτό αναπτύσσεται μια προσέγγιση pairwise scores. Πρώτα επιλέγεται ένα μικρό σύνολο αντιπροσωπευτικών ημερών. Για κάθε ημέρα κατασκευάζεται το συνολικό καθαρό φορτίο της κοινότητας, δηλαδή το άθροισμα κατανάλωσης μείον παραγωγής όλων των χρηστών σε 96 δεκαπεντάλεπτα. Επιλέγονται ρητά δύο ακραίες ημέρες:

μία ημέρα με μέγιστη συνολική θετική καθαρή ζήτηση, που αντιστοιχεί σε high-import condition, και μία ημέρα με μέγιστο συνολικό πλεόνασμα, που αντιστοιχεί σε high-surplus condition. Από τις υπόλοιπες ημέρες επιλέγονται 12 medoids με k-medoids clustering πάνω στα standardized ημερήσια προφίλ καθαρού φορτίου. Το τελικό representative set έχει 14 ημέρες: 12 medoids και 2 ακραίες ημέρες. Κάθε medoid έχει βάρος ίσο με το πλήθος των ημερών του cluster του, ενώ οι ακραίες ημέρες έχουν βάρος 1.

Για κάθε ζεύγος χρηστών (i, j) υπολογίζεται το σταθμισμένο pairwise coordination score

$$q_{ij} = \sum_{d \in \mathcal{D}_{\text{rep}}} w_d \left(C_i^{\text{solo}}(d) + C_j^{\text{solo}}(d) - C_{\{i,j\}}(d) \right).$$

Η τιμή αυτή εκτιμά πόσο ωφέλιμο είναι να τοποθετηθούν οι δύο χρήστες στον ίδιο υποσυνασπισμό. Το score ενός υποσυνασπισμού S προσεγγίζεται ως το άθροισμα όλων των εσωτερικών pairwise scores:

$$Q(S) = \sum_{\{i,j\} \subseteq S} q_{ij}.$$

Η παραδοχή είναι ότι ομάδες με υψηλότερα εσωτερικά pairwise scores θα έχουν και υψηλότερο πραγματικό ετήσιο συνεργατικό κέρδος. Η εργασία το ελέγχει με validation πάνω σε μεγάλο σύνολο υποψήφιων συνασπισμών μεγέθους 4. Τα pairwise scores εμφανίζουν πολύ ισχυρή σχέση με τα πραγματικά κέρδη, με Pearson correlation 0.956 και Spearman correlation 0.947.

Η επιλογή ομάδων δεν γίνεται απλώς μεγιστοποιώντας το συνολικό score. Αν γινόταν αυτό, θα μπορούσαν να δημιουργηθούν λίγες πολύ ισχυρές ομάδες και μία ασθενής ομάδα με χαμηλή δυνατότητα συνεργασίας. Για τον λόγο αυτό χρησιμοποιείται λεξικογραφικός στόχος. Πρώτη προτεραιότητα είναι να μεγιστοποιηθεί το score του ασθενέστερου υποσυνασπισμού, δηλαδή $\min_k Q(S_k)$. Δεύτερη προτεραιότητα, με δεδομένο αυτό το επίπεδο, είναι να μεγιστοποιηθεί το συνολικό εσωτερικό score όλων των ομάδων. Έτσι η partition είναι ταυτόχρονα fairness-aware και αποδοτική.

Για μικρά μεγέθη, ο στόχος αυτός διατυπώνεται ως exact MILP με δυαδικές μεταβλητές ανάθεσης χρήστη σε ομάδα και βοηθητικές δυαδικές μεταβλητές που δείχνουν αν δύο χρήστες βρίσκονται στην ίδια ομάδα. Το μοντέλο επιβάλλει κάθε χρήστης να ανήκει ακριβώς σε μία ομάδα και κάθε ομάδα να έχει το απαιτούμενο μέγεθος. Όμως το πρόβλημα είναι έντονα συνδυαστικό. Για παράδειγμα, ο χωρισμός 20 χρηστών σε πέντε ομάδες των τεσσάρων αντιστοιχεί περίπου σε $3.05 \cdot 10^{11}$ δυνατές partitions, ενώ ο χωρισμός 30 χρηστών σε πέντε ομάδες των έξι αντιστοιχεί περίπου σε $1.43 \cdot 10^{19}$ partitions.

Για τα μεγαλύτερα πειράματα αναπτύσσεται multi-start local-search heuristic. Η μέθοδος ξεκινά από πολλές τυχαίες εφικτές partitions και βελτιώνει κάθε partition με ανταλλαγές χρηστών μεταξύ ομάδων. Μια ανταλλαγή γίνεται αποδεκτή αν βελτιώνει λεξικο-

γραφικά τον στόχο: πρώτα αυξάνοντας το ελάχιστο group score ή, αν αυτό μένει ίδιο, αυξάνοντας το συνολικό score. Για το πείραμα των 30 χρηστών χρησιμοποιούνται 5000 random starts, μέχρι 150 improvement passes ανά start και early stopping μετά από 1000 starts χωρίς βελτίωση. Η τελική λύση βρίσκεται μετά από 2773 ολοκληρωμένα starts και απαιτεί 230.69 δευτερόλεπτα. Οι πέντε ομάδες των έξι χρηστών έχουν scores από 139.19 έως 140.51 και τυπική απόκλιση μόλις 0.48, άρα είναι πολύ ισορροπημένες ως προς τον pairwise στόχο.

Τα κύρια ετήσια αποτελέσματα δείχνουν ότι οι υποσυνασπισμοί ανακτούν μεγάλο μέρος του οφέλους του μεγάλου συνασπισμού. Στην περίπτωση 16 χρηστών, με exact MILP και τέσσερις ομάδες των τεσσάρων, το κέρδος του μεγάλου συνασπισμού είναι 221.81 EUR και το κέρδος των υποσυνασπισμών 203.15 EUR, δηλαδή ανάκτηση 91.59%. Στην περίπτωση 20 χρηστών, με πέντε ομάδες των τεσσάρων, η exact partition ανακτά 259.84 EUR από 297.47 EUR, δηλαδή 87.35%. Στην περίπτωση 30 χρηστών, με local search και πέντε ομάδες των έξι, η μέθοδος ανακτά 312.72 EUR από 331.81 EUR, δηλαδή 94.25%.

Τα πειράματα των 50 χρηστών δίνουν επιπλέον ερμηνεία. Στην αρχική σύνθεση 10 prosumers και 40 consumers, πέντε ομάδες των δέκα ανακτούν 350.65 EUR από 355.21 EUR, δηλαδή 98.72% του οφέλους του μεγάλου συνασπισμού. Στην ισορροπημένη σύνθεση 25 prosumers και 25 consumers, το απόλυτο κέρδος του μεγάλου συνασπισμού είναι πολύ μεγαλύτερο, 601.59 EUR, επειδή υπάρχει περισσότερη τοπική παραγωγή. Όμως η ανάκτηση είναι χαμηλότερη όταν η κοινότητα χωρίζεται: πέντε ομάδες των δέκα ανακτούν 570.32 EUR ή 94.80%, ενώ δέκα ομάδες των πέντε ανακτούν 538.12 EUR ή 89.45%. Αυτό δείχνει ότι όσο μικρότερες γίνονται οι ομάδες, τόσο περισσότερο περιορίζεται η δυνατότητα να ταιριάξει πλεόνασμα και ζήτηση μεταξύ πολλών χρηστών.

Η σύγκριση με balanced random partitions είναι κρίσιμη. Τα random benchmarks δεν είναι πλήρως τυχαία, αλλά ισορροπούν τους prosumers όσο γίνεται ανάμεσα στις ομάδες. Στην περίπτωση 16 χρηστών, η score-based μέθοδος ανακτά 91.59%, ενώ ο μέσος balanced random benchmark ανακτά 83.73%. Στην περίπτωση 20 χρηστών, οι δύο μέθοδοι είναι πολύ κοντά, 87.35% έναντι 87.00%. Στην περίπτωση 30 χρηστών, ο balanced random μέσος όρος είναι 94.71%, ελαφρώς πάνω από το 94.25% του local search. Στην περίπτωση 50 χρηστών με 10P/40C, η score-based μέθοδος ανακτά 98.72%, πολύ κοντά στον random μέσο όρο 98.31%. Όμως στην ισορροπημένη περίπτωση 25P/25C με δέκα ομάδες των πέντε, η score-based μέθοδος ανακτά 89.45%, ενώ ο balanced random μέσος όρος μόνο 85.41%.

Το συμπέρασμα είναι ότι η απλή ισορροπία prosumers-consumers είναι παράγοντας πρώτης τάξης. Όταν κάθε ομάδα είναι αρκετά μεγάλη και περιέχει αρκετή παραγωγή και ζήτηση, πολλές ισορροπημένες partitions πετυχαίνουν ήδη μεγάλο μέρος του διαθέσι-

μου οφέλους. Τότε το pairwise score λειτουργεί ως δεύτερης τάξης βελτίωση. Όταν όμως οι ομάδες είναι μικρότερες ή πιο περιοριστικές, η ακριβής ταύτιση των ατομικών προφίλ γίνεται πολύ πιο σημαντική και η score-based μέθοδος υπερέχει καθαρά.

Η ερμηνεία αυτή επιβεβαιώνεται από τη διάγνωση internal matching στα 50-player πειράματα. Στην περίπτωση 10P/40C, το συνολικό ετήσιο prosumer surplus είναι περίπου 10155 kWh και η local-search partition ταιριάζει εσωτερικά 10059.74 kWh, δηλαδή 99.06% αξιοποίηση. Γι' αυτό ανακτά σχεδόν όλο το όφελος του μεγάλου συνασπισμού. Στην περίπτωση 25P/25C, το ετήσιο surplus είναι περίπου 20985 kWh. Με πέντε ομάδες των δέκα ταιριάζονται 17544.97 kWh, δηλαδή 83.61%, ενώ με δέκα ομάδες των πέντε ταιριάζονται 16000.55 kWh, δηλαδή 76.25%. Όσο περισσότερη παραγωγή υπάρχει και όσο μικρότερες γίνονται οι ομάδες, τόσο μεγαλύτερο surplus μένει αταίριαστο.

1.6 Σταθερότητα υποσυνασπισμών μετά την κατανομή κέρδους

Το κεφάλαιο της ευστάθειας συνδέει τον σχηματισμό υποσυνασπισμών με τον κανόνα διαμοιρασμού κέρδους. Η βασική ερώτηση δεν είναι πλέον μόνο αν μια partition έχει υψηλό συνολικό όφελος, αλλά αν οι χρήστες έχουν κίνητρο να παραμείνουν στις ομάδες που τους ανατέθηκαν αφού μοιραστεί το συνεργατικό bonus. Η εργασία εξετάζει ένα πρακτικό accepted-migration criterion. Ένας χρήστης μπορεί να επιθυμεί να μετακινηθεί σε άλλη ομάδα, αλλά η μετακίνηση θεωρείται αξιόπιστη μόνο αν η ομάδα υποδοχής την αποδέχεται, δηλαδή αν κανένα από τα τωρινά μέλη της δεν χειροτερεύει.

Για κάθε τρέχοντα υποσυνασπισμό S_k, υπολογίζεται το ετήσιο bonus κάθε χρήστη με τον προτεινόμενο positive leave-one-out κανόνα. Ο οικονομικά ασθενέστερος χρήστης της ομάδας είναι εκείνος με το μικρότερο ετήσιο allocated bonus. Η επιλογή αυτή είναι συντηρητική: αν κάποιος είναι πιθανότερο να θέλει να φύγει, είναι ο χρήστης που λαμβάνει το χαμηλότερο bonus στην ομάδα του. Για κάθε τέτοιον χρήστη εξετάζονται πιθανές μετακινήσεις προς κάθε άλλη ομάδα.

Μια μετακίνηση από πηγή S_k προς στόχο S_ℓ θεωρείται blocking accepted migration μόνο αν ισχύουν δύο συνθήκες. Πρώτον, ο μετακινούμενος χρήστης πρέπει να βελτιώνεται αυστηρά, δηλαδή το ετήσιο bonus του στη νέα ομάδα $S_\ell \cup \{w_k\}$ να είναι μεγαλύτερο από το τωρινό του bonus. Δεύτερον, κάθε τωρινό μέλος της ομάδας υποδοχής πρέπει να μη χειροτερεύει μετά την αποδοχή του νέου χρήστη. Η πηγή δεν έχει δικαίωμα veto σε αυτό το test. Η έννοια είναι ασθενέστερη από πλήρη coalition-proof stability, αλλά είναι λειτουργικά σημαντική γιατί ελέγχει αν υπάρχουν προφανείς ατομικές αποκλίσεις που θα γίνονταν δεκτές.

Το πείραμα γίνεται στην περίπτωση των 20 χρηστών και στις πέντε ομάδες των τεσ-

σάρων:

$$S_1 = \{1, 6, 9, 14\}, \quad S_2 = \{2, 7, 15, 19\}, \quad S_3 = \{3, 8, 13, 18\},$$

$$S_4 = \{4, 5, 11, 16\}, \quad S_5 = \{10, 12, 17, 20\}$$

Η αξιολόγηση καλύπτει τις 343 έγκυρες ημέρες του 2020. Για κάθε ημέρα λύνεται η συνεργατική βελτιστοποίηση κάθε υποσυνασπισμού και το ημερήσιο κέρδος μοιράζεται με τον positive leave-one-out κανόνα. Στη συνέχεια τα ημερήσια bonuses αθροίζονται σε ετήσιες τιμές.

Οι ασθενέστεροι χρήστες είναι ο 9 στον πρώτο υποσυνασπισμό, ο 19 στον δεύτερο, ο 8 στον τρίτο, ο 11 στον τέταρτο και ο 20 στον πέμπτο. Τα bonus τους αντιστοιχούν αντίστοιχα στο 21.49%, 69.80%, 48.35%, 80.96% και 60.79% του μέσου bonus της ομάδας τους. Επειδή υπάρχουν πέντε υποψήφιοι μετακινούμενοι και τέσσερις πιθανές ομάδες-στόχοι για τον καθένα, εξετάζονται 20 candidate migrations.

Το αποτέλεσμα είναι ότι δεν υπάρχει καμία blocking accepted migration. Από τις 20 υποψήφιες μετακινήσεις, σε 6 ο μετακινούμενος χρήστης θα βελτιωνόταν προσωπικά. Όμως σε όλες αυτές τις περιπτώσεις τουλάχιστον ένα τωρινό μέλος της ομάδας υποδοχής θα έχανε bonus, άρα η ομάδα υποδοχής δεν θα αποδεχόταν τη μετακίνηση. Η μεγαλύτερη προσωπική βελτίωση εμφανίζεται για τον χρήστη 9 από τον υποσυνασπισμό 1 προς τον υποσυνασπισμό 5: το ετήσιο bonus του θα αυξανόταν κατά 79.64%. Όμως η ίδια μετακίνηση θα μείωνε το bonus μελών της ομάδας υποδοχής, με τη χειρότερη μεταβολή να είναι -34.64%, οπότε απορρίπτεται.

Αυτό το αποτέλεσμα συμπληρώνει τα προηγούμενα κεφάλαια. Οι υποσυνασπισμοί δεν είναι μόνο αποδοτικοί σε συνολικό επίπεδο, αλλά δεν δημιουργούν, υπό το συγκεκριμένο κριτήριο, αποδεκτή ατομική απόκλιση για τα πιο αδύναμα μέλη τους. Η εργασία δεν ισχυρίζεται ότι έχει αποδείξει πλήρη ευστάθεια έναντι όλων των πιθανών συλλογικών αποκλίσεων ή ανταλλαγών πολλών παικτών. Δείχνει όμως ότι ο συνδυασμός score-based formation και positive marginal-contribution profit sharing περνά ένα πρακτικό και ουσιαστικό τεστ ατομικών κινήτρων.

1.7 Απόκριση ζήτησης ως στοχευμένη μετατόπιση φορτίου

Το κεφάλαιο της απόκρισης ζήτησης εξετάζει μια διαφορετική μορφή ευελιξίας. Εδώ ο στόχος δεν είναι ο σχηματισμός συνασπισμών ή η κατανομή κέρδους, αλλά η εκτίμηση του πρόσθετου οφέλους που μπορεί να δώσει ένα περιορισμένο σχήμα load shifting. Η ζήτηση δεν μειώνεται μόνιμα. Ένα μέρος της κατανάλωσης μεταφέρεται από μια ακριβή χρονική περίοδο σε μια ευνοϊκότερη περίοδο, ώστε η συνολική ημερήσια ενέργεια να

παραμένει ίδια.

Το πείραμα χρησιμοποιεί το ίδιο καθαρισμένο Irish dataset, την ίδια χρονική ανάλυση 15 λεπτών, 20 χρήστες και 343 έγκυρες ημέρες. Ως benchmark χρησιμοποιείται κεντρική ιδανική μπαταρία με ημερήσια αρχική και τελική κατάσταση φόρτισης ίση με μηδέν, ώστε τα αποτελέσματα να είναι συγκρίσιμα ανά χωρητικότητα και να απομονώνεται το πρόσθετο αποτέλεσμα της απόκρισης ζήτησης.

Η μέθοδος έχει δύο βήματα. Πρώτα επιλέγεται για κάθε ημέρα ένα activation window διάρκειας μίας ώρας, δηλαδή τέσσερα συνεχόμενα δεκαπεντάλεπτα. Το παράθυρο επιλέγεται ως αυτό με το μεγαλύτερο aggregate import cost:

$$\sum_{t \in A} \pi_t^{\text{buy}} \max\{L_t, 0\},$$

όπου L_t είναι το συνολικό καθαρό φορτίο της κοινότητας. Άρα η παρέμβαση στοχεύει ακριβώς στην περίοδο όπου οι εισαγωγές από το δίκτυο είναι ακριβότερες για την κοινότητα.

Στη συνέχεια εφαρμόζεται cap στην κατανάλωση κάθε χρήστη μέσα στο activation window. Υπολογίζεται η μέση κατανάλωση όλων των χρηστών στο παράθυρο και κάθε χρήστης περιορίζεται στο 80% αυτής της μέσης τιμής. Μόνο οι χρήστες που ξεπερνούν το cap επηρεάζονται. Η περίσσεια ενέργεια αφαιρείται αναλογικά από το activation window και προστίθεται ξανά σε recovery window διάρκειας δύο ωρών. Η τελική εκδοχή του μοντέλου χρησιμοποιεί battery-aware recovery rule: η περίοδος ανάκτησης δεν επιλέγεται απλώς με βάση τη χαμηλότερη τιμή αγοράς, αλλά με βάση το μετα-μπαταριακό οριακό κόστος, ώστε να λαμβάνεται υπόψη ότι η μπαταρία μπορεί ήδη να έχει εκμεταλλευτεί ορισμένες ευκαιρίες arbitrage.

Τα αποτελέσματα δείχνουν ότι η στοχευμένη απόκριση ζήτησης έχει θετικό αλλά περιορισμένο οικονομικό αποτέλεσμα στην οικιακή κοινότητα. Χωρίς μπαταρία, το ετήσιο κόστος μειώνεται από 6758.31 EUR σε 6623.30 EUR. Η εξοικονόμηση είναι 135.01 EUR, δηλαδή 2.00% του αρχικού κόστους. Στις 343 ημέρες, ο μηχανισμός μετατοπίζει συνολικά 2834.50 kWh και επηρεάζει κατά μέσο όρο 8.74 χρήστες ανά ημέρα. Η μέση αξία της μετατοπισμένης ενέργειας χωρίς μπαταρία είναι 0.0476 EUR/kWh.

Όταν υπάρχει μπαταρία, το πρόσθετο όφελος της απόκρισης ζήτησης μειώνεται. Με κεντρική μπαταρία 20 kWh, η εξοικονόμηση από demand response είναι 104.16 EUR ή 1.65%. Με 40 kWh είναι 85.87 EUR ή 1.42%. Με 60 kWh είναι 71.58 EUR ή 1.23%. Με 80 kWh είναι 58.31 EUR ή 1.03%. Η αξία ανά μετατοπισμένη kWh πέφτει από 0.0476 EUR/kWh χωρίς μπαταρία σε 0.0206 EUR/kWh με 80 kWh μπαταρία. Αυτό συμβαίνει επειδή η μπαταρία και η απόκριση ζήτησης ανταγωνίζονται εν μέρει για την ίδια ευκαιρία arbitrage: και τα δύο μέσα προσπαθούν να μειώσουν κατανάλωση ή εισαγωγές σε

ακριβές περιόδους και να τις μεταφέρουν σε ευνοϊκότερες περιόδους.

Η εργασία συγκρίνει επίσης τα μικρά battery cases με μια μεγάλη μπαταρία αναφοράς 185 kWh. Χωρίς μπαταρία, το κόστος είναι 6758.31 EUR. Με no battery + demand response γίνεται 6623.30 EUR, δηλαδή επιτυγχάνεται 2.00% μείωση και 8.27% του κέρδους που θα έδινε η μπαταρία των 185 kWh. Μια μπαταρία 80 kWh μόνη της μειώνει το κόστος σε 5667.00 EUR, δηλαδή 16.15% σε σχέση με τη no-battery βάση και 66.82% του κέρδους της μεγάλης μπαταρίας. Με 80 kWh μπαταρία και demand response, το κόστος γίνεται 5608.68 EUR, δηλαδή 17.01% μείωση και 70.39% του κέρδους της 185 kWh αναφοράς. Άρα το demand response κλείνει μέρος του κενού προς τη μεγάλη μπαταρία, αλλά δεν υποκαθιστά τη σημαντική αξία της μεγαλύτερης αποθήκευσης.

Η ερμηνεία είναι ότι στο οικιακό επίπεδο η διαθέσιμη ευέλικτη ζήτηση είναι σχετικά περιορισμένη. Επιπλέον, ένα πραγματικό πρόγραμμα demand response θα έπρεπε να λάβει υπόψη άνεση χρηστών, διαθεσιμότητα συσκευών, rebound effects και αποζημίωση για την ενόχληση. Επομένως, το τεχνικό δυναμικό που εκτιμάται εδώ πιθανώς υπερεκτιμά αυτό που θα ήταν πρακτικά διαθέσιμο χωρίς κατάλληλο μηχανισμό κινήτρων. Από την πλευρά όμως ενός operator ή aggregator, η ίδια ιδέα μπορεί να έχει μεγαλύτερη αξία, επειδή η κλίμακα είναι μεγαλύτερη, μπορεί να υπάρχουν αρνητικές τιμές ενέργειας, imbalance costs και ανάγκη για ευελιξία σε επίπεδο MWh όταν η αποθήκευση μεγάλης κλίμακας δεν είναι διαθέσιμη ή οικονομικά ελκυστική.

1.8 Συνολικά συμπεράσματα, περιορισμοί και μελλοντική εργασία

Τα συνολικά αποτελέσματα υποστηρίζουν την κεντρική θέση της διπλωματικής: η συνεργασία σε οικιακές ενεργειακές κοινότητες μπορεί να δημιουργήσει μετρήσιμη οικονομική αξία, αλλά η πρακτική αξιοποίησή της απαιτεί ταυτόχρονα λειτουργικό μοντέλο, κανόνα κατανομής, μέθοδο σχηματισμού ομάδων και έλεγχο κινήτρων. Η πλήρης συνεργασία όλων των χρηστών είναι χρήσιμο benchmark, αλλά δεν είναι πάντα η πιο πρακτική μορφή οργάνωσης. Μικρότεροι υποσυνασπισμοί μπορούν να διατηρήσουν μεγάλο μέρος της αξίας του μεγάλου συνασπισμού, αρκεί να σχηματιστούν με βάση τη συμπληρωματικότητα των προφίλ και την ισορροπία μεταξύ παραγωγής και ζήτησης.

Στο επίπεδο λειτουργίας, το βασικό εύρημα είναι ότι η ατομική μπαταρία και η συνεργασία δημιουργούν διαφορετικές και αθροιστικές πηγές αξίας. Στο 20-user ετήσιο πείραμα, το solo scheduling μειώνει το κόστος κατά 8.11% σε σχέση με την παθητική βάση. Η συνεργασία όλων των χρηστών ανεβάζει τη συνολική μείωση σε 12.21%, δηλαδή προσθέτει 4.10 ποσοστιαίες μονάδες σε σχέση με το solo scheduling. Αυτό δείχνει ότι η αξία της ενεργειακής κοινότητας δεν εξαντλείται στην ατομική ευελιξία κάθε νοι-

κοκυριού, αλλά περιλαμβάνει και την εσωτερική αντιστοίχιση πλεονάσματος, ζήτησης και αποθήκευσης μεταξύ διαφορετικών χρηστών.

Στο επίπεδο κατανομής, ο προτεινόμενος positive leave-one-out κανόνας αποτελεί έναν πρακτικό συμβιβασμό. Είναι πολύ φθηνότερος από τον ακριβή Shapley value, καθώς στο 8-player πείραμα απαιτεί μόνο 6.64% των coalition-value evaluations του Shapley, δηλαδή 93.36% λιγότερες optimization calls. Μοιράζει όλο το συνεργατικό πλεόνασμα, δίνει μη αρνητικά bonuses και συνδέει την αποζημίωση με την οριακή σημασία του χρήστη στον πραγματοποιημένο συνασπισμό. Δεν έχει τις πλήρεις αξιωματικές εγγυήσεις του Shapley value ή του nucleolus, αλλά είναι ερμηνεύσιμος και κλιμακώσιμος.

Στο επίπεδο σχηματισμού υποσυνασπισμών, η pairwise-score μέθοδος δείχνει ότι μεγάλο μέρος της αξίας του μεγάλου συνασπισμού μπορεί να διατηρηθεί με μικρότερες ομάδες. Τα score-based partitions ανακτούν από 87.35% έως 98.72% του grand-coalition coordination gain στα κύρια πειράματα, ενώ στην balanced 50-player περίπτωση με δέκα ομάδες των πέντε ανακτούν 89.45%, ξεπερνώντας καθαρά τον balanced random μέσο όρο 85.41%. Η ερμηνεία είναι ότι η ισορροπία prosumer-consumer είναι θεμελιώδης, αλλά όταν οι ομάδες μικραίνουν η συγκεκριμένη αντιστοίχιση χρηστών γίνεται κρίσιμη.

Στο επίπεδο ευστάθειας, το accepted-migration test δεν βρίσκει καμία αποδεκτή blocking migration στο 20-player ετήσιο πείραμα. Παρότι 30.00% των υποψήφιων μετακινήσεων θα βελτίωναν τον μετακινούμενο χρήστη, καμία δεν γίνεται αποδεκτή από την ομάδα υποδοχής, επειδή πάντα τουλάχιστον ένα υπάρχον μέλος της ομάδας υποδοχής χάνει bonus. Άρα, υπό αυτό το πρακτικό κριτήριο, οι επιλεγμένοι υποσυνασπισμοί δεν δημιουργούν προφανή ατομικά κίνητρα αποχώρησης για τα πιο αδύναμα μέλη.

Στο επίπεδο demand response, η στοχευμένη μετατόπιση φορτίου μειώνει το κόστος, αλλά με περιορισμένη ένταση στο οικιακό περιβάλλον. Η εξοικονόμηση είναι 2.00% χωρίς μπαταρία και πέφτει σε 1.03% με 80 kWh μπαταρία, επειδή η μπαταρία έχει ήδη αξιοποιήσει μέρος της ίδιας ευκαιρίας. Η απόκριση ζήτησης επομένως λειτουργεί περισσότερο ως συμπληρωματικός πόρος ευελιξίας παρά ως υποκατάστατο της αποθήκευσης.

Οι περιορισμοί της εργασίας δείχνουν και τις βασικές κατευθύνσεις συνέχειας. Πρώτον, τα μοντέλα είναι ντετερμινιστικά και χρησιμοποιούν παρατηρημένα προφίλ ζήτησης, παραγωγής και τιμών. Αυτό είναι κατάλληλο για την εκτίμηση του δυναμικού συνεργασίας σε κοινό ιστορικό dataset, αλλά αντιστοιχεί σε perfect-information setting. Μελλοντική εργασία θα μπορούσε να εισαγάγει stochastic ή robust scheduling με σφάλματα πρόβλεψης. Δεύτερον, το μοντέλο δεν περιλαμβάνει ρητά περιορισμούς δικτύου διανομής, όπως όρια τάσης, συμφόρηση γραμμών ή απώλειες. Η ενσωμάτωση τέτοιων περιορισμών θα επέτρεπε να αξιολογηθούν οι συνασπισμοί όχι μόνο οικονομικά αλλά και φυσικά. Τρίτον, ο έλεγχος ευστάθειας είναι περιορισμένος: εξετάζει μόνο τον ασθενέ-

στερο χρήστη κάθε ομάδας και μόνο μονομερείς μετακινήσεις που αποδέχεται ο στόχος. Μελλοντικές επεκτάσεις θα μπορούσαν να εξετάσουν όλες τις μονομερείς μετακινήσεις, ανταλλαγές παικτών ή συλλογικές αποκλίσεις. Τέταρτον, το demand-response μοντέλο είναι ιδεατό και aggregate, χωρίς appliance-level constraints, comfort preferences ή μηχανισμό αποζημίωσης. Τέλος, το πλαίσιο θα πρέπει να δοκιμαστεί σε περισσότερα datasets, τιμολογιακά περιβάλλοντα και συνθέσεις κοινοτήτων.

Συνολικά, η διπλωματική δείχνει ότι οι ενεργειακές κοινότητες πρέπει να μελετώνται ως συντονισμένα συστήματα και όχι ως απλές συλλογές ανεξάρτητων παγίων. Οι μπαταρίες, η φωτοβολταϊκή παραγωγή, η ευέλικτη ζήτηση και η εσωτερική ανταλλαγή ενέργειας αλληλεπιδρούν. Η αξία τους εξαρτάται από τον χρόνο, τη συμπληρωματικότητα των προφίλ, τη δομή των ομάδων και τους κανόνες με τους οποίους μοιράζεται το αποτέλεσμα. Η κύρια συνεισφορά της εργασίας είναι ότι συνδέει αυτά τα στοιχεία σε ένα ενιαίο, υπολογιστικά κλιμακώσιμο και ερμηνεύσιμο πλαίσιο για τον σχεδιασμό συνεργατικών ενεργειακών κοινοτήτων.

Chapter 2

Introduction

The increasing penetration of distributed renewable generation, residential battery storage and smart metering is changing the way electricity systems are organised at the distribution level. Households are no longer only passive consumers of electricity. They can also produce energy through photovoltaic installations, store energy locally, adjust part of their demand and interact with other users. This transition has motivated the development of new market and coordination structures for prosumers, namely users who both consume and produce electricity [1].

Renewable energy communities are one of the most important frameworks through which this transition can be organised. Their purpose is not simply to place several individual users under a common label, but to enable local coordination between heterogeneous resources. A household with photovoltaic surplus may be able to support another household that is importing from the grid at the same time. A battery can shift energy from periods of local surplus to periods of higher demand or higher prices. If these actions are coordinated, part of the energy that would otherwise be exported at a low price or imported at a higher price can instead be used within the community.

This local matching opportunity is the central economic motivation of the thesis. The benefit of an energy community depends not only on the amount of photovoltaic generation or storage installed, but also on the timing of demand, production and flexibility across different users. Recent reviews therefore emphasise that energy communities must be studied as socio-technical systems, where technical operation, ownership, market design and incentive structures all affect their performance [2], [3], [4]. Optimisation studies also show that community composition, renewable capacity, storage investment and sharing rules can significantly influence both economic and technical outcomes [5], [6], [7], [8].

In this setting, the grand coalition, where all users coordinate their operation together, provides a natural benchmark. It represents the maximum coordination opportunity available to the community, because the full group can reproduce the operation of any collection of smaller independent groups. However, the grand coalition is not always the most practical structure. As the number of users increases, solving one large optimisation problem can become computationally demanding. It can also require a level of information sharing and organisational agreement that may be difficult to achieve in practice. This creates another tension addressed in the thesis: full coordination is economically attractive, but smaller cooperative groups may

be easier to compute, operate and govern.

2.1 Previous Work

The literature on peer-to-peer energy trading provides an important foundation for this problem. Peer-to-peer platforms have been proposed as a way to coordinate distributed energy resources and allow prosumers to trade directly with one another [9], [10], [11], [12]. Game-theoretic models have been used to describe strategic interactions between buyers and sellers in local energy markets [13], while scheduling studies show that combining battery operation with local energy exchange can improve participant outcomes compared with independent operation [14], [15], [16]. These studies support the idea that local exchange can create real economic value, especially when production and consumption profiles are complementary.

Coalition formation is closely related to this idea. If cooperation is valuable, then it matters which users cooperate with each other. Previous work has studied prosumer coalitions for peer-to-peer trading with storage [17], [18], the formation of valuable trading relationships [19], and cooperative-game mechanisms for stabilising a grand prosumer coalition [20]. Other studies also consider the interaction between local energy exchange and operational network constraints [21]. These contributions show that local cooperation must be evaluated not only as an engineering problem, but also as a question of incentives, participation and group formation.

Once a coalition creates a cost reduction, this gain must be allocated among its members. Cooperative game theory provides the standard language for this question. The Shapley value is a central fairness benchmark because it distributes value according to average marginal contribution [22]. The nucleolus provides a different stability-oriented perspective by focusing on coalitional dissatisfaction [23]. In energy-community applications, related allocation ideas have been used for storage cost sharing, prosumer cooperation and community self-consumption [24], [25], [26], [27], [28]. The common message is that an aggregate saving is not sufficient by itself. A cooperative scheme also needs a credible and interpretable way to share the benefit.

The difficulty is that the most attractive allocation and coalition-formation concepts can be computationally expensive. The exact Shapley value requires the value of every possible coalition [29], [30]. Exact partitioning of a community into subcoalitions also becomes combinatorial as the number of users grows [31]. This difficulty is especially important when coalition values are not simple formulas, but are obtained by solving operational scheduling problems with batteries, grid exchanges and internal energy flows.

Representative-period methods offer one way to reduce the temporal dimension of energy optimisation. Time-series aggregation and representative-day selection can reduce computational requirements, but they must preserve the operational features that matter for the problem being solved, such as peaks, net-load structure and storage-relevant dynamics [32], [33], [34],

[35], [36], [37]. For this reason, representative days are useful only if they are connected to the final operational question. In this thesis they are used as a search aid for subcoalition formation, while the final evaluation is still performed over the full available annual horizon.

The research gap addressed in this thesis lies in the connection between these elements. Energy-community optimisation, peer-to-peer energy sharing, cooperative-game allocation, representative-day reduction and coalition formation are often studied separately. This thesis combines them into a single computational framework. Coalition values are obtained from an operational scheduling model rather than assumed abstractly. The cooperation gain is allocated using a scalable marginal-contribution rule. Smaller subcoalitions are formed using optimisation-derived pairwise scores, and their performance is tested against the grand coalition over the full year. Finally, the resulting groups are tested from an individual-incentive perspective through a migration-based stability criterion.

2.2 Aim of the Thesis

The aim of this thesis is to study how a residential renewable energy community can be coordinated in a way that is economically beneficial, computationally scalable and compatible with individual participation incentives. The empirical setting is based on the Irish energy-community dataset, which provides high-resolution household consumption, photovoltaic production, grid exchange and battery-related measurements [38]. The analysis uses 15-minute profiles from the year 2020 and focuses on the days for which complete household and price data are available.

The thesis is organised around three connected questions. First, it quantifies the value created by moving from passive operation to individual battery scheduling and then to cooperative scheduling. Second, it studies whether the coordination gain of a coalition can be allocated in a way that is interpretable and much cheaper to compute than exact cooperative-game benchmarks. Third, it examines whether a larger community can be decomposed into smaller subcoalitions while retaining most of the value of the grand coalition.

These questions are deliberately linked. It is not enough to show that a grand coalition reduces total cost. A practical framework must also explain how the gain is distributed, how smaller groups can be selected, and whether users have incentives to remain in those groups after the benefits have been allocated.

2.3 Methodology

The first layer of the methodology is an optimisation model for daily energy-community operation. Three operating modes are compared. The baseline represents passive operation, where users do not schedule batteries and do not exchange energy internally. The solo case allows

each user to optimise its own battery independently. The cooperative case allows a coalition of users to coordinate battery operation and internal energy exchange before interacting with the external grid. The coordination gain of a coalition is therefore measured relative to the already optimised solo operation of its members. This distinction is important: cooperation is credited only with the additional value created by coordination, not with the value that users could already obtain independently.

The second layer is profit allocation. The thesis proposes a positive leave-one-out marginal contribution rule. The rule asks how much cooperation surplus is lost when each user is removed from the realised coalition. Users with positive marginal importance receive a proportional share of the cooperation gain, while the full surplus is distributed among the coalition members. This rule is less demanding than the exact Shapley value because it only requires coalition evaluations around the realised group, rather than all possible subcoalitions.

The third layer is subcoalition formation. Instead of evaluating every possible subcoalition directly, the thesis constructs pairwise coordination scores. A small set of representative days is selected using k-medoids clustering of aggregate community net-load profiles, together with explicitly selected high-import and high-surplus days. For each pair of users, the cooperative value of placing them together is estimated on this representative set. These pairwise scores are then used to form equal-size subcoalitions.

The partitioning objective is designed to be both efficient and balanced. It first tries to maximise the weakest subcoalition score, so that no group is left with very low coordination potential. Subject to that, it improves the total score of the partition. Exact mixed-integer optimisation is used where the problem size is still manageable, and a multi-start local-search heuristic is used for larger cases. After the groups are formed, their performance is evaluated on the full annual horizon and compared with the grand coalition and with balanced random partitions.

The final layer is a stability test. After the cooperation gain is allocated within the selected subcoalitions, the thesis checks whether the weakest member of each group has a credible incentive to migrate to another group. A migration is considered blocking only if the moving user improves and the receiving group would accept the user because none of its current members is made worse off. This test does not claim full coalition-proof stability, but it provides a practical indication of whether the proposed grouping and allocation procedure creates obvious individual deviations.

2.4 Contributions

The contribution of the thesis is the development and evaluation of an integrated framework for cooperative energy-community operation. More specifically, the thesis:

1. formulates a layered optimisation framework for passive baseline operation, individual battery scheduling, grand-coalition coordination and subcoalition operation;

2. quantifies cooperation as the additional cost reduction achieved beyond stand-alone battery scheduling;

3. proposes a scalable marginal-contribution allocation rule and compares it with the exact Shapley value in a tractable benchmark case;

4. develops a representative-day pairwise scoring method for estimating user compatibility in cooperative scheduling;

5. forms balanced subcoalitions through a lexicographic max-min objective and a scalable local-search heuristic;

6. evaluates the selected subcoalitions over the full annual horizon against the grand coalition and balanced random partitions;

7. examines whether the resulting subcoalitions remain stable under an accepted single-player migration test.

2.5 Main Findings

The results show that both battery scheduling and cooperation matter. In the 20-user annual evaluation, individual battery scheduling reduces the passive baseline cost by about 8.1%. Allowing the same users to coordinate as a grand coalition increases the total saving to about 12.2% relative to the baseline. This confirms that individual battery flexibility already creates value, but that additional value is obtained when users are allowed to share energy internally and coordinate their storage operation.

The allocation experiment shows that the proposed marginal-contribution rule remains close to the exact Shapley benchmark in the small case where the Shapley value can still be computed, while requiring far fewer coalition-value evaluations. This supports its use as a scalable allocation rule for the larger annual experiments.

The subcoalition experiments show that smaller groups can retain a large share of the grand-coalition coordination gain. Across the main annual cases, the selected subcoalitions recover roughly 87% to 99% of the grand-coalition coordination benefit. The 30-user local-search partition recovers about 94.3%, while the original 50-user case recovers about 98.7%. In the balanced 50-user case with equal numbers of prosumers and consumers, five groups of ten users recover about 94.8%, while ten smaller groups of five users recover about 89.5%. This

shows that smaller groups are more restrictive, but can still preserve most of the value of full coordination.

The comparison with balanced random partitions adds an important interpretation. When groups are large and each group contains a reasonable mix of prosumers and consumers, simple balance already captures much of the available coordination value. When groups become smaller, the exact matching of users becomes more important. In the balanced 50-user case with ten groups of five, the score-based method clearly outperforms the balanced random benchmark, recovering about 89.5% of the grand-coalition gain compared with about 85.4% on average for balanced random partitions.

Finally, the accepted-migration tests find no blocking migrations in the examined annual cases. This means that, under the proposed profit-sharing rule and the tested migration criterion, the weakest members of the selected subcoalitions do not have a credible accepted move to another group. The resulting partitions are therefore not only effective in aggregate, but also stable under this practical individual-deviation test.

2.6 Organization

The remainder of the thesis follows the structure of the proposed framework. The next chapter presents the data preparation and the optimisation model used to describe baseline operation, individual battery scheduling and cooperative scheduling. The following chapter introduces the profit-allocation rule and compares it with cooperative game-theoretic benchmarks. The subcoalition-formation chapter then develops the representative-day selection procedure, the pairwise-score approximation and the lexicographic grouping method. The annual evaluation chapter compares the selected subcoalitions with the grand coalition and with balanced random partitions. The stability chapter examines accepted single-player migrations after profit allocation. The thesis concludes by summarising the main findings, discussing limitations and outlining possible directions for future work.

Chapter 3

Optimization Model

3.1 Problem Setting and Notation

This chapter presents the optimization framework used to model the operation of individual users and coalitions of users in a renewable energy community. The analysis is developed in three layers. First, a baseline operating mode without battery scheduling is defined. Second, each user independently optimizes the operation of its own battery. Third, groups of users are allowed to coordinate battery operation and exchange energy internally, thereby forming coalitions that can reduce their total energy cost.

To make these layers easier to follow, consider a small running example with three users. User A is a prosumer with photovoltaic generation and often has surplus energy around midday. User B is mainly a consumer whose demand is higher in the evening. User C has a more flexible profile and can shift part of the energy balance through a battery. The sections below ask the same question in increasingly richer settings: what happens if the users simply follow their raw demand and generation profiles, what improves when each user schedules only its own battery, and what additional value appears when the users are allowed to coordinate as a group.

Let $\mathcal{N} = \{1, 2, \ldots, n\}$ denote the set of users participating in the community, and let $\mathcal{T} = \{1, 2, \ldots, T\}$ denote the set of time periods in the optimization horizon. In this thesis, one day is divided into $T = 96$ time intervals of equal duration $\Delta t = 0.25$ h, corresponding to a temporal resolution of 15 minutes.

For each user $i \in \mathcal{N}$ and time period $t \in \mathcal{T}$, the following exogenous inputs are assumed to be known:

- $d_{i,t}$: electricity demand of user i during period t, expressed as average power in kW,
- $g_{i,t}$: local renewable generation of user i during period t, expressed as average power in kW,
- π_t^{buy}: electricity import price from the grid in EUR/kWh,
- π_t^{sell}: electricity export price to the grid in EUR/kWh.

Throughout the chapter, the following price ordering is assumed:

$$\pi_t^{\text{buy}} \geq \pi_t^{\text{sell}}, \quad \forall t \in \mathcal{T}. \tag{3.1}$$

This reflects the common situation in which importing electricity from the external grid is more expensive than exporting surplus energy to it.

The optimization variables are expressed as average power in kW, whereas prices are given in EUR/kWh. Therefore, whenever a power variable $x_{i,t}$ appears in a cost term, it is multiplied by the duration of the time step, Δt, in order to obtain the corresponding energy quantity in kWh.

Each user may be equipped with a battery energy storage system. For user i, the battery is described by the following parameters:

- B_i: nominal (nameplate) battery energy capacity in kWh,
- DoD_i: admissible depth of discharge (maximum fraction of B_i that can be used without harming the battery),
- $\bar{E}_i = B_i \text{DoD}_i$: usable battery energy capacity in kWh (energy that can be charged/discharged in operation),
- $\bar{p}_i^{\text{ch}}$: maximum charging power in kW (upper limit on how fast the battery can be charged),
- $\bar{p}_i^{\text{dis}}$: maximum discharging power in kW (upper limit on how fast the battery can supply power),
- η_i^{rt}: round-trip efficiency (fraction of charged energy that can be recovered after a full charge–discharge cycle),
- η_i^{ch} and η_i^{dis}: charging and discharging efficiencies (loss factors during charging/discharging),
- $E_{i,0}$: initial state of charge in kWh (stored energy at the beginning of the horizon),
- c_i^{deg}: battery degradation cost coefficient in EUR/kWh (monetary cost assigned per kWh of energy throughput),
- α^{deg}: weight applied to the degradation term (tuning parameter that scales the importance of degradation costs).

Following the common symmetric representation of charging and discharging losses, the charging and discharging efficiencies are defined as

$$\eta_i^{\text{ch}} = \eta_i^{\text{dis}} = \sqrt{\eta_i^{\text{rt}}}. \tag{3.2}$$

To avoid end-of-horizon artefacts, the state of charge is defined for $t = 0, 1, \ldots, T$, while all power variables are defined for $t \in \mathcal{T}$. The terminal state of charge is constrained to be equal to the initial one.

3.2 Baseline Operation

The baseline answers the simplest possible question: what would each user's cost be if no one made any scheduling or sharing decision? In the running example, if user A produces more photovoltaic energy than needed at noon, that surplus is exported to the grid. If user B needs energy in the evening, that demand is imported from the grid. No battery is charged or discharged, and no energy is sent directly from A to B or C. The baseline therefore represents the passive reference case against which all later improvements are measured. We now formalize this passive operating mode by defining the corresponding grid import and export quantities.

In the baseline, local generation is first used to cover local demand. Any remaining deficit is imported from the grid, while any surplus generation is exported to the grid.

For each user $i \in \mathcal{N}$ and time period $t \in \mathcal{T}$, the baseline grid import and export powers are defined as

$$p_{i,t}^{\text{imp,base}} = \max\{0,\, d_{i,t} - g_{i,t}\}, \quad \forall i \in \mathcal{N},\, \forall t \in \mathcal{T}, \tag{3.3}$$

and

$$p_{i,t}^{\text{exp,base}} = \max\{0,\, g_{i,t} - d_{i,t}\}, \quad \forall i \in \mathcal{N},\, \forall t \in \mathcal{T}. \tag{3.4}$$

The baseline cost of user i is then computed as

$$C_i^{\text{base}} = \sum_{t \in \mathcal{T}} \left(\pi_t^{\text{buy}} p_{i,t}^{\text{imp,base}} \Delta t - \pi_t^{\text{sell}} p_{i,t}^{\text{exp,base}} \Delta t \right) + F_i, \tag{3.5}$$

where F_i denotes a fixed cost, if applicable.

For any subset of users $S \subseteq \mathcal{N}$, the aggregate baseline cost is

$$C^{\text{base}}(S) = \sum_{i \in S} C_i^{\text{base}}. \tag{3.6}$$

The baseline is not the solution of an optimization problem. It is only used as a reference case against which the benefits of battery scheduling and cooperation are evaluated.

3.3 Individual Battery Scheduling

The individual scheduling layer adds one form of intelligence: each user can use its own battery to reduce its own cost, but still acts alone. In the running example, user A may store part of

the midday surplus and use it later instead of exporting it immediately. User B may charge the battery during cheaper periods and discharge it during expensive evening periods. However, A's surplus cannot be assigned to B, and C's flexibility cannot be used for the benefit of the other users. This layer therefore isolates the value of private flexibility before any cooperation is introduced. We now formulate this setting as a stand-alone optimization problem for each user.

Each user $i \in \mathcal{N}$ independently optimizes the operation of its own battery while still interacting only with the external grid. For each time period $t \in \mathcal{T}$, the decision variables are:

- $p_{i,t}^{\text{ch}} \geq 0$: battery charging power,
- $p_{i,t}^{\text{dis}} \geq 0$: battery discharging power,
- $E_{i,t}$: battery state of charge, for $t = 0, 1, \ldots, T$,
- $p_{i,t}^{\text{imp}} \geq 0$: grid import power,
- $p_{i,t}^{\text{exp}} \geq 0$: grid export power.

A binary variable $u_{i,t}^{\text{bat}} \in \{0, 1\}$ is introduced to prevent simultaneous charging and discharging:

$$0 \leq p_{i,t}^{\text{ch}} \leq \bar{p}_i^{\text{ch}} u_{i,t}^{\text{bat}}, \quad \forall i \in \mathcal{N},\ \forall t \in \mathcal{T}, \tag{3.7}$$

$$0 \leq p_{i,t}^{\text{dis}} \leq \bar{p}_i^{\text{dis}} \left(1 - u_{i,t}^{\text{bat}}\right), \quad \forall i \in \mathcal{N},\ \forall t \in \mathcal{T}. \tag{3.8}$$

If explicit prevention of simultaneous grid import and export is desired, an additional binary variable $u_{i,t}^{\text{grid}} \in \{0, 1\}$ can be used:

$$0 \leq p_{i,t}^{\text{imp}} \leq \bar{p}_i^{\text{imp}} u_{i,t}^{\text{grid}}, \quad \forall i \in \mathcal{N},\ \forall t \in \mathcal{T}, \tag{3.9}$$

$$0 \leq p_{i,t}^{\text{exp}} \leq \bar{p}_i^{\text{exp}} \left(1 - u_{i,t}^{\text{grid}}\right), \quad \forall i \in \mathcal{N},\ \forall t \in \mathcal{T}. \tag{3.10}$$

The power balance of user i at each time period is

$$p_{i,t}^{\text{imp}} + p_{i,t}^{\text{dis}} + g_{i,t} = d_{i,t} + p_{i,t}^{\text{ch}} + p_{i,t}^{\text{exp}}, \quad \forall i \in \mathcal{N},\ \forall t \in \mathcal{T}. \tag{3.11}$$

The terms in the power balance can be read as:

- sources (left-hand side): grid import, battery discharge, and local generation,
- sinks (right-hand side): demand, battery charge, and grid export.

The battery state evolves according to

$$E_{i,t} = E_{i,t-1} + \eta_i^{\text{ch}} p_{i,t}^{\text{ch}} \Delta t - \frac{1}{\eta_i^{\text{dis}}} p_{i,t}^{\text{dis}} \Delta t, \quad \forall i \in \mathcal{N},\ \forall t \in \mathcal{T}. \tag{3.12}$$

The state update consists of:

- charged energy added to the state of charge: $\eta_i^{\text{ch}} p_{i,t}^{\text{ch}} \Delta t$,
- discharged energy removed from the state of charge: $\frac{1}{\eta_i^{\text{dis}}} p_{i,t}^{\text{dis}} \Delta t$.

The state of charge is bounded by the usable capacity:

$$0 \leq E_{i,t} \leq \bar{E}_i, \quad \forall i \in \mathcal{N},\ \forall t = 0, 1, \ldots, T. \tag{3.13}$$

To avoid artificial discharge or charge at the end of the horizon, the following periodicity constraint is imposed:

$$E_{i,T} = E_{i,0}, \quad \forall i \in \mathcal{N}. \tag{3.14}$$

The optimization problem solved independently by each user is

$$C_i^{\text{solo}} = \min \sum_{t \in \mathcal{T}} \left[\pi_t^{\text{buy}} p_{i,t}^{\text{imp}} \Delta t - \pi_t^{\text{sell}} p_{i,t}^{\text{exp}} \Delta t + \alpha^{\text{deg}} \frac{c_i^{\text{deg}}}{2} \left(p_{i,t}^{\text{ch}} + p_{i,t}^{\text{dis}} \right) \Delta t \right] + F_i, \tag{3.15}$$

(minimum achievable individual cost with optimal battery scheduling)

The objective function in (3.15) consists of the following terms:

- grid import cost: $\pi_t^{\text{buy}} p_{i,t}^{\text{imp}} \Delta t$,
- grid export revenue (subtracted): $\pi_t^{\text{sell}} p_{i,t}^{\text{exp}} \Delta t$,
- battery degradation cost: $\alpha^{\text{deg}} \frac{c_i^{\text{deg}}}{2} (p_{i,t}^{\text{ch}} + p_{i,t}^{\text{dis}}) \Delta t$,
- fixed cost: F_i.

subject to the operational constraints above.

The optimized individual cost C_i^{solo} is used later as the stand-alone reference for measuring the additional value of cooperation.

3.4 Cooperative Coalition Optimization

The cooperative layer removes the restriction that each user must solve the problem alone. A coalition is treated as a small coordinated community: energy can be exchanged internally, and the battery schedules of all members are chosen together. In the running example, user A's surplus can help cover user B's demand, while user C's battery can create additional flexibility for the group. The objective is not to minimize each bill separately at each step, but to minimize the total cost of the coalition. Any question about how the resulting gain is divided among users

is handled later by the profit-allocation rule. We now define the coalition decision variables and constraints.

Consider a coalition of users $S \subseteq \mathcal{N}$. Members of S coordinate battery operation and may exchange energy internally before settling any net deficit or surplus with the external grid.

For each ordered pair of distinct users $i, j \in S$, and each time period $t \in \mathcal{T}$, let

$$q_{i,j,t} \geq 0 \tag{3.16}$$

denote the power transferred internally from user i to user j.

Under this notation, the power balance of user $i \in S$ becomes

$$p_{i,t}^{\text{imp}} + p_{i,t}^{\text{dis}} + g_{i,t} + \sum_{\substack{j \in S \\ j \neq i}} q_{j,i,t} = d_{i,t} + p_{i,t}^{\text{ch}} + p_{i,t}^{\text{exp}} + \sum_{\substack{j \in S \\ j \neq i}} q_{i,j,t}, \quad \forall i \in S,\ \forall t \in \mathcal{T}. \tag{3.17}$$

The additional terms relative to the individual balance are the internal transfers:

- $\sum_{j \in S,\, j \neq i} q_{j,i,t}$ is the total power received by user i from other coalition members at time t,
- $\sum_{j \in S,\, j \neq i} q_{i,j,t}$ is the total power sent by user i to other members at time t.

The battery constraints remain identical to those of the individual scheduling problem, but are now enforced for every user in the coalition.

The total cooperative cost of coalition S is obtained by solving

$$\begin{aligned} C^{\text{coal}}(S) = \min \sum_{i \in S} \sum_{t \in \mathcal{T}} \Big[\pi_t^{\text{buy}} p_{i,t}^{\text{imp}} \Delta t - \pi_t^{\text{sell}} p_{i,t}^{\text{exp}} \Delta t + \alpha^{\text{deg}} \frac{c_i^{\text{deg}}}{2} \big(p_{i,t}^{\text{ch}} + p_{i,t}^{\text{dis}} \big) \Delta t \Big] \\ + \sum_{i \in S} F_i + \varepsilon^{\text{tr}} \sum_{i \in S} \sum_{\substack{j \in S \\ j \neq i}} \sum_{t \in \mathcal{T}} q_{i,j,t} \end{aligned} \tag{3.18}$$

(minimum total cost of coalition S under coordinated operation)

subject to all power balance, battery dynamics, power limits and state-of-charge constraints.

The objective function in (3.18) consists of the following terms:

- grid import cost: $\pi_t^{\text{buy}} p_{i,t}^{\text{imp}} \Delta t$,
- grid export revenue (subtracted): $\pi_t^{\text{sell}} p_{i,t}^{\text{exp}} \Delta t$,
- battery degradation cost: $\alpha^{\text{deg}} \frac{c_i^{\text{deg}}}{2} (p_{i,t}^{\text{ch}} + p_{i,t}^{\text{dis}}) \Delta t$,
- fixed cost: F_i,

- internal-transfer regularization: $\varepsilon^{\text{tr}} \sum q_{i,j,t}$.

Here, ε^{tr} is a very small regularization coefficient that discourages redundant internal exchanges without materially changing the economic objective.

3.5 Coalition Value and Economic Interpretation

The economic value created by cooperation is measured relative to the optimized stand-alone operation of the users. Accordingly, the value of coalition $S \subseteq \mathcal{N}$ is defined as

$$v(S) = \sum_{i \in S} C_i^{\text{solo}} - C^{\text{coal}}(S), \tag{3.19}$$

(additional saving from coordinating within S vs. acting alone)

This quantity captures the extra saving achieved when the users in S coordinate their operation instead of optimizing independently.

For the grand coalition $\mathcal{N}$, three economic indicators are of particular interest:

$$G^{\text{sched}}(\mathcal{N}) = C^{\text{base}}(\mathcal{N}) - \sum_{i \in \mathcal{N}} C_i^{\text{solo}}, \tag{3.20}$$

(value of individual scheduling vs. baseline)

$$G^{\text{coord}}(\mathcal{N}) = \sum_{i \in \mathcal{N}} C_i^{\text{solo}} - C^{\text{coal}}(\mathcal{N}), \tag{3.21}$$

(extra value from coordination beyond solo scheduling)

$$G^{\text{total}}(\mathcal{N}) = C^{\text{base}}(\mathcal{N}) - C^{\text{coal}}(\mathcal{N}), \tag{3.22}$$

(total value of the coordinated solution vs. baseline)

The first term quantifies the value of individual battery scheduling relative to the passive baseline. The second quantifies the additional value of cooperation beyond stand-alone battery optimization. The third gives the total improvement of the coordinated solution relative to the baseline.

3.6 Implementation Remarks

In the implementation used in this thesis, all optimization problems are formulated as mixed-integer linear programming problems because they include both continuous variables and binary variables enforcing mutually exclusive operating modes. The models are solved with the SCIP solver through the OR-Tools optimization interface.

For the experiments presented later, the selected households are modeled with identical battery parameters: nominal capacity of 10 kWh, maximum charging and discharging power of 3.3 kW, round-trip efficiency of 0.95, usable capacity equal to $0.85 \times 10 = 8.5$ kWh, and initial state of charge equal to 50% of the usable capacity. Grid import and export limits are set to 20 kW unless stated otherwise.

3.7 Experimental Evaluation

3.7.1 Experimental Setup

The optimization model was evaluated on the Irish household dataset for the 20 users considered in this chapter. The experiment covers the year 2020 and uses all days for which complete 15-minute profiles were available for the selected users and for the market prices. Out of the full calendar year, 343 days were solved successfully, while 23 days were skipped because at least one required time series did not contain the expected 96 periods. The skipped days were mainly caused by missing household profiles or daylight-saving-time inconsistencies in the price data.

The purpose of the experiment is to compare the three operating modes introduced in this chapter: (i) the passive baseline, where users do not schedule batteries and do not exchange energy internally; (ii) individual battery scheduling, where each user minimizes its own daily cost independently; and (iii) the grand coalition, where all 20 users coordinate battery operation and internal energy exchanges in a single cooperative optimization problem.

For each valid day, the experiment solves the following sequence of problems:

1. the baseline cost of each user without battery scheduling,
2. the solo battery-scheduling problem of each user,
3. the cooperative grand-coalition optimization problem for all 20 users.

The daily costs and gains are then aggregated over all valid days. Therefore, the reported values represent annual totals over the 343 solved days rather than a single representative day.

3.7.2 Results

Table 3.1 summarizes the annual costs and gains of the three operating modes, while Figures 3.1 and 3.2 show the cumulative gains over the year and the monthly cost comparison, respectively.

The results show that individual battery scheduling already provides a substantial economic improvement compared with the passive baseline. The total annual cost decreases from 7259.68 EUR in the baseline case to 6670.89 EUR when users optimize their batteries independently. This corresponds to a saving of 588.79 EUR, or 8.11% of the baseline cost.

Table 3.1: Annual cost comparison for baseline operation, solo scheduling and grand-coalition coordination.

Metric	Value
Evaluation period	2020
Valid solved days	343
Skipped days	23
Number of users	20
Baseline annual cost	7259.68 EUR
Solo scheduling annual cost	6670.89 EUR
Grand-coalition annual cost	6373.42 EUR
Solo scheduling gain vs. baseline	588.79 EUR
Coordination gain vs. solo scheduling	297.47 EUR
Total gain vs. baseline	886.26 EUR
Solo scheduling gain vs. baseline	8.11%
Coordination gain vs. solo scheduling	4.46%
Total gain vs. baseline	12.21%

Allowing the users to cooperate as a grand coalition produces an additional improvement. The cooperative solution reduces the annual cost further to 6373.42 EUR. Compared with the solo-scheduling case, this gives an additional coordination gain of 297.47 EUR, equal to 4.46% of the solo cost. Relative to the original baseline, the total annual saving of the grand coalition is 886.26 EUR, corresponding to 12.21%.

These results confirm the economic role of both layers of the model. Battery scheduling alone captures the value of shifting each user's own energy over time, while cooperation captures the additional value of sharing energy between users with different demand and generation patterns. The grand coalition therefore provides the lowest total cost among the three operating modes, because it combines individual battery flexibility with coordinated internal energy exchange across all users.

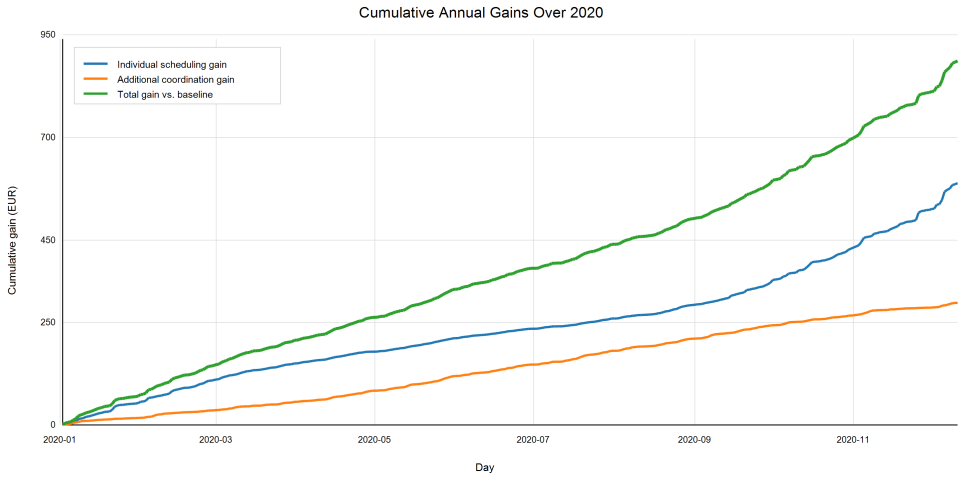


Figure 3.1: Cumulative annual gains from individual battery scheduling and grand-coalition coordination over the 343 solved days of 2020.

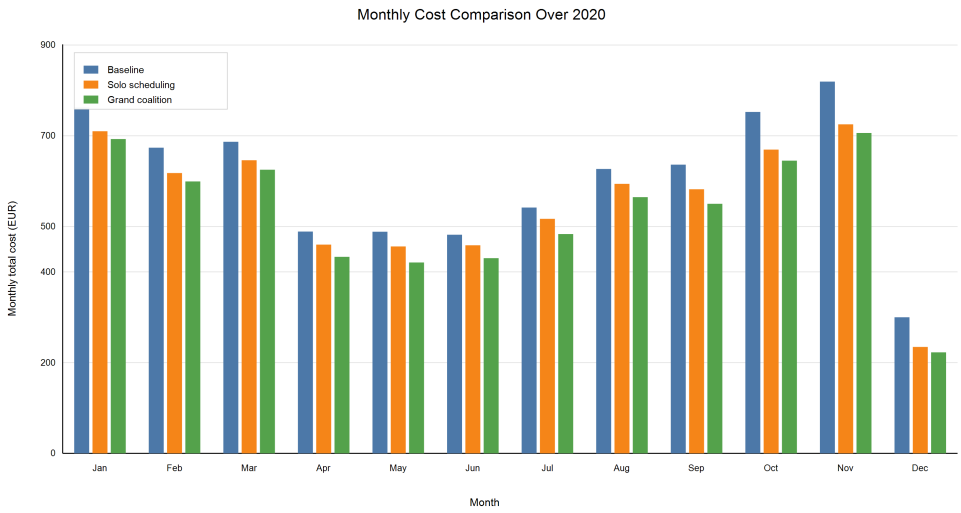


Figure 3.2: Monthly total cost comparison between baseline operation, individual battery scheduling and grand-coalition coordination over the solved days of 2020.

Chapter 4

Profit Allocation

4.1 Profit Allocation Based on Grand-Coalition Marginal Contribution

After defining the baseline, the stand-alone battery scheduling problem, and the cooperative coalition optimization problem, the next question is how the total economic benefit of cooperation should be distributed among the participants. In this thesis, the allocation rule separates two conceptually different sources of value: (i) the benefit each user can obtain individually by optimizing its own battery, and (ii) the additional value created only when users coordinate their operation inside a coalition. The mechanism preserves the former at the user level and distributes only the latter according to each user's contribution to the grand coalition.

4.1.1 Cooperative Game Formulation

Let $S \subseteq \mathcal{N}$ denote a coalition of users. Recall that the baseline cost of coalition S is

$$C^{\text{base}}(S) = \sum_{i \in S} C_i^{\text{base}}, \tag{4.1}$$

while the optimized coalition cost is denoted by $C^{\text{coal}}(S)$ and is obtained by solving the cooperative scheduling model for the members of S.

A natural transferable-utility cooperative game can therefore be defined through the total cost saving of coalition S with respect to the baseline:

$$\begin{gathered} v(S) := C^{\text{base}}(S) - C^{\text{coal}}(S), \qquad \forall S \subseteq \mathcal{N}, \\ \text{(total saving of coalition } S \text{ vs. baseline)} \end{gathered} \tag{4.2}$$

where:

- $C^{\text{base}}(S)$ is the baseline (no-coordination) cost of coalition S,
- $C^{\text{coal}}(S)$ is the optimized cost when the members of S coordinate.

with $v(\emptyset) = 0$.

For every user $i \in \mathcal{N}$, let

$$C_i^{\text{solo}} := C^{\text{coal}}(\{i\}) \tag{4.3}$$

denote the optimized cost achieved when user i schedules its battery independently, without any cooperation. The stand-alone optimized value of player i is then

$$s_i := C_i^{\text{base}} - C_i^{\text{solo}} = v(\{i\}). \tag{4.4}$$

Hence, s_i measures the benefit of solo battery scheduling relative to the baseline.

The allocation rule proposed in this thesis is not based directly on the full game v, but on the *cooperation-surplus game*

$$g(S) := v(S) - \sum_{i \in S} s_i, \qquad \forall S \subseteq \mathcal{N}, \tag{4.5}$$

(saving created only by cooperation, beyond stand-alone benefits)

where:

- $v(S)$ is the total saving of coalition S relative to the baseline,
- s_i is the stand-alone saving of player i (achievable without cooperation).

Using the definition of $v(S)$ and the additivity of the baseline cost, this can be rewritten as

$$g(S) = \sum_{i \in S} C_i^{\text{solo}} - C^{\text{coal}}(S), \tag{4.6}$$

(extra saving of coalition S vs. members acting alone)

where:

- C_i^{solo} is the optimized cost of player i when scheduling independently,
- $C^{\text{coal}}(S)$ is the optimized cost when the members of S coordinate.

Therefore, $g(S)$ captures only the additional saving that coalition S creates beyond what its members can already achieve through stand-alone optimization. Moreover, the cooperation-surplus game is superadditive. For any two disjoint coalitions $A, B \subseteq \mathcal{N}$, the joint optimization for $A \cup B$ can reproduce the independent optimal schedules of A and B by setting all transfers between the two coalitions to zero. Hence,

$$C^{\text{coal}}(A \cup B) \leq C^{\text{coal}}(A) + C^{\text{coal}}(B), \tag{4.7}$$

and therefore,

$$g(A \cup B) \geq g(A) + g(B). \tag{4.8}$$

Since $g(\{i\}) = C_i^{\text{solo}} - C^{\text{coal}}(\{i\}) = 0$, this also implies that every leave-one-out marginal contribution of the grand coalition is theoretically non-negative:

$$g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\}) \geq 0. \tag{4.9}$$

In particular, for the grand coalition $\mathcal{N}$,

$$\begin{gathered} G(\mathcal{N}) := g(\mathcal{N}) = \sum_{i \in \mathcal{N}} C_i^{\text{solo}} - C^{\text{coal}}(\mathcal{N}), \\ \text{(total cooperation surplus of the grand coalition)} \end{gathered} \tag{4.10}$$

where $C^{\text{coal}}(\mathcal{N})$ is the optimized cost of the grand coalition. This quantity is the total cooperation surplus.

4.1.2 Allocation Rule

The proposed rule follows a two-stage logic.

First, each user is guaranteed its stand-alone optimized value s_i.

Second, the total cooperation surplus $G(\mathcal{N})$ is distributed according to a contribution weight that measures how much cooperation surplus is lost when that user is removed from the grand coalition. Specifically, for each $i \in \mathcal{N}$, define the leave-one-out marginal contribution

$$\begin{gathered} m_i := g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\}), \\ \text{(surplus lost when player } i \text{ is removed)} \end{gathered} \tag{4.11}$$

Expanding the definition of g, this becomes

$$\begin{gathered} m_i = C_i^{\text{solo}} + C^{\text{coal}}(\mathcal{N} \setminus \{i\}) - C^{\text{coal}}(\mathcal{N}), \\ \text{(equivalent expression in terms of optimized costs)} \end{gathered} \tag{4.12}$$

where:

- $C^{\text{coal}}(\mathcal{N} \setminus \{i\})$ is the optimized cost when all players except i coordinate,
- $C^{\text{coal}}(\mathcal{N})$ is the optimized cost of the grand coalition.

Thus, m_i measures the reduction in cooperation surplus caused by removing user i from the grand coalition.

Because the cooperation-surplus game is superadditive, these leave-one-out marginal contributions are non-negative in the exact mathematical model. The allocation weight is therefore defined as

$$w_i := m_i, \tag{4.13}$$
(grand-coalition leave-one-out contribution weight)

In the numerical implementation, very small negative values of m_i may still be clipped to zero as a safeguard against solver tolerances or numerical noise. This clipping is not part of the theoretical rule, but only a practical implementation safeguard. If $\sum_{j\in\mathcal{N}} w_j > 0$, the cooperation bonus allocated to user i is

$$b_i := G(\mathcal{N})\frac{w_i}{\sum_{j\in\mathcal{N}} w_j}, \tag{4.14}$$
(allocate cooperation surplus proportionally to w_i)

where:

- w_i is player i's (nonnegative) contribution weight,
- $\sum_{j\in\mathcal{N}} w_j$ is the total weight used for normalization.

Hence, the final savings allocated to user i are

$$x_i := s_i + b_i, \tag{4.15}$$
(stand-alone saving plus cooperation bonus)

Because the optimization model of this thesis is naturally written in cost form, it is also convenient to define the equivalent final allocated cost

$$\alpha_i := C_i^{\text{base}} - x_i, \tag{4.16}$$
(final cost share of player i after allocation)

Using the definition of s_i, this simplifies to

$$\alpha_i = C_i^{\text{solo}} - b_i, \tag{4.17}$$
(solo optimized cost minus the cooperation bonus)

Therefore, the rule can be interpreted equivalently as either

- allocating total savings x_i to each player, or
- allocating the grand-coalition optimized cost $C^{\text{coal}}(\mathcal{N})$ through the cost shares α_i.

In the degenerate case where $G(\mathcal{N}) > 0$ but $\sum_{j \in \mathcal{N}} w_j = 0$, the current implementation uses an equal split of the cooperation surplus:

$$b_i = \frac{G(\mathcal{N})}{|\mathcal{N}|}, \qquad \forall i \in \mathcal{N}, \tag{4.18}$$

(fallback: split the surplus equally)

4.1.3 Interpretation and Justification

The rule is designed to reward only *genuine cooperative value creation*. This distinction is important in renewable energy communities. A user may achieve a large improvement relative to the baseline because its own battery shifts energy efficiently across time, even if that user contributes little to coalition-level energy sharing. Conversely, another user may contribute strongly to the coalition because its demand, generation, and flexibility profile complement those of the other users, even if its stand-alone gain is modest.

By assigning each user first its stand-alone optimized value s_i, the mechanism guarantees that nobody loses the benefit of individual battery scheduling by joining the coalition. By distributing only $G(\mathcal{N})$, the mechanism isolates the incremental value of cooperation. Finally, by defining the weights from the leave-one-out reduction in grand-coalition cooperation surplus, the rule rewards users that are most important for maintaining the additional benefit created by coordination.

This also clarifies the difference between the proposed rule and the Shapley value. The Shapley value averages a player's marginal contribution over *all* possible sub-coalitions. By contrast, the present rule uses only one marginal term, namely the player's leave-one-out contribution to the grand coalition. This design choice yields a transparent and computationally lightweight mechanism tailored to energy-community settings; it is intentionally distinct from the Shapley value and targets a narrower set of axioms.

4.1.4 A Working Example

Before turning to the formal properties of the rule, it is instructive to see the mechanism operate on a small numerical example. Consider a community of three users. Player 1 is a prosumer with a battery whose stand-alone scheduling is quite effective; player 2 is a prosumer with a more modest individual benefit; player 3 is a consumer whose battery (or lack thereof) offers no stand-alone improvement. All costs are daily costs in EUR. The baseline and stand-alone

optimized costs are:

$$\begin{aligned} C_1^{\text{base}} &= 10, \quad C_1^{\text{solo}} = 8, \quad s_1 = 2, \\ C_2^{\text{base}} &= 12, \quad C_2^{\text{solo}} = 11, \quad s_2 = 1, \\ C_3^{\text{base}} &= 9, \quad C_3^{\text{solo}} = 9, \quad s_3 = 0. \end{aligned} \tag{4.19}$$

Solving the cooperative scheduling problem for the grand coalition and for the three leave-one-out coalitions yields the optimized costs

$$C^{\text{coal}}(\{1,2,3\}) = 24, \quad C^{\text{coal}}(\{2,3\}) = 19, \quad C^{\text{coal}}(\{1,3\}) = 15, \quad C^{\text{coal}}(\{1,2\}) = 16. \tag{4.20}$$

The first step is to compute the cooperation surplus of each of these coalitions, using $g(S) = \sum_{i \in S} C_i^{\text{solo}} - C^{\text{coal}}(S)$:

$$\begin{aligned} G(\mathcal{N}) = g(\{1,2,3\}) &= (8+11+9) - 24 = 4, \\ g(\{2,3\}) &= (11+9) - 19 = 1, \\ g(\{1,3\}) &= (8+9) - 15 = 2, \\ g(\{1,2\}) &= (8+11) - 16 = 3. \end{aligned} \tag{4.21}$$

In words: the full community saves 4 EUR per day *beyond* what its members already achieve on their own, while any two-player coalition saves strictly less.

The second step computes the leave-one-out marginal contributions, i.e. how much of this 4 EUR would be lost if each player left:

$$\begin{aligned} m_1 &= g(\mathcal{N}) - g(\{2,3\}) = 4 - 1 = 3, \\ m_2 &= g(\mathcal{N}) - g(\{1,3\}) = 4 - 2 = 2, \\ m_3 &= g(\mathcal{N}) - g(\{1,2\}) = 4 - 3 = 1. \end{aligned} \tag{4.22}$$

Player 1 is the most important for the cooperation surplus: without them, three quarters of the surplus disappears. Player 3 is the least critical, but still contributes: removing them costs the community 1 EUR of surplus per day.

With weights $w_i = m_i$ and total weight $W = 3 + 2 + 1 = 6$, the surplus $G(\mathcal{N}) = 4$ is distributed proportionally:

$$b_1 = 4 \cdot \tfrac{3}{6} = 2, \qquad b_2 = 4 \cdot \tfrac{2}{6} = \tfrac{4}{3} \approx 1.33, \qquad b_3 = 4 \cdot \tfrac{1}{6} = \tfrac{2}{3} \approx 0.67. \tag{4.23}$$

Finally, each player receives their stand-alone value plus their bonus, $x_i = s_i + b_i$, or equivalently pays the final cost $\alpha_i = C_i^{\text{solo}} - b_i$. Table 4.1 summarizes the computation.

The example illustrates the two defining features of the rule. First, every player is at least as well off as under stand-alone operation: player 3, who gains nothing alone, still receives a

Table 4.1: Worked three-player example of the proposed allocation rule (all values in EUR per day).

Player	C_i^{base}	C_i^{solo}	s_i	$m_i = w_i$	b_i	$x_i = s_i + b_i$	α_i
1	10	8	2	3	2.00	4.00	6.00
2	12	11	1	2	1.33	2.33	9.67
3	9	9	0	1	0.67	0.67	8.33
Total	31	28	3	6	4.00	7.00	24.00

positive share of the cooperation surplus because their presence enables internal energy sharing. Second, the totals confirm efficiency: the allocated savings sum to $\sum_i x_i = 7 = C^{\text{base}}(\mathcal{N}) - C^{\text{coal}}(\mathcal{N}) = v(\mathcal{N})$, and the final costs sum to $\sum_i \alpha_i = 24 = C^{\text{coal}}(\mathcal{N})$, so the entire grand-coalition saving is distributed with nothing left over and nothing created.

4.1.5 Properties of the Allocation Rule

Before stating the formal properties of the allocation rule, it is useful to clarify the scope of the analysis. The rule introduced above is a *pragmatic, contribution-based allocation mechanism*, designed specifically for the renewable energy community setting considered in this thesis. It is not proposed as a full axiomatic solution concept in the sense of cooperative game theory, and it does not aim to reproduce the Shapley value or the nucleolus. Instead, the rule is built around two practical requirements: (i) every participant should be guaranteed at least their stand-alone optimized outcome, and (ii) the additional surplus created by cooperation should be distributed using a single, computationally cheap, and economically interpretable contribution measure, namely the leave-one-out marginal contribution to the grand-coalition cooperation surplus. The remainder of this subsection first situates the rule relative to two well-known solution concepts—the Shapley value and the nucleolus—and to the naive equal-split benchmark, and then establishes a sequence of mathematical propositions that characterize precisely which properties the rule satisfies, which it satisfies only in a weaker form, and which it does not satisfy at all.

Roadmap. The formal results below are organized in two groups. Lemma 4.1 and Propositions 4.1–4.8 establish what the rule *guarantees*: the cooperation surplus is never negative, the full saving of the grand coalition is distributed exactly (efficiency), no participant is ever worse off than under stand-alone operation (individual rationality), and players with equal or larger contribution weights receive equal or larger bonuses. Propositions 4.9–4.11 then document, through explicit counterexamples, the classical axioms the rule deliberately does *not* target: additivity, core stability, and population monotonicity. A reader who is not interested in the proofs can retain the summary of Table 4.2 and the worked example of Section 4.1.4, and proceed directly to Section 4.1.6 without loss of continuity.

Comparative overview. Table 4.2 summarizes how the proposed rule compares with the Shapley value, the nucleolus, and the equal split of the cooperation surplus across a set of standard allocation criteria. The table is intended as an orientation device; the properties relevant to the proposed rule are then proved formally in the propositions that follow.

As Table 4.2 indicates, the Shapley value offers a broad axiomatic foundation (efficiency, symmetry, dummy, and additivity), but it is computationally demanding in large communities. The nucleolus prioritizes stability, selecting a core allocation whenever the core is non-empty, albeit through a more involved construction. The equal split benchmark is maximally simple and transparent, but it disregards heterogeneity in contributions. The proposed rule sits between these benchmarks: it retains efficiency and individual rationality while distributing the cooperation surplus using a single, interpretable leave-one-out contribution signal, and it remains practical to compute at community scale; the subsequent propositions make precise which classical axioms it satisfies, and which it deliberately does not target.

Guarantees and Limitations

Guarantees. The following results record the main guarantees provided by the proposed rule (e.g., budget balance and individual rationality with respect to stand-alone optimization), as well as a few additional structural properties of the cooperation bonus.

Lemma 4.1 (Non-negativity of the cooperation surplus)**.** *For every coalition $S \subseteq \mathcal{N}$,*

$$g(S) \geq 0, \tag{4.24}$$

and in particular $G(\mathcal{N}) = g(\mathcal{N}) \geq 0$.

Intuitively, a coalition can always ignore cooperation and let each member run its own stand-alone schedule; joint optimization can therefore only match or improve on this option, never worsen it.

Proof. Fix a coalition $S \subseteq \mathcal{N}$. By construction of the cooperative scheduling model, the feasible region of the joint optimization problem defining $C^{\text{coal}}(S)$ contains, as a particular (generally suboptimal) feasible point, the policy in which every member $i \in S$ independently follows its own stand-alone-optimal schedule. The cost of this particular feasible point is $\sum_{i \in S} C_i^{\text{solo}}$. Since $C^{\text{coal}}(S)$ is obtained by minimizing cost over a feasible region that contains this point,

$$C^{\text{coal}}(S) \leq \sum_{i \in S} C_i^{\text{solo}}. \tag{4.25}$$

Table 4.2: Comparison of the proposed allocation rule with the Shapley value, the nucleolus, and the equal split of the cooperation surplus.

Property / Criterion	Proposed rule	Shapley value	Nucleolus	Equal split
Efficiency / budget balance	Yes	Yes	Yes	Yes
Individual rationality	Yes	Not guaranteed in general	Yes, when the core is non-empty	Yes
Treatment of stand-alone value	Explicitly secured first, then surplus distributed	Embedded implicitly in marginal contributions	Embedded implicitly in coalition values	Stand-alone value not isolated; surplus split blindly
Contribution awareness	Yes, via leave-one-out marginal contribution to $g(\mathcal{N})$	Yes, via all marginal contributions	Indirect, via minimization of coalition excesses	No
Symmetry	Weak: equal s_i and equal w_i imply equal x_i	Full Shapley symmetry axiom	Satisfies symmetry	Trivially symmetric (equal shares)
Dummy / zero contribution	If $m_i = 0$ and total weight is positive, then $b_i = 0$ and $x_i = s_i$	Satisfies dummy axiom	No simple contribution-based interpretation	Generally violated; non-contributors still receive a share
Additivity	No	Yes	No	Yes
Core stability	Not guaranteed	Not guaranteed in general	In the core when the core is non-empty	Not guaranteed
Population monotonicity	Not guaranteed	Not guaranteed in general	Not guaranteed in general	Not guaranteed
Computational cost	Low: $1 + 2\lvert\mathcal{N}\rvert$ optimizations	High: $2^{\lvert\mathcal{N}\rvert}$ coalition evaluations	High: sequential linear programs over coalitions	Trivial
Interpretability for energy communities	High; directly tied to leave-one-out contribution	Conceptually rich but harder to explain to non-experts	Stability-oriented, less intuitive economically	Very high, but ignores contribution

Using the equivalent expression for the cooperation-surplus game,

$$g(S) = \sum_{i \in S} C_i^{\text{solo}} - C^{\text{coal}}(S) \geq 0. \tag{4.26}$$

Taking $S = \mathcal{N}$ gives $G(\mathcal{N}) = g(\mathcal{N}) \geq 0$. □

Proposition 4.1 (Efficiency and budget balance)**.** *Suppose the cooperation surplus is fully distributed among the players. Then*

$$\sum_{i \in \mathcal{N}} x_i = v(\mathcal{N}), \tag{4.27}$$

and equivalently

$$\sum_{i \in \mathcal{N}} \alpha_i = C^{\text{coal}}(\mathcal{N}). \tag{4.28}$$

In plain terms, the rule neither creates nor destroys money: whatever the grand coalition saves is handed out in full, no more and no less.

Proof. By definition, $x_i = s_i + b_i$ for every $i \in \mathcal{N}$, so

$$\sum_{i \in \mathcal{N}} x_i = \sum_{i \in \mathcal{N}} s_i + \sum_{i \in \mathcal{N}} b_i. \tag{4.29}$$

If $\sum_{j \in \mathcal{N}} w_j > 0$, the proportional bonus rule gives

$$\sum_{i \in \mathcal{N}} b_i = \sum_{i \in \mathcal{N}} G(\mathcal{N}) \frac{w_i}{\sum_{j \in \mathcal{N}} w_j} = G(\mathcal{N}) \frac{\sum_{i \in \mathcal{N}} w_i}{\sum_{j \in \mathcal{N}} w_j} = G(\mathcal{N}). \tag{4.30}$$

If instead $\sum_{j \in \mathcal{N}} w_j = 0$, the fallback equal split gives $\sum_{i \in \mathcal{N}} b_i = |\mathcal{N}| \cdot \frac{G(\mathcal{N})}{|\mathcal{N}|} = G(\mathcal{N})$ as well. In either case, $\sum_{i \in \mathcal{N}} b_i = G(\mathcal{N}) = g(\mathcal{N})$. Therefore,

$$\sum_{i \in \mathcal{N}} x_i = \sum_{i \in \mathcal{N}} s_i + g(\mathcal{N}). \tag{4.31}$$

By the definition of the cooperation-surplus game, $g(\mathcal{N}) = v(\mathcal{N}) - \sum_{i \in \mathcal{N}} s_i$, and substituting yields

$$\sum_{i \in \mathcal{N}} x_i = \sum_{i \in \mathcal{N}} s_i + v(\mathcal{N}) - \sum_{i \in \mathcal{N}} s_i = v(\mathcal{N}). \tag{4.32}$$

For the cost-equivalent statement, recall $\alpha_i = C_i^{\text{base}} - x_i$, so

$$\sum_{i \in \mathcal{N}} \alpha_i = \sum_{i \in \mathcal{N}} C_i^{\text{base}} - \sum_{i \in \mathcal{N}} x_i = C^{\text{base}}(\mathcal{N}) - v(\mathcal{N}). \tag{4.33}$$

Since $v(\mathcal{N}) = C^{\text{base}}(\mathcal{N}) - C^{\text{coal}}(\mathcal{N})$ by definition, it follows that $\sum_{i \in \mathcal{N}} \alpha_i = C^{\text{coal}}(\mathcal{N})$. □

Proposition 4.2 (Individual rationality with respect to stand-alone optimization)**.** *For every* $i \in \mathcal{N}$*,*

$$x_i \geq s_i, \tag{4.34}$$

and equivalently

$$\alpha_i \leq C_i^{\text{solo}}. \tag{4.35}$$

This is the participation guarantee: joining the coalition can never leave a user worse off than optimizing their own battery alone, because the cooperation bonus is always non-negative.

Proof. By the superadditivity argument above, $m_i = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\}) \geq 0$ for every i, and since $w_i = m_i$, it follows that $w_i \geq 0$. By Lemma 4.1, $G(\mathcal{N}) \geq 0$. If $\sum_{j \in \mathcal{N}} w_j > 0$, then since $G(\mathcal{N}) \geq 0$ and $w_i / \sum_j w_j \geq 0$, the bonus satisfies $b_i = G(\mathcal{N})\, w_i / \sum_j w_j \geq 0$. If $\sum_{j \in \mathcal{N}} w_j = 0$, the fallback rule gives $b_i = G(\mathcal{N})/|\mathcal{N}| \geq 0$ as well, since $G(\mathcal{N}) \geq 0$. In both cases, $b_i \geq 0$. Since $x_i = s_i + b_i$, it follows immediately that $x_i \geq s_i$. The cost-equivalent inequality follows from $\alpha_i = C_i^{\text{solo}} - b_i$ and $b_i \geq 0$. □

Proposition 4.3 (Non-negativity of the cooperation bonus)**.** *For every* $i \in \mathcal{N}$*,* $b_i \geq 0$*.*

Proof. By Lemma 4.1, $G(\mathcal{N}) \geq 0$, and by superadditivity $w_i = m_i \geq 0$ for every $i \in \mathcal{N}$. This is precisely the intermediate claim established in the proof of Proposition 4.2: under the proportional rule, $b_i = G(\mathcal{N})\, w_i / \sum_j w_j$ is a ratio of non-negative quantities whenever $\sum_j w_j > 0$, and under the fallback rule $b_i = G(\mathcal{N})/|\mathcal{N}|$ is non-negative since $G(\mathcal{N}) \geq 0$. Hence $b_i \geq 0$ in both cases. □

Proposition 4.4 (Weak symmetry)**.** *If* $i, j \in \mathcal{N}$ *satisfy* $s_i = s_j$ *and* $w_i = w_j$*, then*

$$x_i = x_j. \tag{4.36}$$

Proof. Suppose $\sum_{k \in \mathcal{N}} w_k > 0$. Since $w_i = w_j$, the proportional bonus formula gives

$$b_i = G(\mathcal{N}) \frac{w_i}{\sum_{k \in \mathcal{N}} w_k} = G(\mathcal{N}) \frac{w_j}{\sum_{k \in \mathcal{N}} w_k} = b_j. \tag{4.37}$$

If instead $\sum_{k \in \mathcal{N}} w_k = 0$, the fallback rule assigns $b_i = b_j = G(\mathcal{N})/|\mathcal{N}|$ directly. In either case $b_i = b_j$. Combined with the assumption $s_i = s_j$, this gives $x_i = s_i + b_i = s_j + b_j = x_j$. □

Remark 4.1 (Relation to the Shapley symmetry axiom)**.** *Proposition 4.4 is a restricted symmetry statement: it applies only to players who already coincide in both their stand-alone value* s_i *and their grand-coalition contribution weight* w_i*. The rule does not establish the full Shapley symmetry axiom, which requires equal payoffs whenever two players are interchangeable across* all *coalitions, that is, whenever* $g(S \cup \{i\}) = g(S \cup \{j\})$ *for every* $S \subseteq \mathcal{N} \setminus \{i, j\}$*. Since the proposed rule never evaluates* g *on coalitions other than* $\mathcal{N}$ *and the leave-one-out coalitions*

$\mathcal{N} \setminus \{i\}$, *two players could in principle be Shapley-symmetric without satisfying* $w_i = w_j$, *or could satisfy* $w_i = w_j$ *without being Shapley-symmetric. The property proved above is therefore a consequence of the construction of the rule, not an instance of the Shapley symmetry axiom.*

Proposition 4.5 (Equal bonus symmetry)**.** *If* $i, j \in \mathcal{N}$ *satisfy* $w_i = w_j$, *then*

$$b_i = b_j. \tag{4.38}$$

Proof. This is precisely the intermediate result already established in the proof of Proposition 4.4: under the proportional rule the equality $w_i = w_j$ directly yields $b_i = b_j$, and under the fallback rule $b_i = b_j = G(\mathcal{N})/|\mathcal{N}|$ holds independently of the weights. □

Proposition 4.6 (Zero-contribution / dummy-like property)**.** *If* $m_i = 0$ *and* $\sum_{j \in \mathcal{N}} w_j > 0$, *then* $w_i = 0$ *and* $b_i = 0$. *Consequently,*

$$x_i = s_i \qquad \textit{and} \qquad \alpha_i = C_i^{\text{solo}}. \tag{4.39}$$

In other words, a player whose departure would cost the community nothing receives no share of the cooperation surplus—but still keeps their full stand-alone benefit.

Proof. By definition $w_i = m_i$. If $m_i = 0$, then $w_i = 0$. Under the proportional rule, $b_i = G(\mathcal{N})\, w_i / \sum_j w_j = 0$ whenever $\sum_j w_j > 0$, since the numerator vanishes. If instead all weights are zero, the fallback equal split applies and need not assign zero bonus to a zero-marginal player; the zero-contribution conclusion is therefore stated for the generic case in which at least one player has positive weight. Given $b_i = 0$, it follows directly that $x_i = s_i + b_i = s_i$, and $\alpha_i = C_i^{\text{solo}} - b_i = C_i^{\text{solo}}$. □

Remark 4.2 (Relation to the Shapley dummy axiom)**.** *Proposition 4.6 establishes a dummy-like property with respect to the* cooperation-surplus game g, *not with respect to the original saving game* v. *The Shapley dummy axiom states that if a player's marginal contribution* $v(S \cup \{i\}) - v(S)$ *equals* $v(\{i\})$ *for every* $S \subseteq \mathcal{N} \setminus \{i\}$, *then that player should receive exactly* $v(\{i\})$. *The condition used here,* $m_i = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\}) = 0$, *only inspects the player's leave-one-out effect on* g *at the grand coalition, and is silent about the player's marginal contributions to* v *at smaller coalitions. The two notions coincide in spirit—a player who adds nothing to the grand coalition's cooperation surplus is excluded from the bonus when other positive contribution weights exist—but they are not formally the same axiom.*

Proposition 4.7 (Marginal-contribution awareness)**.** *The contribution weight* w_i *used by the rule is constructed from the leave-one-out marginal contribution*

$$m_i = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\}), \tag{4.40}$$

so that the cooperation bonus b_i rewards player i in proportion to the surplus lost when i is removed from the grand coalition. However, the rule does not satisfy the full Shapley marginality axiom.

The rule looks at a single question—"how much surplus is lost if player i leaves the grand coalition?"—whereas the Shapley value asks this question for every possible sub-coalition and averages the answers. The proof below makes this distinction precise.

Proof. The first claim follows directly from the definitions $w_i = m_i$ and $b_i = G(\mathcal{N})\, w_i / \sum_j w_j$ (or the fallback equal split): under the proportional rule, b_i is a (weakly) increasing function of m_i for fixed weights of the other players, so the bonus is determined by the single marginal term m_i.

For the second claim, recall that the Shapley marginality axiom requires a player's payoff to depend only on the full vector of marginal contributions

$$\big(g(S \cup \{i\}) - g(S)\big)_{S \subseteq \mathcal{N} \setminus \{i\}}, \tag{4.41}$$

taken over *all* subcoalitions $S \subseteq \mathcal{N} \setminus \{i\}$, not only $S = \mathcal{N} \setminus \{i\}$. The weight w_i defined above uses exclusively the single term corresponding to $S = \mathcal{N} \setminus \{i\}$, namely $m_i = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\})$, and discards all other terms $g(S \cup \{i\}) - g(S)$ for $S \subsetneq \mathcal{N} \setminus \{i\}$. Consequently, two cooperation-surplus games that induce the same value of m_i but differ in the marginal contributions of i to smaller coalitions $S \subsetneq \mathcal{N} \setminus \{i\}$ produce identical weights w_i, and hence identical bonuses b_i (for fixed weights of the other players), even though the Shapley value would in general treat them differently. Therefore, the rule does not satisfy the full marginality axiom. □

Proposition 4.8 (Contribution monotonicity within a fixed grand coalition)**.** *Assume $W := \sum_{j \in \mathcal{N}} w_j > 0$. If $w_i \geq w_j$ for some $i, j \in \mathcal{N}$, then*

$$b_i \geq b_j. \tag{4.42}$$

Proof. Under the assumption $W > 0$, the proportional bonus formula gives

$$b_i - b_j = G(\mathcal{N})\frac{w_i}{W} - G(\mathcal{N})\frac{w_j}{W} = \frac{G(\mathcal{N})}{W}\,(w_i - w_j). \tag{4.43}$$

By Lemma 4.1, $G(\mathcal{N}) \geq 0$, and $W > 0$ by assumption, so the prefactor $G(\mathcal{N})/W$ is non-negative. Combined with $w_i - w_j \geq 0$, this yields $b_i - b_j \geq 0$, that is, $b_i \geq b_j$. □

Remark 4.3 (Scope of the monotonicity property)**.** *Proposition 4.8 concerns only the cooperation bonus b_i, not the final allocated saving $x_i = s_i + b_i$. Since x_i also depends on the stand-alone term s_i, a player with a smaller contribution weight w_i can still receive a larger final allocation*

x_i if its stand-alone saving s_i is sufficiently larger than that of the other player. The monotonicity property is therefore a statement about the cooperative component of the allocation only, not about the total payoff.

Limitations (properties not targeted). The next propositions document classical axioms from cooperative game theory that the proposed rule does not attempt to satisfy in general. This is a deliberate design trade-off: the rule uses only grand-coalition and leave-one-out information to remain transparent and computationally light, which is incompatible with axioms that inherently depend on the full collection of subcoalition values.

Proposition 4.9 (Not additive in general)**.** *The proposed rule is not additive in general: the bonus vector obtained by applying the rule to two cooperation-surplus games separately does not, in general, equal the bonus vector obtained by applying the rule once to the sum of the two games.*

The intuition behind the counterexample is that proportional sharing depends on the *ratios* of the weights, and ratios are not preserved under addition: a player who dominates scenario A and is dominated in scenario B looks merely average in the combined scenario.

Proof. Consider two players, 1 and 2, and two independent cooperation-surplus scenarios, A and B, with cooperation surpluses

$$G^A = 100, \qquad G^B = 10, \tag{4.44}$$

and contribution-weight vectors

$$w^A = (9, 1), \qquad w^B = (1, 9). \tag{4.45}$$

Applying the rule separately to each scenario yields

$$b_1^A = 100 \cdot \frac{9}{10} = 90, \qquad b_2^A = 100 \cdot \frac{1}{10} = 10, \tag{4.46}$$

$$b_1^B = 10 \cdot \frac{1}{10} = 1, \qquad b_2^B = 10 \cdot \frac{9}{10} = 9, \tag{4.47}$$

so that the sum of the two separately computed bonus vectors is

$$b^A + b^B = (90 + 1,\ 10 + 9) = (91, 19). \tag{4.48}$$

Now consider the combined scenario $A + B$, obtained by summing the cooperation surpluses and the weights:

$$G^{A+B} = G^A + G^B = 110, \qquad w^{A+B} = w^A + w^B = (10, 10). \tag{4.49}$$

Applying the rule once to this combined scenario gives

$$b_1^{A+B} = 110 \cdot \frac{10}{20} = 55, \qquad b_2^{A+B} = 110 \cdot \frac{10}{20} = 55, \tag{4.50}$$

that is, $b^{A+B} = (55, 55)$. Since

$$b^{A+B} = (55, 55) \neq (91, 19) = b^A + b^B, \tag{4.51}$$

the rule does not satisfy additivity. □

Proposition 4.10 (Core membership is not guaranteed in general)**.** *The proposed rule is not guaranteed to lie in the core of the cooperation-surplus game.*

In plain terms, the counterexample constructs a pair of players who jointly generate 10 units of surplus but are allocated only 9.6 under the rule; they would therefore prefer to leave and split their own surplus, which is exactly what the core forbids.

Proof. Consider a three-player cooperation-surplus game with

$$g(\{1\}) = g(\{2\}) = g(\{3\}) = 0, \tag{4.52}$$

$$g(\{1,2\}) = 10, \qquad g(\{1,3\}) = 8, \qquad g(\{2,3\}) = 8, \tag{4.53}$$

$$g(\{1,2,3\}) = 12. \tag{4.54}$$

The leave-one-out marginal contributions are

$$m_1 = g(\{1,2,3\}) - g(\{2,3\}) = 12 - 8 = 4, \tag{4.55}$$

$$m_2 = g(\{1,2,3\}) - g(\{1,3\}) = 12 - 8 = 4, \tag{4.56}$$

$$m_3 = g(\{1,2,3\}) - g(\{1,2\}) = 12 - 10 = 2, \tag{4.57}$$

so that $w = (4, 4, 2)$ and $W = \sum_j w_j = 10$. The allocated bonuses are therefore

$$b = \left(12 \cdot \frac{4}{10},\ 12 \cdot \frac{4}{10},\ 12 \cdot \frac{2}{10}\right) = (4.8,\ 4.8,\ 2.4). \tag{4.58}$$

The coalition $\{1, 2\}$ receives, under this allocation,

$$b_1 + b_2 = 4.8 + 4.8 = 9.6 < 10 = g(\{1,2\}), \tag{4.59}$$

so players 1 and 2 are jointly allocated less than the surplus their coalition can generate on its own. Hence, $\{1, 2\}$ can profitably block the grand-coalition allocation by leaving and sharing

$g(\{1,2\}) = 10$ between themselves instead. This violates the core condition, and therefore the rule does not in general guarantee core stability. □

Proposition 4.11 (Population monotonicity is not guaranteed in general)**.** *The proposed rule does not in general satisfy population monotonicity: the entry of a new player into the grand coalition can decrease the cooperation bonus of an existing player, even if the total cooperation surplus increases.*

Intuitively, when a highly complementary new member joins, the pie grows, but the newcomer's weight can grow so much faster that existing members receive thinner slices of the larger pie.

Proof. Let $\mathcal{N}$ be a player set and $\mathcal{N}' = \mathcal{N} \cup \{k\}$ the player set obtained by adding a new player k. In general, the cooperation-surplus game itself changes when the player set changes, since coalition values are now defined over a different family of subsets and the optimized coalition costs $C^{\text{coal}}(S)$ for $S \subseteq \mathcal{N}'$ need not bear any fixed relationship to the costs $C^{\text{coal}}(S)$ for $S \subseteq \mathcal{N}$. Consequently, for an existing player $i \in \mathcal{N}$, the leave-one-out marginal contribution computed in the enlarged game,

$$m_i' = g(\mathcal{N}') - g(\mathcal{N}' \setminus \{i\}), \tag{4.60}$$

need not equal the marginal contribution $m_i = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\})$ computed before player k joined, since both terms on the right-hand side are evaluated over coalitions that now include k. Similarly, the total weight $W' = \sum_{j \in \mathcal{N}'} w_j'$ differs from $W = \sum_{j \in \mathcal{N}} w_j$, both because a new term w_k' is added and because the existing weights w_j' for $j \in \mathcal{N}$ may themselves change.

As a concrete instance, suppose that before player k joins, two players have weights $w_1 = w_2 = 10$ and total surplus $G(\mathcal{N}) = 100$, so that $b_1 = 100 \cdot \frac{10}{20} = 50$. Suppose that after k joins, the total surplus increases to $G(\mathcal{N}') = 110$, but the new player's marginal contribution dominates the others, giving weights $w_1' = 10$, $w_2' = 10$, $w_k' = 90$ (for instance because k's flexible load profile is highly complementary to the rest of the coalition). Then

$$b_1' = 110 \cdot \frac{10}{10 + 10 + 90} = 110 \cdot \frac{10}{110} = 10, \tag{4.61}$$

so player 1's bonus falls from 50 to 10, a strict decrease, even though the total cooperation surplus increased from 100 to 110. This demonstrates that the bonus (and hence the final allocation $x_1 = s_1 + b_1$) of an existing player can decrease upon the entry of a new player, so the rule does not satisfy population monotonicity. Individual rationality is nonetheless preserved throughout, since $b_1' = 10 \geq 0$ still guarantees $x_1' \geq s_1$. □

Proposition 4.12 (Translation invariance with respect to stand-alone values)**.** *Let $\delta_i \in \mathbb{R}$ for each $i \in \mathcal{N}$, and consider a modified game in which the stand-alone value of player i increases*

by δ_i while, for every coalition $S \ni i$, the coalition value increases by the same constant:

$$\tilde{v}(S) = v(S) + \delta_i \quad \textit{for every } S \subseteq \mathcal{N} \textit{ with } i \in S, \tag{4.62}$$

with $\tilde{v}(S) = v(S)$ for $S \not\ni i$, and with $\tilde{s}_i = s_i + \delta_i$. Then the cooperation-surplus game is unchanged, $\tilde{g}(S) = g(S)$ for every $S \subseteq \mathcal{N}$, and consequently $G(\mathcal{N})$, m_i, w_i, and b_i are all unchanged, while the final allocation becomes

$$x'_i = x_i + \delta_i. \tag{4.63}$$

Proof. By definition, for any coalition S with $i \in S$,

$$\tilde{g}(S) = \tilde{v}(S) - \sum_{k \in S} \tilde{s}_k. \tag{4.64}$$

Since $i \in S$, the sum $\sum_{k \in S} \tilde{s}_k$ contains the term $\tilde{s}_i = s_i + \delta_i$ in place of s_i, and all other terms $\tilde{s}_k = s_k$ for $k \in S \setminus \{i\}$ are unchanged. Hence

$$\sum_{k \in S} \tilde{s}_k = \sum_{k \in S} s_k + \delta_i. \tag{4.65}$$

Substituting $\tilde{v}(S) = v(S) + \delta_i$ and the above expression,

$$\tilde{g}(S) = \big(v(S) + \delta_i\big) - \Big(\sum_{k \in S} s_k + \delta_i\Big) = v(S) - \sum_{k \in S} s_k = g(S). \tag{4.66}$$

For any coalition S with $i \notin S$, both $\tilde{v}(S) = v(S)$ and $\tilde{s}_k = s_k$ for all $k \in S$, so trivially $\tilde{g}(S) = g(S)$ as well. Hence $\tilde{g}(S) = g(S)$ for every $S \subseteq \mathcal{N}$, including $S = \mathcal{N}$ and $S = \mathcal{N} \setminus \{i\}$. It follows immediately that

$$G(\mathcal{N}) = \tilde{g}(\mathcal{N}) = g(\mathcal{N}), \qquad m_i = \tilde{g}(\mathcal{N}) - \tilde{g}(\mathcal{N} \setminus \{i\}) = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\}), \tag{4.67}$$

and the same identity holds for every other player's marginal contribution m_j, $j \neq i$, since the corresponding leave-one-out coalitions either contain i (in which case the δ_i shift cancels as shown above) or do not involve i at all. Consequently w_j and b_j are unchanged for every $j \in \mathcal{N}$, including $j = i$.

Finally, the new final allocation for player i is

$$x'_i = \tilde{s}_i + b_i = (s_i + \delta_i) + b_i = (s_i + b_i) + \delta_i = x_i + \delta_i, \tag{4.68}$$

while $x'_j = s_j + b_j = x_j$ for every $j \neq i$, since neither s_j nor b_j is affected by the shift. □

Proposition 4.13 (Computational complexity)**.** *Computing the allocation* $(x_i)_{i\in\mathcal{N}}$ *requires solving exactly* $|\mathcal{N}|$ *stand-alone optimization problems,* 1 *grand-coalition optimization problem, and* $|\mathcal{N}|$ *leave-one-out coalition optimization problems, that is,* $2|\mathcal{N}|+1$ *optimization problems in total. This is exponentially cheaper than the exact computation of the Shapley value, which requires the coalition value* $v(S)$ *(or* $g(S)$*) for every one of the* $2^{|\mathcal{N}|}$ *subsets* $S\subseteq\mathcal{N}$*.*

Proof. By the construction of the rule, the only quantities required are: (i) C_i^{solo} for every $i\in\mathcal{N}$, each obtained from one stand-alone optimization, giving $|\mathcal{N}|$ problems; (ii) $C^{\text{coal}}(\mathcal{N})$, obtained from a single grand-coalition optimization; and (iii) $C^{\text{coal}}(\mathcal{N}\setminus\{i\})$ for every $i\in\mathcal{N}$, each obtained from one leave-one-out coalition optimization, giving a further $|\mathcal{N}|$ problems. No other coalition value is used anywhere in the construction of s_i, g, m_i, w_i, b_i, or x_i. Summing these contributions gives $|\mathcal{N}|+1+|\mathcal{N}|=2|\mathcal{N}|+1$ optimization problems.

By contrast, the exact Shapley value of player i is defined as an average of marginal contributions over all subsets $S\subseteq\mathcal{N}\setminus\{i\}$,

$$\phi_i=\sum_{S\subseteq\mathcal{N}\setminus\{i\}}\frac{|S|!\,(|\mathcal{N}|-|S|-1)!}{|\mathcal{N}|!}\Big(g(S\cup\{i\})-g(S)\Big),\tag{4.69}$$

which requires knowledge of $g(S)$ (equivalently $C^{\text{coal}}(S)$) for every $S\subseteq\mathcal{N}$, that is, for all $2^{|\mathcal{N}|}$ subsets. Since $2|\mathcal{N}|+1$ grows linearly in $|\mathcal{N}|$ while $2^{|\mathcal{N}|}$ grows exponentially, the proposed rule is computationally far less expensive than the exact Shapley value for any non-trivial number of players. □

4.1.6 Comparison with Shapley Value, Nucleolus, and Equal Split

Having established the formal properties of the rule, it is useful to discuss, in narrative terms, how the proposed mechanism relates to the three classical benchmarks introduced above.

Proposed rule versus the Shapley value. The Shapley value remains the most theoretically complete benchmark, since it satisfies efficiency, symmetry, the dummy axiom, additivity, and the full marginality axiom simultaneously. The price of this completeness is computational: the exact Shapley value requires the coalition value to be evaluated on every one of the $2^{|\mathcal{N}|}$ possible coalitions, which becomes impractical as the number of participants in a renewable energy community grows. The proposed rule sacrifices additivity and the full marginality and symmetry axioms, retaining instead only a single, intuitively interpretable marginal-contribution term per player—the leave-one-out effect on the grand-coalition cooperation surplus. This makes the rule considerably more scalable, and arguably more directly interpretable for the prosumers participating in an energy community, who can be told in plain terms that their bonus depends on how much the community's joint savings would fall if they left the coalition.

Proposed rule versus the nucleolus. The nucleolus is explicitly stability-oriented: it is constructed by lexicographically minimizing the vector of coalition excesses, and therefore, whenever the core is non-empty, it always selects a core allocation. This stability guarantee, however, comes at the cost of a more abstract and computationally demanding construction, typically requiring the solution of a sequence of linear programs over the coalition values, and it does not admit a simple, direct contribution-based interpretation analogous to a marginal-contribution weight. The proposed rule, in contrast, is considerably simpler and more transparent: it is directly tied to a single, easily explained quantity, the leave-one-out cooperation surplus of each player, at the cost of providing no guarantee that the resulting allocation lies in the core, as demonstrated explicitly in Proposition 4.10.

Proposed rule versus the equal split of the cooperation surplus. The equal split of the cooperation surplus is the simplest possible benchmark: it is efficient and trivially symmetric, but it is entirely insensitive to how much each player actually contributes to the coalition, and it therefore generally fails any meaningful notion of the dummy/zero-contribution property. The proposed rule preserves the simplicity and low computational cost of the equal split while explicitly correcting this shortcoming: the cooperation bonus is distributed in proportion to each player's leave-one-out marginal contribution rather than uniformly, so that, when at least one player has positive contribution weight, players with zero leave-one-out contribution receive no cooperation bonus, while still being guaranteed their stand-alone optimized outcome.

In summary, the proposed rule occupies a deliberate middle ground: it is less axiomatically complete than the Shapley value and offers no stability guarantee comparable to the nucleolus, but it is markedly simpler, more transparent, and computationally far cheaper than either, while still being meaningfully more contribution-aware than a naive equal split.

The rule is considered appropriate for the purposes of this thesis for several reasons. First, it cleanly separates the gains a prosumer can already obtain through solo battery scheduling from the genuinely cooperative gains created by coordination within the coalition. Second, by guaranteeing each participant at least their stand-alone optimized outcome, it supports voluntary participation in the energy community, since no rational participant has an incentive to schedule independently rather than join the coalition. Third, by weighting the cooperation bonus according to the leave-one-out marginal contribution to the grand coalition, the rule rewards complementarity among prosumers' demand, generation, and flexibility profiles, which is precisely the mechanism through which cooperative value is created in this setting. Finally, as shown in Proposition 4.13, the rule remains computationally practical even as the number of participants grows, requiring only $2|\mathcal{N}| + 1$ optimization problems rather than an exponential number of coalition evaluations, which makes it well suited to the renewable energy community experiments carried out in this thesis.

4.1.7 Computational Procedure

The allocation can be computed through the following steps:

1. Compute each user's baseline cost C_i^{base}.
2. Solve each user's stand-alone optimization problem to obtain C_i^{solo} and $s_i = C_i^{\text{base}} - C_i^{\text{solo}}$.
3. Solve the grand-coalition optimization problem to obtain $C^{\text{opt}}(\mathcal{N})$ and the cooperation surplus $G(\mathcal{N})$.
4. For every user $i \in \mathcal{N}$, solve the leave-one-out coalition problem for $\mathcal{N} \setminus \{i\}$.
5. Compute $m_i = g(\mathcal{N}) - g(\mathcal{N} \setminus \{i\})$ and set $w_i = m_i$; in the numerical implementation, tiny negative values may be clipped to zero as a safeguard against solver tolerances.
6. Distribute $G(\mathcal{N})$ proportionally to the weights to obtain b_i, and finally compute $x_i = s_i + b_i$ or $\alpha_i = C_i^{\text{solo}} - b_i$.

The method is exact for the proposed rule and requires only the grand coalition and the $|\mathcal{N}|$ leave-one-out coalitions, in addition to the stand-alone problems. Therefore, it is substantially cheaper than the exact Shapley value, which requires the value of every coalition and thus scales exponentially with the number of players.

4.1.8 Experimental Comparison with the Exact Shapley Value

The proposed allocation rule was also compared experimentally with the exact Shapley value. The purpose of this experiment is not to replace the theoretical discussion above, but to examine how close the proposed grand-marginal allocation is to the Shapley benchmark in a case where the exact Shapley value can still be computed.

This comparison is computationally demanding because the exact Shapley value requires the value of all coalitions. For a coalition of n players, this corresponds to 2^n coalition evaluations. For this reason, the experiment was carried out on a reduced set of 8 players rather than on the full community. Even for 8 players, the exact Shapley benchmark requires $2^8 = 256$ coalition values per day. Over the 30 days of June 2020, this corresponds to 7680 coalition-value evaluations. By contrast, the proposed rule requires only $2n + 1 = 17$ optimization problems per day, or 510 optimization problems over the same monthly horizon. For 20 players, the exact Shapley value would require $2^{20} = 1{,}048{,}576$ coalition values per day, which makes an exact full-scale comparison impractical.

The experiment uses the players

$$\{1, 3, 5, 17, 14, 18, 19, 20\},$$

and covers the period from 2020-06-01 to 2020-06-30. All 30 days were solved successfully. For each day, the baseline costs, the stand-alone optimized costs, the grand-coalition cost, the proposed grand-marginal allocation, and the exact Shapley allocation were computed. The monthly cooperation surplus was then aggregated over all days.

Table 4.3: Monthly allocation experiment for the 8-player Shapley comparison.

Metric	Value
Evaluation period	June 2020
Valid solved days	30
Number of players	8
Baseline monthly cost	159.46 EUR
Solo scheduling monthly cost	152.26 EUR
Grand-coalition monthly cost	138.36 EUR
Monthly cooperation surplus	13.90 EUR
Sum of proposed allocated gains	13.90 EUR
Sum of exact Shapley gains	13.90 EUR
Total absolute difference between methods	2.66 EUR
Mean absolute difference	0.33 EUR/player

Both methods distribute the same total cooperation surplus, as required by efficiency. The difference between them is therefore not in the total amount distributed, but in how this surplus is assigned to individual users. Table 4.4 reports the monthly cooperation gain allocated to each player under the proposed rule and under the exact Shapley value, both in EUR and as a percentage of the total monthly cooperation surplus.

Table 4.4: Player-level allocation of the monthly cooperation surplus.

Player	Type	Proposed (EUR)	Shapley (EUR)	Proposed (%)	Shapley (%)
1	Prosumer	5.27	4.54	37.88	32.66
3	Prosumer	1.27	1.20	9.13	8.62
5	Prosumer	1.40	1.24	10.09	8.93
17	Prosumer	3.08	2.71	22.18	19.50
14	Consumer	0.85	1.18	6.10	8.51
18	Consumer	1.13	1.41	8.11	10.17
19	Consumer	0.58	0.95	4.16	6.80
20	Consumer	0.33	0.67	2.35	4.80

The results show a clear and interpretable difference between the two allocation methods. Under the proposed grand-marginal rule, the four prosumers receive 79.28% of the monthly cooperation surplus, compared with 69.71% under the exact Shapley value. Conversely, the four consumers receive 20.72% under the proposed rule, compared with 30.29% under the Shapley value. In absolute terms, the proposed rule shifts 1.33 EUR of the monthly cooperation surplus from the consumer group to the prosumer group relative to the Shapley benchmark.

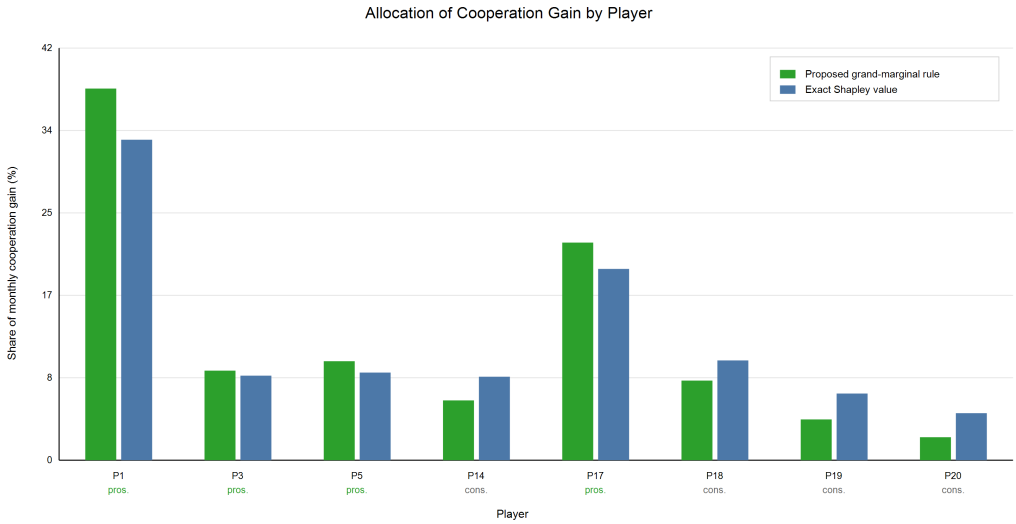


Figure 4.1: Player-level share of the monthly cooperation surplus under the proposed grand-marginal rule and under the exact Shapley value.

This behaviour is consistent with the design of the proposed rule. The grand-marginal allocation rewards users according to the loss of cooperation surplus caused by their removal from the grand coalition. In this experiment, the users classified as prosumers are also the main net providers of internal energy during the month. Players 1, 3, 5 and 17 export substantially more internal energy to the community than they receive, whereas players 14, 18, 19 and 20 are mainly net receivers of internal energy. As a result, the leave-one-out marginal rule assigns a larger share of the cooperation surplus to the users whose presence is most important for maintaining the coalition's internal energy-sharing benefit.

This does not imply that the consumer-side participants are treated unfairly. First, the allocation rule preserves the stand-alone optimized outcome of every participant before distributing the cooperation surplus. Second, all users still receive a non-negative cooperation bonus. Third, the total cooperation surplus is fully distributed. The difference relative to the Shapley value should instead be interpreted as a modelling choice: the proposed rule gives stronger emphasis to the marginal importance of users in the grand coalition, while the Shapley value averages marginal contributions over all possible subcoalitions.

Overall, the experiment supports the intended role of the proposed allocation rule. It remains efficient and individually rational, is much cheaper to compute than the exact Shapley value, and produces an allocation that is economically interpretable in the energy-community context. In the tested month, the rule assigns a larger share of the cooperation surplus to prosumers, which is reasonable in a setting where these users provide most of the internal energy that enables cooperation. At the same time, the difference from the Shapley benchmark remains moderate in absolute monetary terms, with a mean absolute difference of 0.33 EUR per player over the month.

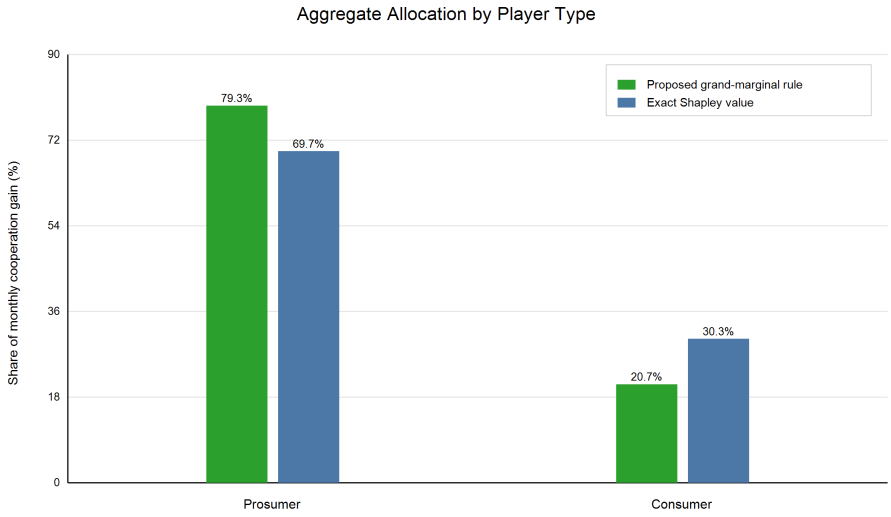


Figure 4.2: Aggregate allocation of the monthly cooperation surplus to prosumers and consumers under the two allocation methods.

4.1.9 Discussion

The main advantage of the proposed mechanism is that it is economically intuitive and computationally lightweight. It preserves the users' stand-alone optimized benefits and rewards only the incremental value that truly emerges from cooperation. This is particularly suitable for renewable energy communities in which users may differ significantly in their demand profiles, generation patterns, and battery flexibility.

At the same time, the rule should be interpreted as a pragmatic contribution-based allocation rather than as an axiomatic cooperative-game solution such as the Shapley value or the nucleolus. It is efficient and individually rational with respect to stand-alone optimization, but it does not generally inherit the additivity or core-stability guarantees of more demanding solution concepts. For this reason, the rule is best viewed as a transparent and scalable allocation mechanism that prioritizes implementability and contribution awareness over full cooperative-game-theoretic fairness and stability.

Chapter 5

Subcoalition Formation

5.1 Motivation

The previous chapters assumed that cooperation can be organized either for a given coalition or for the whole community. This chapter changes the question: if many users are available, which users should actually be placed together? In a larger version of the running example, the community may contain several A-type prosumers with midday surplus, several B-type consumers with evening demand, and several C-type users with useful battery flexibility. A good subcoalition should combine profiles that complement one another, so that local surplus, demand, and flexibility can meet inside the same group.

Although the grand-coalition model provides the maximum attainable coordination benefit (since all users optimize together), forming one large coalition may not always be desirable. A large coalition increases the computational burden of daily operation, requires broader information sharing between participants, and may be less attractive from an organizational perspective. We therefore study how to decompose a community into smaller subcoalitions while retaining as much as possible of the grand-coalition benefit.

The subcoalition-formation step does not replace the cooperative scheduling model; it selects the groups before daily scheduling is solved. Once the groups have been formed, each subcoalition operates as a smaller cooperative community. The quality of a partition is evaluated by comparing the annual coordination gain of the resulting subcoalitions with the coordination gain of the grand coalition.

Before presenting the formation method, it is important to clarify the composition of the user population used in the larger experiments. The first 20 players correspond to cleaned household profiles from the Irish smart-meter dataset. Experiments with more than 20 players are extended with synthetic users: players 21 onward were generated with the `LoadProfileGenerator` tool and converted to the same 15-minute CSV structure as the measured households. These synthetic profiles are used to study scalability and larger community sizes, and should therefore be interpreted as plausible additional household profiles rather than independently metered Irish households.

5.2 Problem Formulation

A partition is simply a grouping plan: one set of users operates as the first subcoalition, another set as the second, and so on, until every user belongs to exactly one group. We now formalize this idea and define how the quality of a grouping plan is measured.

Let N denote the set of users in the community (so the grand coalition is N itself). A subcoalition partition divides N into disjoint groups

$$\mathcal{P} = \{S_1, S_2, \ldots, S_K\},$$

such that

$$S_k \cap S_\ell = \emptyset \quad \forall k \neq \ell, \qquad \bigcup_{k=1}^{K} S_k = N.$$

In this work, all subcoalitions are constrained to have equal size. If m denotes the target subcoalition size, then

$$|S_k| = m \quad \forall k.$$

For a given day, the grand-coalition coordination gain is defined as the reduction in total cost obtained by allowing all users to cooperate relative to the case where each user optimizes individually. If C_i^{solo} is the stand-alone optimized cost of player i and C_N is the optimized cost of the grand coalition, then the grand-coalition coordination gain is

$$G_N = \sum_{i \in N} C_i^{\text{solo}} - C_N.$$

For a partition $\mathcal{P}$, the corresponding subcoalition coordination gain is

$$G_{\mathcal{P}} = \sum_{k=1}^{K} \left(\sum_{i \in S_k} C_i^{\text{solo}} - C_{S_k} \right),$$

where C_{S_k} is the optimized cooperative cost of subcoalition S_k. The fraction of grand-coalition coordination gain recovered by the partition is then

$$R_{\mathcal{P}} = 100 \cdot \frac{G_{\mathcal{P}}}{G_N}.$$

A value close to 100% means that the partition captures most of the benefit that would have been achieved by coordinating all users together.

5.3 Representative-Day Selection

The pairwise scores used for subcoalition formation are not computed over the full year directly. Doing so would require solving a two-player cooperative scheduling problem for every pair of users and for every valid day of the year, which would be computationally expensive. Instead, the annual behaviour of the community is approximated using a small set of representative days. These days are selected so that they capture both typical operating conditions and important extreme conditions.

For each valid day d, an aggregate net-load profile is constructed by summing the 15-minute net load of all selected users:

$$L_d(t) = \sum_{i \in N} \left(c_i(d,t) - p_i(d,t) \right), \qquad t = 1, \ldots, 96,$$

where $c_i(d,t)$ and $p_i(d,t)$ denote the consumption and photovoltaic production of user i at time period t of day d. Therefore, each day is represented by a 96-dimensional vector

$$\mathbf{L}_d = \left(L_d(1), L_d(2), \ldots, L_d(96) \right).$$

This representation focuses the clustering procedure on the daily shape of the community's aggregate net-load profile, which is the main driver of internal energy-sharing opportunities. When the experiment includes synthetic users, their `LoadProfileGenerator` profiles enter the aggregate net-load calculation in exactly the same way as the measured profiles, after conversion to the common 15-minute data format.

Two extreme days are first selected explicitly. The first is the day with the largest total positive aggregate net load,

$$d^{\text{imp}} = \arg\max_d \sum_{t=1}^{96} \max\{L_d(t), 0\},$$

which corresponds to a high-import condition. The second is the day with the largest total aggregate surplus,

$$d^{\text{sur}} = \arg\max_d \sum_{t=1}^{96} \max\{-L_d(t), 0\},$$

which corresponds to a high-surplus condition. These days are included separately because they are important for evaluating the value of internal energy sharing but may not be selected by a clustering method focused on typical days.

After removing the two extreme days, k-medoids clustering is applied to the remaining daily profiles. Before clustering, each of the 96 time-period features is standardized, so that periods

with larger numerical variance do not dominate the distance calculation. The distance between two days is then computed using Euclidean distance between their standardized aggregate net-load profiles. The number of medoid clusters is set to 12. To reduce sensitivity to initialization, the k-medoids algorithm is run with 50 different k-medoids++ initializations, and the solution with the lowest total assignment distance is retained.

Each selected medoid represents all days assigned to its cluster. Therefore, its weight is equal to the corresponding cluster size:

$$w_d = |\{d' : d' \text{ is assigned to medoid } d\}|.$$

Each extreme day is assigned weight 1. The final representative set therefore contains 14 days: 12 k-medoids representative days and 2 extreme days. The sum of all weights is equal to the number of valid clean days used in the annual experiment. Consequently, the weighted pairwise scores approximate full-year behaviour while requiring only a small fraction of the daily pairwise optimizations.

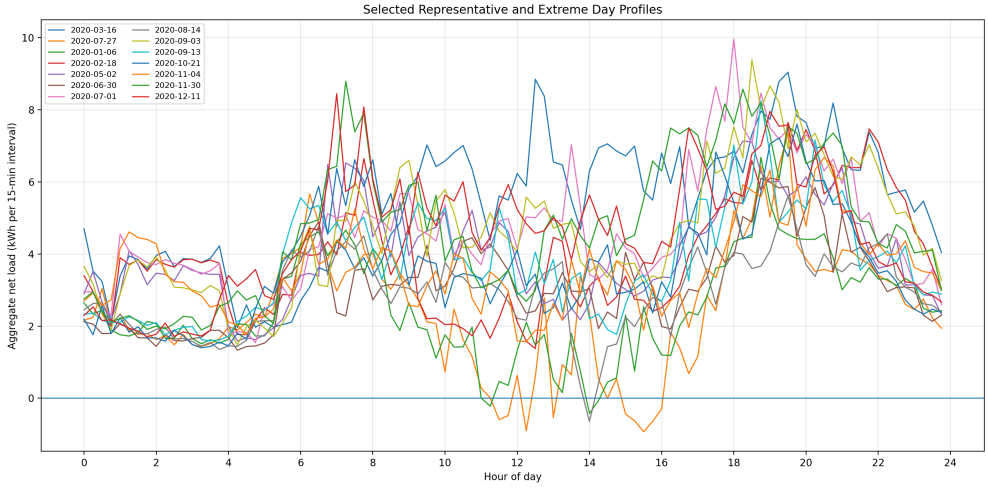


Figure 5.1: Aggregate net-load profiles of the representative days selected for the 30-player experiment. The set combines k-medoids medoid days with explicitly selected import and surplus extreme days.

5.4 Pairwise Score Approximation

Solving the full cooperative scheduling model for every possible candidate subcoalition is computationally expensive. To obtain a scalable proxy for coalition quality, this work constructs a pairwise score matrix. For each pair of users (i, j), a pairwise coordination gain is computed on the representative-day set described in Section 5.3.

Let $\mathcal{D}_{\text{rep}}$ denote the selected representative days and let w_d be the weight of representative

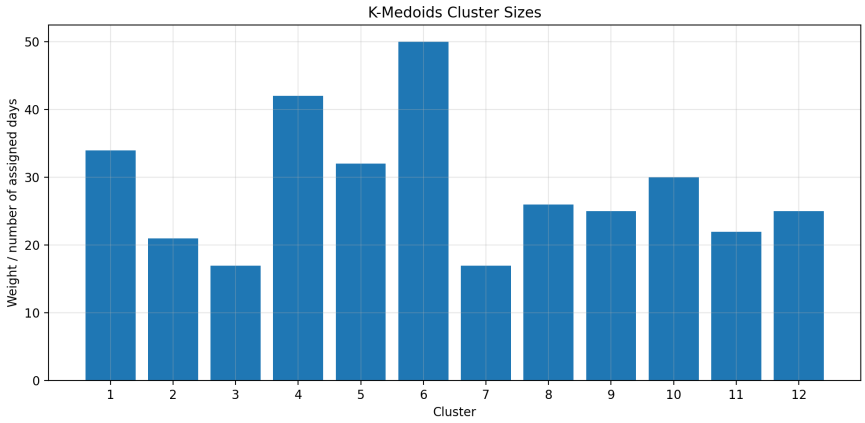


Figure 5.2: Cluster sizes of the k-medoids representative days for the 30-player experiment. Each cluster size is used as the weight of the corresponding medoid day in the pairwise-score calculation.

day d. For a pair of users (i, j), the weighted pairwise score is

$$q_{ij} = \sum_{d \in \mathcal{D}_{\text{rep}}} w_d \left(C_i^{\text{solo}}(d) + C_j^{\text{solo}}(d) - C_{\{i,j\}}(d) \right).$$

The value q_{ij} measures how beneficial it is to place users i and j in the same coalition, according to their observed ability to create internal coordination gains on representative days.

Given the pairwise score matrix, the score of a candidate subcoalition S is approximated as the sum of all pairwise scores inside the subcoalition:

$$Q(S) = \sum_{\{i,j\} \subseteq S} q_{ij}.$$

The score of a full partition is then represented by the vector

$$(Q(S_1), Q(S_2), \ldots, Q(S_K)).$$

The central assumption is that coalitions with larger internal pairwise scores are more likely to achieve larger actual annual coordination gains. This assumption was tested experimentally by comparing pairwise-score predictions with actual coalition gains across a large set of candidate size-4 coalitions. Figure 5.3 shows that the pairwise scores are informative: coalitions with higher score tend to produce higher realized gain. In this validation experiment, the Pearson and Spearman correlations were 0.956 and 0.947, respectively. The relationship is not perfect, because the true cooperative scheduling problem includes multi-user battery and exchange interactions, but the score provides a useful and much cheaper approximation.

Figure 5.4 shows an example of the resulting pairwise score matrix for the 30-player experiment. Brighter entries correspond to user pairs with stronger estimated coordination potential.

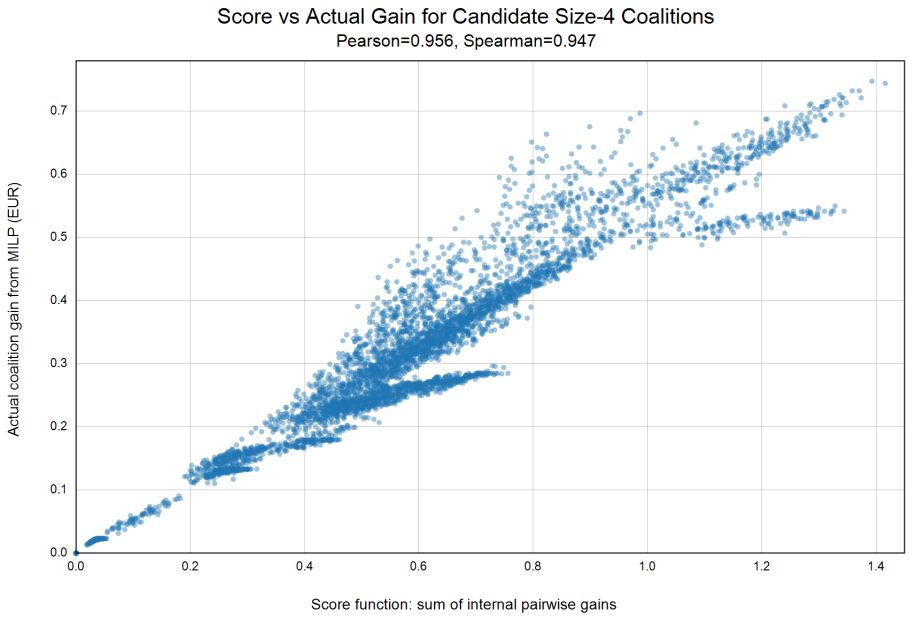


Figure 5.3: Validation of the pairwise-score approximation over a large set of candidate size-4 coalitions: candidate coalition score versus actual realized coordination gain.

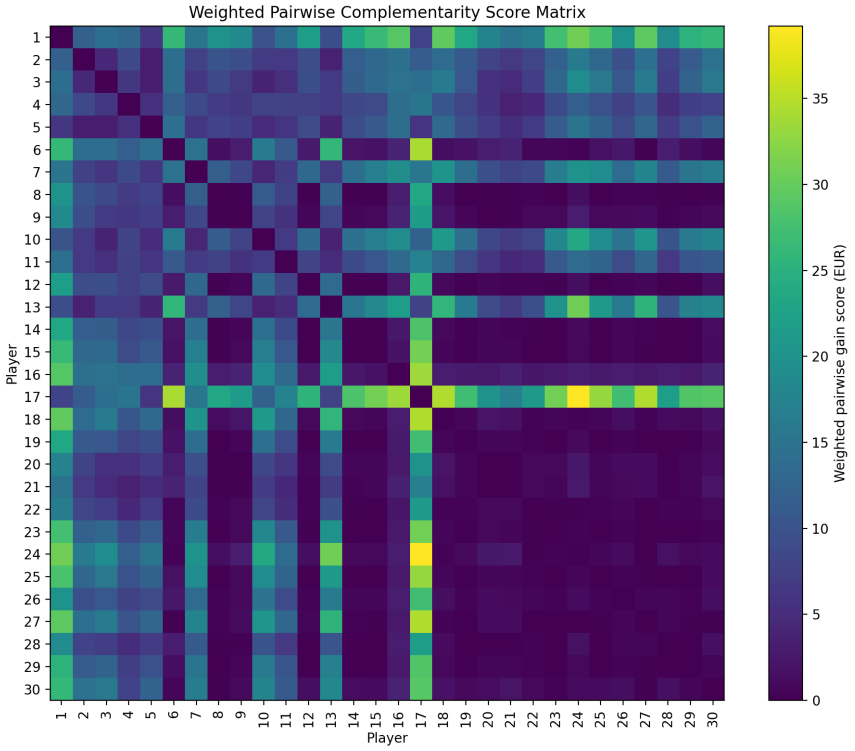


Figure 5.4: Weighted pairwise coordination score matrix for the 30-player experiment.

5.5 Lexicographic Fairness Objective

If the partitioning objective were only to maximize the total score

$$\sum_{k=1}^{K} Q(S_k),$$

the resulting partition could place most of the coordination value in a few strong groups while leaving another group with a poor score. This would be undesirable, because each subcoalition should have a reasonable opportunity to benefit from internal coordination.

For this reason, the proposed objective is lexicographic. The first priority is to maximize the weakest subcoalition score:

$$\max_{\mathcal{P}} \min_{k=1,\dots,K} Q(S_k).$$

Subject to this maximum fairness level, the second priority is to maximize the total internal score:

$$\max_{\mathcal{P}} \sum_{k=1}^{K} Q(S_k).$$

This objective reflects the idea that the partition should be both fair and efficient. The minimum group score prevents one subcoalition from being left behind, while the total score uses the remaining degrees of freedom to improve aggregate coordination potential.

5.6 Exact MILP Formulation

The exact formulation uses binary assignment variables

$$x_{ig} = \begin{cases} 1, & \text{if player } i \text{ is assigned to group } g, \\ 0, & \text{otherwise,} \end{cases}$$

and auxiliary binary variables

$$y_{ijg} = \begin{cases} 1, & \text{if players } i \text{ and } j \text{ are both assigned to group } g, \\ 0, & \text{otherwise.} \end{cases}$$

Each player must be assigned to exactly one group:

$$\sum_{g=1}^{K} x_{ig} = 1 \qquad \forall i \in N,$$

and each group must have the required size:

$$\sum_{i \in N} x_{ig} = m \qquad \forall g = 1, \ldots, K.$$

The pairwise variables are linked to the assignment variables through the standard linearization:

$$y_{ijg} \leq x_{ig}, \qquad y_{ijg} \leq x_{jg}, \qquad y_{ijg} \geq x_{ig} + x_{jg} - 1.$$

The score of group g is then

$$Q_g = \sum_{i<j} q_{ij} y_{ijg}.$$

The lexicographic objective is implemented in two stages. In the first stage, the model maximizes a variable z such that

$$Q_g \geq z \qquad \forall g.$$

After the optimal or best available value of z is obtained, the second stage maximizes the total score subject to the constraint that every group remains above this minimum score level.

This formulation is exact, but it becomes difficult to solve as the number of players increases. The number of possible equal-size partitions is

$$\frac{n!}{(m!)^K K!},$$

where n is the number of players, m is the group size and $K = n/m$. For example, partitioning 20 players into five groups of four already gives approximately $3.05 \cdot 10^{11}$ possible partitions. Partitioning 30 players into five groups of six gives approximately $1.43 \cdot 10^{19}$ possible partitions. The solver does not enumerate these partitions explicitly, but the size of the search space explains why the exact MILP becomes impractical for larger cases.

5.7 Scalable Local-Search Heuristic

To preserve the same pairwise-score logic while avoiding the computational cost of exact optimization, a multi-start local-search heuristic was developed. The heuristic starts from many random feasible partitions and improves each partition by swapping players between groups.

For each initial partition, the algorithm computes the score of every group and evaluates the lexicographic objective

$$\left(\min_k Q(S_k), \sum_k Q(S_k) \right).$$

It then considers pairwise swaps of players belonging to different groups. A swap is accepted if

it improves the lexicographic objective: first by increasing the minimum group score, or, if the minimum score remains unchanged, by increasing the total score. The search continues until no improving swap exists or until a maximum number of passes has been reached. Repeating this process from many random starts reduces the probability of being trapped in a poor local optimum.

For the 30-player experiment, the strong local-search run used 5000 random starts, at most 150 improvement passes per start, and early stopping after 1000 starts without improvement of the global best solution. The final solution was found after 2773 completed starts and required 230.69 seconds. The selected subcoalitions are shown in Table 5.1. The same scalable procedure was also applied to the 50-player cases. For the original 50-player composition with 10 prosumers and 40 consumers, the local search required 1222.31 seconds. For the balanced 50-player composition with 25 prosumers and 25 consumers, the corresponding runs required 1350.36 seconds for five groups of ten and 1507.56 seconds for ten groups of five. These runtimes are much smaller than the time required by the exact MILP on the larger instances, while preserving the same pairwise-score objective.

Table 5.1: Subcoalitions selected by the 30-player pairwise local-search heuristic.

Group	Members	Pairwise score
1	1, 2, 8, 15, 19, 21	139.42
2	3, 10, 18, 23, 29, 30	139.30
3	4, 7, 11, 16, 25, 28	140.51
4	5, 6, 12, 13, 20, 24	139.40
5	9, 14, 17, 22, 26, 27	139.19

The minimum group score is 139.19, while the maximum group score is 140.51. The standard deviation of the group scores is only 0.48, which indicates that the local-search heuristic produced a highly balanced partition according to the pairwise objective. Figure 5.5 shows the internal score of each selected subcoalition.

5.8 Experimental Setup

The subcoalition experiments use the cleaned Irish household profiles for players 1–20 and the same daily cooperative scheduling model as the annual evaluation of the grand coalition. Larger cases are constructed by adding synthetic `LoadProfileGenerator` profiles for players 21 onward. Therefore, the 16- and 20-player experiments use only measured dataset users, the 30-player experiment uses 20 measured and 10 synthetic users, and the 50-player experiments use 20 measured and 30 synthetic users. All profiles are stored in the same processed 15-minute format and are evaluated under the same battery, tariff and cooperative-scheduling assump-

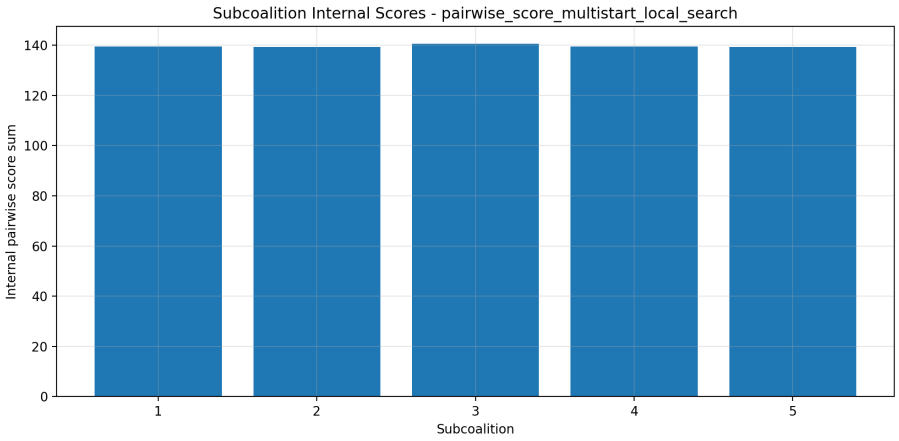


Figure 5.5: Internal pairwise score of the subcoalitions selected by the 30-player local-search heuristic.

tions. All annual evaluations cover the 343 valid days of 2020 for which complete 15-minute household and price data are available.

Two different 50-player compositions are considered. The first contains 10 prosumers and 40 consumers, which corresponds to the original larger-scale setting used to test whether large subcoalitions can preserve the grand-coalition value. The second contains 25 prosumers and 25 consumers. This balanced-prosumer setting is included to test whether the conclusions change when local production is much more widely distributed across the population. For the balanced 25P/25C setting, two group sizes are evaluated: five subcoalitions of ten players and ten subcoalitions of five players.

The experiments compare three types of subcoalition formation:

1. exact lexicographic MILP partitioning for cases where the MILP is computationally tractable;
2. pairwise-score local search for the scalable 30- and 50-player cases;
3. balanced random benchmarks, where prosumers are distributed as evenly as possible across the groups and the remaining consumers are assigned randomly.

The balanced random benchmark is intentionally stronger than a completely random partition. It incorporates the simple but important design rule that each subcoalition should contain both producers and consumers. This makes it a meaningful baseline for testing whether pairwise-score optimization adds value beyond prosumer-consumer balance.

5.9 Annual Results

Table 5.2 reports the main annual results for the score-based partitions. The recovered grand-coalition coordination gain is the most important metric. It measures how much of the co-

ordination benefit of the grand coalition is retained after the community is split into smaller groups.

Table 5.2: Annual performance of score-based subcoalition formation.

Case	Groups	Method	Grand gain (EUR)	Subcoalition gain (EUR)	Recovered (%)
16 players	4 groups of 4	Exact MILP	221.81	203.15	91.59
20 players	5 groups of 4	Exact MILP	297.47	259.84	87.35
30 players	5 groups of 6	Local search	331.81	312.72	94.25
50 players, 10P/40C	5 groups of 10	Local search	355.21	350.65	98.72
50 players, 25P/25C	5 groups of 10	Local search	601.59	570.32	94.80
50 players, 25P/25C	10 groups of 5	Local search	601.59	538.12	89.45

The results show that subcoalitions can recover a large share of the grand-coalition coordination gain. In the 16-player case, the exact MILP partition recovers 91.59% of the grand-coalition coordination gain. In the 20-player case, the recovered fraction is 87.35%. For the 30-player case, the scalable local-search heuristic recovers 94.25% of the grand-coalition gain.

The 50-player experiments show two additional effects. In the original 10P/40C setting, five subcoalitions of ten recover 98.72% of the grand-coalition gain. In the balanced 25P/25C setting, the absolute annual coordination gain is much larger, because more players have local production. However, the recovered percentage is lower: five groups of ten recover 94.80%, and ten groups of five recover 89.45%. This reduction is expected because the grand coalition has more freedom to match surplus and demand across many prosumers and consumers. When the same population is divided into smaller groups, part of that exchange potential is restricted.

Figure 5.6 shows the cumulative coordination gain of the 30-player local-search partition compared with the grand coalition. The two curves remain close over the annual horizon, which indicates that most of the value of full coordination is preserved by the selected subcoalitions.

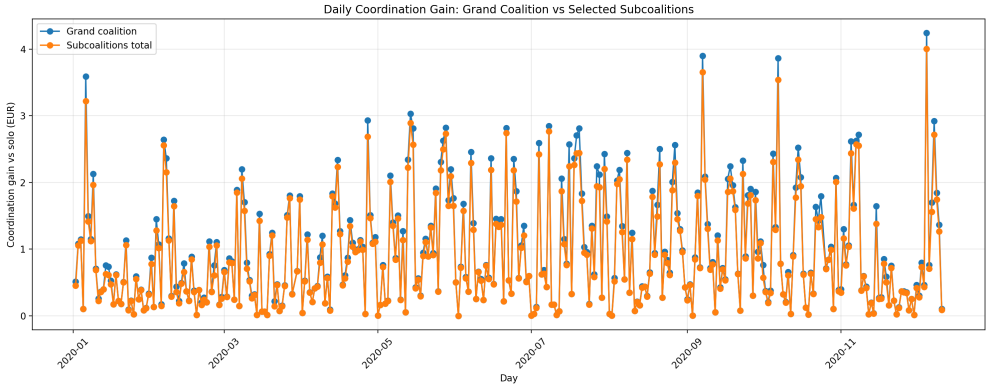


Figure 5.6: Cumulative annual coordination gain for the 30-player grand coalition and the local-search subcoalitions.

5.10 Comparison with Balanced Random Partitions

Table 5.3 compares the score-based results with balanced random benchmarks. The random benchmark is averaged over multiple random groupings. In the 16-player case, the random benchmark distributes the six prosumers as evenly as possible among four groups. In the 20-, 30- and 50-player cases, the random benchmark places the same number of prosumers in each group whenever this is possible.

Table 5.3: Comparison between score-based partitions and balanced random benchmarks.

Case	Score-based recovered (%)	Random mean (%)	Random min (%)	Random max (%)
16 players	91.59	83.73	64.60	91.00
20 players	87.35	87.00	84.71	88.31
30 players	94.25	94.71	93.82	95.35
50 players, 10P/40C, 5x10	98.72	98.31	97.32	98.83
50 players, 25P/25C, 5x10	94.80	94.06	93.56	94.80
50 players, 25P/25C, 10x5	89.45	85.41	84.85	85.86

The comparison reveals an important pattern. For the smaller 16-player case, the score-based method clearly outperforms the balanced random benchmark on average. This suggests that when the number of users is small, the identity of the individual users placed together has a strong effect on coalition quality. In this setting, pairwise-score optimization is valuable because poor pairings cannot be easily absorbed by the rest of the group.

As the number of players and the group size increase, the difference between score-based optimization and balanced random partitions decreases in cases where each subcoalition already contains enough production and demand diversity. In the 20-player case, the score-based partition and the balanced random mean are very close. In the 30-player case, the local-search partition recovers 94.25% of the grand-coalition gain, while the average balanced random partition recovers 94.71%. In the 50-player 10P/40C case, the local-search partition recovers 98.72%, which is slightly above the balanced-random average of 98.31% and very close to the best random draw of 98.83%.

The balanced 25P/25C experiments add a useful contrast. With five groups of ten, the local-search method recovers 94.80%, slightly above the random mean of 94.06%. With ten groups of five, the local-search method recovers 89.45%, while the balanced random mean is only 85.41%. This difference is much larger and shows that the pairwise-score method becomes more useful when the subcoalitions are smaller and the formation decision restricts internal exchanges more strongly.

This behaviour does not invalidate the pairwise-score approach. Instead, it shows that prosumer-consumer balance is a dominant design factor in the mixed measured-synthetic pop-

ulation used here. Once every subcoalition contains enough prosumers and enough consumers, most of the available internal energy-sharing value is already captured. The pairwise score then acts as a second-order refinement. When groups become smaller, however, individual matching matters more, and the score-guided method can clearly outperform balanced random assignment.

5.11 Internal Matching Interpretation

The strong performance of balanced random partitions can be explained by internal energy matching. A subcoalition creates coordination value when local surplus production can be consumed internally by other members instead of being exported to the grid at a lower price. Table 5.4 reports internal matching diagnostics for the three 50-player local-search cases.

Table 5.4: Internal matching diagnostic for the 50-player local-search experiments.

Case	Matched surplus (kWh)	Surplus utilization (%)	Unmatched surplus (kWh)
50 players, 10P/40C, 5x10	10059.74	99.06	95.55
50 players, 25P/25C, 5x10	17544.97	83.61	3439.85
50 players, 25P/25C, 10x5	16000.55	76.25	4984.27

In the 10P/40C case, the total annual prosumer surplus is approximately 10155 kWh, and the local-search partition internally matches 10059.74 kWh of it. This corresponds to 99.06% utilization, which explains why the subcoalition partition recovers almost all of the grand-coalition gain. There is enough consumer demand inside each group to absorb almost every available surplus kWh.

In the 25P/25C case, the total annual prosumer surplus is much larger, approximately 20985 kWh. With five groups of ten, the partition internally matches 17544.97 kWh, corresponding to 83.61% utilization. With ten groups of five, the matched surplus falls to 16000.55 kWh, corresponding to 76.25% utilization. This explains why the recovered grand-coalition percentage is lower in the balanced-prosumer experiments: the grand coalition can pool surplus and demand across all 50 players, while smaller subcoalitions leave more surplus unmatched.

The internal matching diagnostic therefore clarifies the difference between the 50-player cases. The 10P/40C setting has a smaller amount of surplus relative to total demand, so almost all local production can be used internally even after partitioning. The 25P/25C setting creates more absolute coordination value, but it is harder for each smaller subcoalition to absorb all available surplus. This effect becomes stronger when the group size is reduced from ten to five.

5.12 Computational Scalability

The exact MILP is useful for small cases because it directly implements the desired lexicographic objective. However, its scalability is limited by the combinatorial nature of the partitioning problem. For this reason, the final larger-scale experiments use the pairwise local-search heuristic as the scalable formation method.

The local-search heuristic addresses this limitation. For the 30-player case, the strong local-search run completed in 230.69 seconds and produced a well-balanced partition. For the 50-player cases, the local-search runs completed in roughly 20–25 minutes depending on the population composition and group size. The 10P/40C case recovered 98.72% of the grand-coalition coordination gain, while the 25P/25C cases recovered 94.80% with groups of ten and 89.45% with groups of five. This is several orders of magnitude faster than attempting to prove exact optimality with the MILP. The heuristic does not provide a certificate of optimality, but it preserves the same pairwise-score objective and gives high-quality feasible solutions quickly enough to be used at larger scale.

This leads to a practical conclusion. Exact MILP partitioning is appropriate for small communities, where optimality can still be obtained. For larger communities, a score-guided local-search heuristic is more suitable, because it retains the economic logic of the pairwise score while avoiding the computational burden of exact branch-and-bound optimization.

5.13 Discussion

The experimental results lead to two main conclusions. First, pairwise-score-based subcoalition formation is a meaningful and scalable way to construct cooperative groups. The pairwise scores are validated against realized coalition gains, and the selected partitions recover a large share of the grand-coalition coordination benefit.

Second, the experiments show that prosumer-consumer balance is a first-order design principle. In small or constrained cases, the pairwise-score optimization can substantially improve over balanced random assignment. As the number of users and the group size increase, balanced random partitions already contain enough diversity to capture most of the available internal exchange benefit. In those cases, the additional value of score-based optimization becomes more modest, as seen in the 50-player 10P/40C case and in the 25P/25C case with groups of ten.

The 25P/25C experiments show that this conclusion depends strongly on how restrictive the subcoalitions are. When the same 50-player balanced-prosumer population is divided into ten groups of five, the local-search method recovers 89.45% of the grand-coalition gain, while the balanced random benchmark recovers only 85.41% on average. This is a substantial improvement and suggests that pairwise-score formation is especially valuable when subcoalitions are

small, because each individual pairing has a larger effect on the ability of the group to match local surplus with local demand.

This does not mean that the score-based method is unnecessary when balanced random partitions perform well. Rather, it shows that the proposed method is a robust and reliable formation rule. The score-guided procedure produces partitions that are consistently competitive with, and in several cases better than, the balanced random benchmark, without relying on a favourable random draw. When the community is large and each group already contains enough production and demand diversity, balanced random assignment can perform well because the main energetic complementarity is already present. However, the pairwise-score method preserves this balance while also using information about the specific compatibility of individual users. This becomes especially important when subcoalitions are smaller or more restrictive, as in the 50-player 25P/25C case with ten groups of five, where the score-based method clearly outperforms the balanced random benchmark.

Overall, the subcoalition experiments show that most of the grand-coalition coordination gain can be preserved without requiring all users to operate as a single cooperative entity. The proposed pairwise-score framework provides an interpretable way to form such subcoalitions and exposes an important empirical insight: in this mixed measured-synthetic population, the dominant driver of subcoalition performance is not only the exact optimization of pairwise scores, but also the basic energy balance between local production and local demand.

Chapter 6

Subcoalition Stability under Profit Sharing

6.1 Motivation

After subcoalitions have been formed, there is still one practical question left: will users want to remain in the groups assigned to them? In the running example, suppose user C receives only a small cooperation bonus in its current group. C may prefer to join another group where its battery flexibility is more valuable. Such a move is credible only if the receiving group also benefits, or at least does not lose, after accepting C.

This chapter connects subcoalition formation with the proposed profit-sharing rule and evaluates whether the resulting groups are stable under an accepted-migration criterion. The goal is not only to recover a large fraction of the grand-coalition gain, but also to test whether any user has a credible incentive to move to another subcoalition after bonuses have been allocated.

6.2 Problem Formulation

Let $\mathcal{P} = \{S_1, S_2, \ldots, S_K\}$ denote the subcoalition partition obtained from the pairwise-score formation procedure. For each day, every subcoalition is optimized independently, and the coordination gain of that subcoalition is allocated to its members using the proposed positive leave-one-out marginal contribution rule. Over an evaluation horizon $\mathcal{D}$, the total allocated bonus of player i in coalition S is denoted by

$$B_i(S) = \sum_{d \in \mathcal{D}} b_i(S, d),$$

where $b_i(S, d)$ is the daily bonus allocated to player i when coalition S is optimized on day d.

The stability test itself is based on these allocated bonuses. For each current subcoalition S_k, the economically weakest member is

$$w_k = \arg\min_{i \in S_k} B_i(S_k).$$

This choice is intentionally conservative: if a subcoalition has a member who is most likely to be dissatisfied with the current allocation, that member is the first natural candidate for migration.

To present the size of a candidate migration, the bonus change is also expressed relative to the player's current bonus:

$$\Delta_i^{\text{rel}}(S \to S') = 100 \cdot \frac{B_i(S') - B_i(S)}{B_i(S)}.$$

Thus, a value of 50% means that the player would receive 50% more annual bonus after the move than in the current subcoalition. This percentage is used only to report the magnitude of the incentive; the stability decision is still made from the sign of the underlying bonus change.

6.3 Accepted-Migration Stability Criterion

The accepted-migration criterion can be read as a simple consent test. A move is not enough to be attractive for the moving user; it must also be acceptable to the users in the target subcoalition. This mirrors the practical situation in which a participant cannot simply enter another cooperative group if doing so reduces the bonuses of the existing members.

For every weakest player w_k, the experiment considers a possible migration from the current source coalition S_k to every other target coalition S_ℓ, with $\ell \neq k$. The receiving coalition after the move is

$$S'_\ell = S_\ell \cup \{w_k\}.$$

The move is considered a blocking accepted migration if two conditions hold.

First, the moving player must strictly improve:

$$B_{w_k}(S'_\ell) - B_{w_k}(S_k) > 0.$$

Second, every current member of the target coalition must weakly improve after accepting the new player:

$$B_j(S'_\ell) - B_j(S_\ell) \geq 0 \qquad \forall j \in S_\ell.$$

Thus, a player cannot unilaterally enter another subcoalition. The receiving subcoalition must also have no member who is made worse off. The source coalition is not given veto power in this test; the criterion focuses on the incentive of the moving player and the consent of the destination group.

This is a weaker and more operational stability notion than requiring stability against all possible coalition deviations. It is nevertheless meaningful for the present setting because it directly tests whether the proposed profit-sharing rule creates an accepted individual deviation after the subcoalitions have been formed.

6.4 Experimental Setup

The stability experiment was performed on the 20-player case, using the subcoalitions obtained from the pairwise-score formation experiment. The initial partition was:

$$S_1 = \{1, 6, 9, 14\},$$
$$S_2 = \{2, 7, 15, 19\},$$
$$S_3 = \{3, 8, 13, 18\},$$
$$S_4 = \{4, 5, 11, 16\},$$
$$S_5 = \{10, 12, 17, 20\}.$$

The evaluation horizon consisted of the 343 valid days of the year 2020. For each day, the cooperative optimization problem was solved for the current subcoalitions, and the daily gains were allocated using the positive leave-one-out marginal contribution rule. The annual bonus of each player was then computed by summing the allocated daily bonuses over all valid days.

The weakest player in each current subcoalition is shown in Table 6.1. These five players define the candidate movers. Since each of the five players can attempt to move to four other subcoalitions, the experiment evaluates 20 candidate migrations in total.

Table 6.1: Weakest player in each current subcoalition. The last column reports the player's current annual bonus relative to the average annual bonus of the same subcoalition.

Coalition	Weakest player	Current members	Relative to group avg. (%)
1	9	1, 6, 9, 14	21.49
2	19	2, 7, 15, 19	69.80
3	8	3, 8, 13, 18	48.35
4	11	4, 5, 11, 16	80.96
5	20	10, 12, 17, 20	60.79

Figure 6.1 visualizes the same comparison. Player 9 in the first subcoalition receives the smallest bonus relative to the average bonus of their own group, making this player the most natural candidate to seek a different group.

6.5 Results

The annual stability test found no blocking accepted migration. Table 6.2 summarizes the result.

The relative changes show that some rejected moves would be attractive for the moving player. The largest case is player 9 moving from subcoalition 1 to subcoalition 5, which would increase player 9's annual bonus by 79.64% relative to the current bonus. However, the same move would also impose large relative losses on members of the target coalition. Therefore,

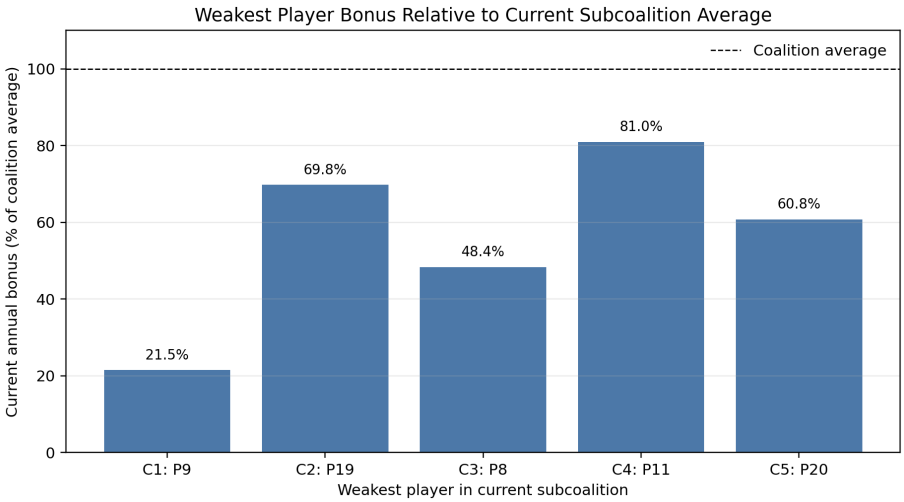


Figure 6.1: Annual bonus of the weakest player in each current subcoalition, expressed relative to the average annual bonus of that subcoalition.

Table 6.2: Summary of the accepted-migration stability experiment.

Quantity	**Value**
Players	20
Initial subcoalitions	5
Valid days evaluated	343
Candidate migrations checked	20
Candidate migrations where the moving player improves	6
Blocking accepted migrations	0
Largest relative moving-player bonus increase (%)	79.64

the relevant question is not only whether the moving player benefits, but whether the receiving group can accept the move without harming any current member.

Figure 6.2 focuses on the six candidate migrations where the moving player would personally improve. For each candidate move, the blue bar shows the moving player's relative annual bonus increase, while the red bar shows the largest relative annual bonus loss among the current members of the target coalition. A move can only be accepted if the red bar is non-negative, because this would mean that no current target member loses bonus.

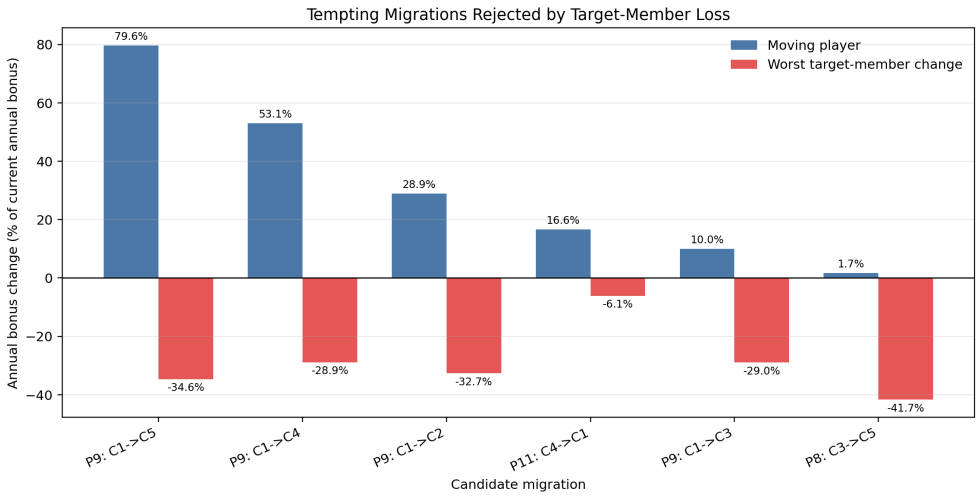


Figure 6.2: Candidate migrations that would improve the moving player's annual bonus. The bars report percentage changes relative to each affected player's current annual bonus. In every case, at least one current member of the target coalition would lose bonus, so the move is rejected.

The figure shows that all potentially attractive moves for the moving player are rejected by the target coalition. The remaining candidate migrations are not attractive to the moving player in the first place.

Table 6.3 lists the six moves that would benefit the moving player. None of them satisfies the target-acceptance condition.

Table 6.3: Candidate migrations that improve the moving player's annual bonus but are rejected by the target coalition. Percentage changes are measured relative to each affected player's current annual bonus.

Player	Source	Target	Player increase (%)	Worst target change (%)
9	1	5	79.64	-34.64
9	1	4	53.10	-28.88
9	1	2	28.91	-32.69
11	4	1	16.61	-6.12
9	1	3	9.97	-28.98
8	3	5	1.67	-41.70

The most attractive rejected move is the migration of player 9 from subcoalition 1 to subcoalition 5. This would increase player 9's annual bonus by 79.64% relative to the current

bonus. However, the move would reduce the annual bonuses of players 12 and 20 in the target coalition by 34.64% and 34.08%, respectively. Therefore, the target coalition would not accept the move. This case is shown in Figure 6.3.

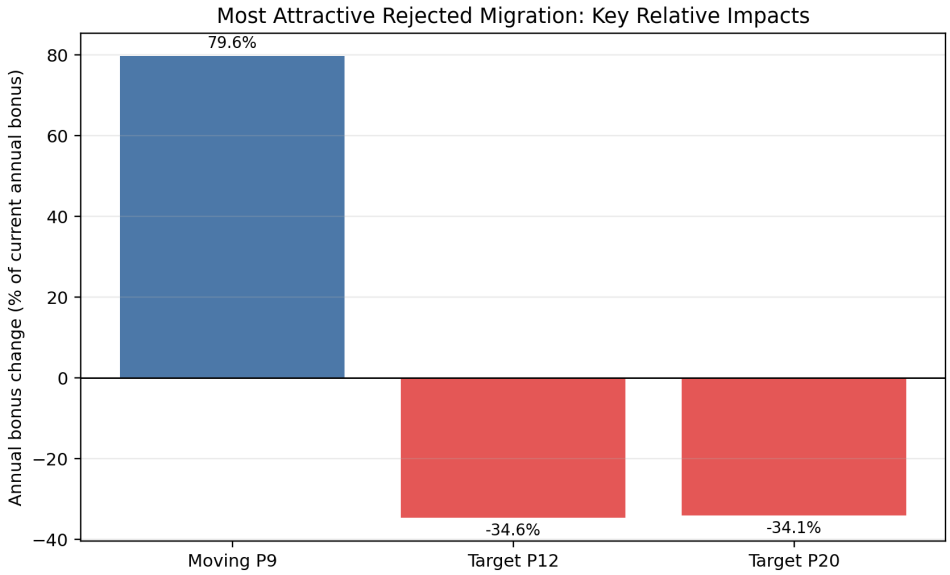


Figure 6.3: Key relative impacts for the most attractive rejected migration, where player 9 moves from subcoalition 1 to subcoalition 5. Although the moving player improves, two current target members lose annual bonus, so the move is not accepted.

6.6 Interpretation

The result suggests that the selected subcoalitions are stable under the accepted single-player migration criterion. This does not mean that the partition is stable against every possible collective deviation. The experiment only tests moves where the weakest player of each current subcoalition attempts to join one other existing subcoalition, and where the receiving subcoalition decides whether to accept the player based on the annual payoff changes of its current members.

Within this criterion, however, the result is strong. There are no accepted blocking migrations over the full annual evaluation horizon. Some players would personally benefit from moving to another subcoalition, and in relative terms some of these improvements are substantial because the current bonus of the moving player is small. Nevertheless, every profitable migration is blocked by the acceptance condition: at least one current member of the receiving coalition would receive a lower annual bonus after accepting the mover. The proposed profit-sharing rule therefore does not create a credible accepted incentive for the economically weakest players to leave their assigned subcoalitions.

This chapter complements the earlier results in two ways. The subcoalition-formation experiments showed that the selected groups preserve a large share of the grand-coalition coordination gain. The present stability experiment adds an individual-incentive perspective: after the gains are allocated, the weakest members do not have a credible accepted migration to another group. Thus, the combination of pairwise-score subcoalition formation and positive marginal-contribution profit sharing produces subcoalitions that are not only efficient in aggregate, but also stable under this migration-based test.

Chapter 7

Demand Response

7.1 Demand Response through Targeted Load Shifting

This section evaluates demand response as a targeted load-shifting mechanism. In contrast to the previous cooperative scheduling experiments, the objective here is not to form coalitions or to design a profit-allocation rule. Instead, we study whether a small amount of controllable demand can complement battery scheduling by moving consumption away from expensive operating periods and recovering it at more favourable times.

In the running example, demand response does not mean that user B consumes less energy over the day. Instead, user B shifts a flexible appliance or process away from an expensive evening interval and recovers the same energy in a more favourable period. User A's photovoltaic surplus and user C's battery remain important sources of flexibility, but the demand profile itself now becomes partially controllable. The relevant question is therefore incremental: after battery scheduling has already captured part of the available arbitrage opportunity, how much additional saving can be obtained by shifting a limited amount of flexible demand?

The demand-response experiment uses the same cleaned Irish household dataset and 15-minute resolution as the previous annual experiments. The analysis is performed for 20 users and for 343 valid days of the available 2020 data. The battery benchmark is a central ideal battery with daily initial and final state of charge equal to zero, keeping results comparable across battery capacities and isolating the effect of adding demand response.

7.2 Targeted Demand-Response Model

The model has two intuitive steps. First, it finds the short time window where consumption is most expensive for the community. Second, it shifts only the excess part of demand from that window to a recovery window where the marginal cost is lower. This preserves the total energy consumed over the day, while testing whether a small change in timing can reduce the scheduled operating cost.

For each day, the algorithm first identifies an activation window of one hour. The activation window is selected as the four consecutive 15-minute periods with the largest aggregate import

cost. If L_t denotes the aggregate net load in period t and π_t^{buy} is the buy price, the activation score of a candidate window A is

$$\sum_{t \in A} \pi_t^{\text{buy}} \max\{L_t, 0\}.$$

The window with the largest score is selected as the daily demand-response activation period. This means that demand response is targeted specifically at the time interval where grid imports are most expensive for the community.

The intervention is implemented as a cap on each user's total consumption inside the activation window. Let $C_i(A)$ be the consumption of user i in the activation window, summed over the four 15-minute periods. The mean activation-window consumption over all users is

$$\bar{C}(A) = \frac{1}{|N|} \sum_{i \in N} C_i(A).$$

In the main experiment, each user is capped at 80% of this mean value:

$$C_i^{\text{cap}}(A) = 0.8\bar{C}(A).$$

Only users whose activation-window consumption exceeds this cap are affected. For such users, the shifted energy is

$$E_i^{\text{shift}} = \max\{0, C_i(A) - C_i^{\text{cap}}(A)\}.$$

This energy is removed proportionally from the user's activation-window consumption profile. Importantly, the energy is not curtailed permanently. The same amount of energy is added back in a recovery window, so the mechanism represents load shifting rather than load shedding.

The recovery window has duration two hours. In the main scenario, it is selected using a battery-aware rule. Rather than choosing only the window with the lowest buy price, the algorithm uses the post-battery marginal consumption cost implied by the baseline battery schedule. Therefore, the recovery period is chosen to be favourable after accounting for the battery's ability to absorb surplus or shift energy on its own. This is important because a low-price period is not necessarily valuable for recovery if the battery has already captured most of the opportunity.

For each battery capacity, the daily evaluation follows three steps. First, the baseline scheduling problem is solved without demand response. Second, the cap-based demand-response modification is applied to the aggregate consumption profiles. Third, the battery scheduling problem is solved again on the modified profiles. The demand-response benefit is the difference between the baseline scheduled cost and the scheduled cost after the demand-response load shift.

7.3 Experimental Setup

Table 7.1 summarizes the main configuration of the experiment. The reported demand-response results focus on the final scenario used in the analysis: a cap ratio of 80%, a one-hour activation window, a two-hour recovery window, and the battery-aware recovery rule.

Table 7.1: Configuration of the targeted demand-response experiment.

Parameter	Value
Number of users	20
Valid solved days	343
Time resolution	15 minutes
Periods per day	96
Activation-window length	1 hour
Recovery-window length	2 hours
Activation criterion	Maximum import cost
Demand-response rule	Cap at 80% of mean activation load
Recovery rule	Battery-aware marginal cost
Small battery capacities	0–80 kWh
Large battery reference	185 kWh

7.4 Demand-Response Results

The results show that targeted demand response creates a measurable reduction in operating cost, but that the magnitude of the additional benefit is modest at the household-community scale. Table 7.2 reports the total annual demand-response saving for selected battery capacities. The amount of shifted energy is the same across battery capacities, because the demand-response rule is applied to the original consumption profiles. However, the economic value of this shifted energy decreases as the battery capacity increases.

Table 7.2: Targeted demand-response results for the cap-80%, battery-aware recovery scenario.

Battery (kWh)	Baseline cost (EUR)	DR cost (EUR)	DR saving (EUR)	Saving (%)	Value (EUR/kWh)
0	6758.31	6623.30	135.01	2.00	0.0476
20	6310.71	6206.55	104.16	1.65	0.0367
40	6037.24	5951.37	85.87	1.42	0.0303
60	5829.87	5758.29	71.58	1.23	0.0253
80	5667.00	5608.68	58.31	1.03	0.0206

Without a battery, the targeted demand-response policy reduces the total cost from 6758.31 EUR to 6623.30 EUR. This corresponds to a saving of 135.01 EUR, or 2.00% of the no-demand-response cost. Across the 343 solved days, the policy shifts 2834.50 kWh of con-

sumption and affects on average 8.74 users per day. The average economic value of the shifted energy in the no-battery case is therefore approximately 0.0476 EUR/kWh.

When battery capacity is added, the incremental value of demand response declines. With an 80 kWh central battery, the baseline scheduled cost is already reduced to 5667.00 EUR. Applying the same demand-response rule further reduces the cost to 5608.68 EUR, but the additional saving is only 58.31 EUR, or 1.03%. The value per shifted kWh falls from 0.0476 EUR/kWh in the no-battery case to 0.0206 EUR/kWh with an 80 kWh battery.

Figure 7.1 shows this trend across the tested small-battery capacities. The curve decreases with battery size because the battery and the demand-response action partly compete for the same arbitrage opportunity. Both mechanisms attempt to reduce imports in expensive periods and move energy to periods with lower marginal cost. As the battery becomes larger, more of this opportunity is already captured by battery scheduling, leaving less additional value for demand response.

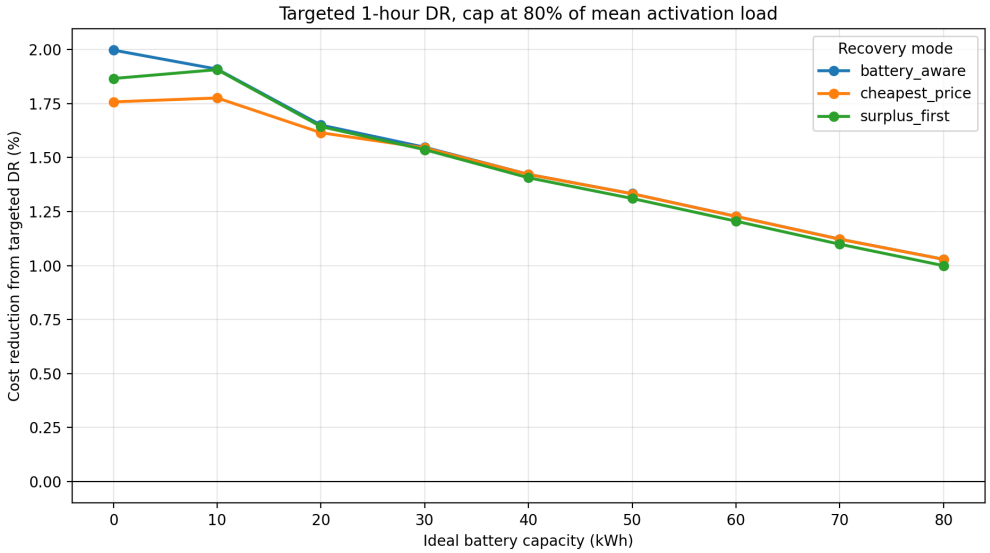


Figure 7.1: Cost reduction from targeted demand response as a function of the central ideal battery capacity, for the cap-80% demand-response rule.

7.5 Comparison with Large-Battery Scheduling

The targeted demand-response policy can also be interpreted as a way to partially compensate for limited storage capacity. Table 7.3 compares the small-battery cases with a large-battery scheduling reference of 185 kWh.

The 80 kWh battery alone achieves a 16.15% reduction relative to the no-battery baseline. Adding targeted demand response increases the total reduction to 17.01%. In terms of the 185 kWh reference, the 80 kWh battery alone captures 66.82% of the large-battery saving, while

Table 7.3: Comparison between small-battery scheduling, targeted demand response, and the 185 kWh scheduling reference.

Case	Cost (EUR)	Saving vs. no battery	Share of 185 kWh gain
No battery	6758.31	0.00%	0.00%
No battery + DR	6623.30	2.00%	8.27%
80 kWh battery	5667.00	16.15%	66.82%
80 kWh battery + DR	5608.68	17.01%	70.39%
185 kWh battery	5125.05	24.17%	100.00%

the combination of 80 kWh battery and demand response captures 70.39%. Thus, demand response closes part of the remaining gap to the large-battery benchmark, but it does not replace the value of substantially larger storage.

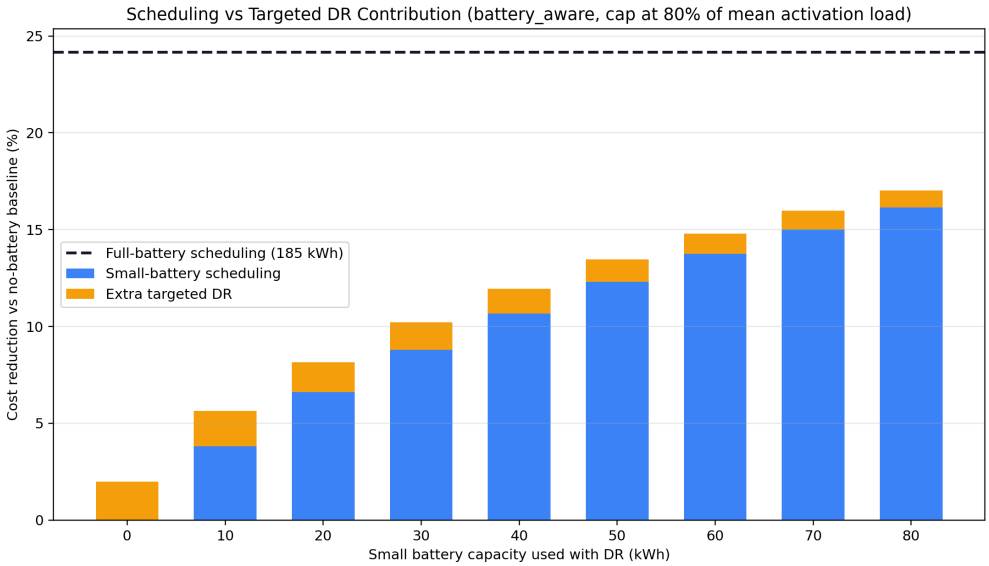


Figure 7.2: Cost reduction from small-battery scheduling and additional targeted demand response, compared with the 185 kWh battery scheduling reference.

Figure 7.3 shows the combined saving as a percentage of the 185 kWh full-battery saving. The combined small-battery and demand-response system approaches the large-battery benchmark as the small battery capacity increases. However, the improvement due specifically to demand response is incremental. Even at 80 kWh, the targeted demand-response action closes only 10.76% of the remaining gap between the 80 kWh battery and the 185 kWh reference.

7.6 Discussion

The results suggest that targeted demand response is technically useful, but its economic impact is limited in the residential community setting considered here. The main reason is that the

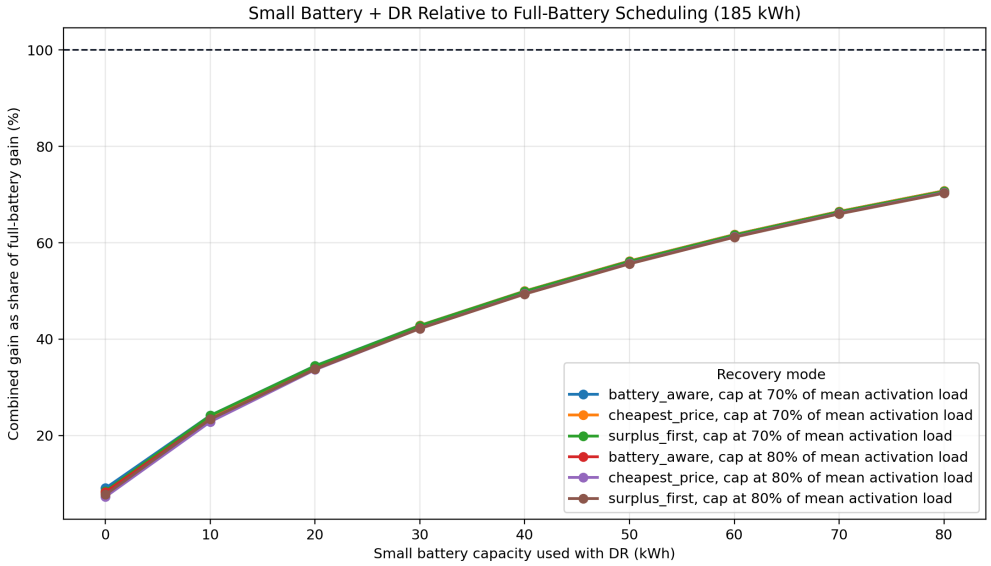


Figure 7.3: Combined gain from small-battery scheduling and targeted demand response as a share of the 185 kWh battery scheduling gain.

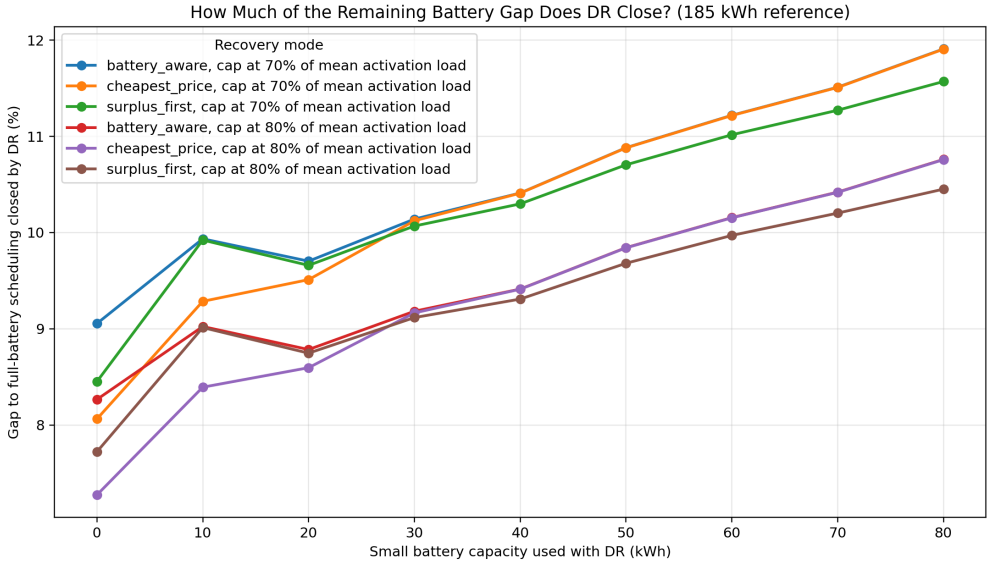


Figure 7.4: Share of the remaining gap to the 185 kWh battery reference that is closed by targeted demand response.

amount of flexible consumption per household is small. The policy shifts 2834.50 kWh over the whole evaluation horizon, but the value of this shift is only 135.01 EUR in the no-battery case and decreases further when storage is available. In addition, real residential demand response would involve comfort constraints, user acceptance, appliance availability, and potentially the need for compensation mechanisms. These practical constraints are not fully represented in the idealized load-shifting experiment, meaning that the real deployable flexibility could be lower than the technical potential estimated here.

Another important observation is that demand response and battery scheduling are partly substitutable. A battery can already move energy from low-value to high-value periods. Therefore, when the battery capacity is increased, the marginal benefit of also shifting demand becomes smaller. This explains why the demand-response saving falls from 2.00% without a battery to 1.03% with an 80 kWh battery. In this setting, demand response should therefore be interpreted as a complementary flexibility resource rather than as a replacement for storage.

The interpretation changes when the same mechanism is considered from the perspective of an operator or aggregator. At the household level, the objective is mainly to reduce the electricity cost of a relatively small group of users. At the operator level, however, demand response can be valuable for additional reasons. First, during periods of negative energy prices, increasing consumption can be profitable because the operator may effectively be paid to consume energy. Second, the relevant scale may be in MWh rather than kWh, while storage assets of that size may not yet be available or economically justified. Third, flexible demand can help reduce exposure to imbalance prices by adjusting consumption when realized demand or generation deviates from the scheduled position.

Consequently, the same targeted load-shifting idea has different value depending on the viewpoint. For residential prosumers, the experiment shows a positive but modest cost reduction. For an operator, the value may be larger because demand response can act as dispatchable flexibility for absorbing surplus energy, avoiding imbalance costs, and reducing the need for large-scale storage. This distinction is important when interpreting the results: the limited savings observed in the household-scale experiment do not imply that demand response is generally weak, but rather that its value depends strongly on scale, market exposure, and the availability of alternative flexibility resources.

Chapter 8

Conclusions and Future Work

This thesis studied cooperative operation, profit allocation, subcoalition formation and demand-side flexibility in a residential renewable energy community. The central question was whether a community of heterogeneous users can be coordinated in a way that creates economic value, remains computationally tractable, and gives participants credible incentives to remain in the selected cooperative groups.

The analysis was built around an operational optimization model rather than an abstract coalition-value function. Coalition values were obtained by solving daily scheduling problems with household demand, photovoltaic production, battery operation, external grid exchange and internal energy sharing. This made it possible to quantify cooperation in a physically meaningful way: cooperation was measured as the additional cost reduction achieved beyond what users could already obtain through individual battery scheduling. Since the absolute monetary values depend on the particular market prices used in the dataset, the conclusions below focus mainly on normalized indicators, such as percentage cost reductions, recovered shares of the grand-coalition gain and percentage-point improvements.

8.1 Summary of the Main Findings

The first conclusion is that both individual flexibility and cooperation create measurable value. In the 20-user annual experiment, individual battery scheduling reduced the passive baseline cost by 8.11%. When the same users were allowed to operate as a grand coalition, the total reduction relative to the passive baseline increased to 12.21%. Thus, cooperation increased the total cost reduction by 4.10 percentage points compared with individual scheduling alone.

This result confirms the value of distinguishing between temporal flexibility and cooperative coordination. Battery scheduling captures the value of moving each user's own energy across time. Coalition operation captures a different source of value: the ability to match surplus and demand across users before interacting with the external grid. The grand coalition combines these two effects and therefore provides the lowest cost among the operating modes considered.

The second conclusion concerns profit allocation. The proposed positive leave-one-out marginal contribution rule was designed as a scalable alternative to exact cooperative-game

allocation methods. In the 8-player Shapley comparison for June 2020, the proposed rule required only 6.64% of the coalition-value evaluations needed by the exact Shapley calculation, corresponding to a 93.36% reduction in optimization calls. Both methods distributed 100% of the cooperation surplus. The difference was therefore not the amount distributed, but its allocation across users. Relative to the Shapley benchmark, the proposed rule shifted 9.57% of the surplus toward the prosumer group, increasing the prosumer share from 69.71% to 79.28%.

The proposed rule therefore preserves the main practical properties needed in this setting. It distributes the full cooperation surplus, gives users non-negative cooperation bonuses, and remains interpretable because users are rewarded according to their marginal importance in the realized coalition. The comparison with the Shapley benchmark also showed that the proposed rule gives more weight to prosumers when they are the main providers of internal energy. This is a reasonable outcome in an energy-community context, because the value of cooperation depends strongly on the availability of local surplus that can be shared.

The third conclusion is that smaller subcoalitions can preserve most of the grand-coalition coordination benefit. The pairwise-score framework used representative days to estimate the compatibility of user pairs and then formed equal-size subcoalitions using a lexicographic objective. This objective first protects the weakest group by maximizing the minimum subcoalition score, and then improves the total score of the partition. Exact MILP optimization was used for smaller cases, while a multi-start local-search heuristic was introduced for larger cases.

Across the annual experiments, the score-based subcoalitions recovered a large fraction of the grand-coalition coordination gain. The 16-player exact MILP partition recovered 91.59% of the grand-coalition gain. The 20-player exact partition recovered 87.35%. The 30-player local-search partition recovered 94.25%. In the original 50-player case with 10 prosumers and 40 consumers, the local-search partition recovered 98.72%. In the balanced 50-player case with 25 prosumers and 25 consumers, five groups of ten users recovered 94.80%, while ten groups of five users recovered 89.45%.

The comparison with balanced random partitions added an important interpretation. In many cases, prosumer-consumer balance is a first-order design principle. When each group contains enough production and demand diversity, many balanced partitions perform close to the score-based solution. However, when groups become smaller and internal exchange opportunities become more restricted, the exact matching of individual users becomes more important. This was most visible in the balanced 50-player case with ten groups of five, where the score-based partition recovered 89.45% of the grand-coalition gain, compared with 85.41% on average for balanced random partitions.

The fourth conclusion concerns stability under profit sharing. The accepted-migration experiment connected the subcoalition-formation procedure with the proposed allocation rule. The test focused on the weakest member of each selected subcoalition and examined whether

Table 8.1: Summary of the main normalized quantitative results.

Result	Value
20-user solo scheduling cost reduction vs. baseline	8.11%
20-user grand-coalition cost reduction vs. baseline	12.21%
Additional reduction from cooperation	4.10 percentage points
Coordination gain relative to solo scheduling cost	4.46%
Proposed allocation evaluations vs. exact Shapley	6.64%
Reduction in allocation optimization calls	93.36%
Prosumer share under proposed rule vs. Shapley	+9.57 percentage points
30-player local-search recovered grand-coalition gain	94.25%
50-player original local-search recovered gain	98.72%
50-player balanced, five groups of ten recovered gain	94.80%
50-player balanced, ten groups of five recovered gain	89.45%
Personally attractive candidate migrations	30.00%
Accepted blocking migrations	0.00%
No-battery targeted demand-response saving	2.00%
Targeted demand-response saving with 80 kWh battery	1.03%
80 kWh battery + demand response vs. no battery	17.01%

that user could profitably move to another group in a way that the receiving group would accept. In the 20-player annual experiment, 20 candidate migrations were checked. In 30.00% of these cases, the moving player would have improved individually. However, none of these cases satisfied the target-acceptance condition. Therefore, the were no accepted blocking-migrations.

This result does not prove full coalition-proof stability against every possible group deviation. However, it does show that the selected subcoalitions are stable under a practical and meaningful individual-migration criterion. The important result is not the absolute size of the rejected incentives, but the fact that every individually attractive move would make at least one current member of the target coalition worse off. Thus, the proposed grouping and allocation procedure did not create an accepted incentive for the weakest users to leave their assigned groups.

The final experimental extension considered demand response as targeted load shifting. The final demand-response model selected a one-hour activation window with high import cost, capped each user's activation-window consumption at 80% of the mean activation load, and recovered the shifted energy in a two-hour battery-aware recovery window. Without a battery, this mechanism reduced the scheduled cost by 2.00%. With an 80 kWh central battery, the additional demand-response saving fell to 1.03%. In other words, adding the same targeted demand-response action to the 80 kWh battery case increased the total cost reduction from 16.15% to 17.01%, an improvement of 0.86 percentage points.

This shows that targeted demand response is technically useful but modest in the residential setting studied here. The mechanism shifts energy from expensive periods to more

favourable periods, but the amount of flexible household consumption is limited. Moreover, battery scheduling and demand response partly compete for the same arbitrage opportunity. As battery capacity increases, the battery already captures more of the available value, leaving less incremental benefit for load shifting. From a household perspective, demand response should therefore be interpreted as a complementary flexibility resource rather than a replacement for storage.

8.2 Overall Interpretation

Taken together, the results support the main premise of the thesis: cooperation in residential energy communities can create value, but practical deployment requires more than simply solving a grand-coalition optimization problem. A useful framework must answer several connected questions. It must quantify the value of cooperation relative to individual flexibility. It must allocate the resulting gain in a scalable and interpretable way. It must form smaller groups when full coordination is impractical. It must also examine whether participants have incentives to remain in those groups after the benefit has been allocated.

The proposed framework addresses these questions as a connected pipeline. The operational optimization model defines coalition values. The marginal-contribution rule allocates cooperation surplus without requiring all possible coalition values. The pairwise-score method uses representative days to estimate user compatibility. The lexicographic grouping objective forms balanced subcoalitions, while the local-search heuristic makes the approach scalable to larger cases. The accepted-migration test then adds an individual-incentive check after the groups have been formed.

An important empirical insight is that energy balance inside the subcoalitions is often as important as the exact optimization of pairwise scores. Groups need enough demand to absorb local surplus and enough production diversity to create internal exchange opportunities. When groups are large enough, simple prosumer-consumer balance can already capture much of the available value. When groups are small, individual matching becomes more important because each missing complementary user has a larger effect on the group's ability to exchange energy internally.

The demand-response extension adds a second interpretation of flexibility. In the residential community model, demand response produces positive but limited savings. However, from the perspective of an operator or aggregator, the same type of load shifting can be more valuable. At that scale, demand response can help absorb energy during negative-price periods, reduce exposure to imbalance prices, and provide flexibility when MWh-scale storage is not available or economically justified. Therefore, the value of demand response depends strongly on scale, market exposure and the availability of alternative flexibility resources.

8.3 Limitations and Future Work

The limitations of the present study also define the main directions for future work. First, the optimization models are deterministic. They use observed demand, photovoltaic production and price profiles rather than forecasts. This is appropriate for evaluating the economic potential of cooperation under a common historical dataset, but it represents a perfect-information setting. A natural extension would be to formulate the scheduling problem under stochastic or robust operation, using forecasts and uncertainty sets for demand, generation and prices. This would make the model closer to actual day-ahead or intraday operation and would allow the value of cooperation to be evaluated under forecast errors.

Second, the cooperative scheduling model focuses on economic exchange with the external grid and internal energy sharing, but it does not explicitly model distribution-network constraints. In real low-voltage networks, voltage limits, line congestion and losses may restrict some internal exchanges or change the value of local coordination. Future work could integrate network constraints into the cooperative scheduling problem, so that coalition and subcoalition structures are evaluated not only economically but also physically. This would also make it possible to examine whether local cooperation reduces network stress or whether some internal trades need to be limited.

Third, the stability analysis is intentionally limited. The accepted-migration test checks whether the weakest current member of each subcoalition can profitably move to another group in a way that the receiving group would accept. This provides a practical and computationally manageable incentive check, but it does not establish full stability against all possible deviations. Future work could broaden the test to all single-player migrations, pairwise swaps between groups, or collective deviations involving several users. Such extensions would provide a stronger view of stability, although at a higher computational cost. They could also be connected with explicit participation contracts or side-payment mechanisms.

Fourth, the demand-response model represents flexible demand in an aggregate and idealized way. It assumes that shifted energy can be recovered within the selected recovery window without modelling appliance-level constraints, comfort preferences, rebound effects or user compensation. A more detailed demand-response model could separate the technical potential of load shifting from the realistically available demand-side flexibility. It would also allow the analysis to test whether the resulting savings are large enough to compensate users for inconvenience.

Finally, the framework should be tested on additional datasets and regulatory settings. The experiments in this thesis are based on the Irish dataset and on a specific set of household profiles, price signals and prosumer-consumer compositions.Communities with higher photovoltaic penetration, more flexible demand or stronger price volatility may produce different balances

between battery scheduling, energy sharing and demand response.

8.4 Closing Remarks

The thesis shows that cooperative energy-community operation can create meaningful economic value, but that the value of cooperation must be supported by scalable computation and credible incentives. Full coordination provides a useful benchmark, but smaller subcoalitions can recover most of its benefit when they are formed with attention to user complementarity and prosumer-consumer balance. Profit allocation based on marginal contribution provides an interpretable way to distribute the cooperation surplus, and the accepted-migration test suggests that the resulting groups do not create obvious individual incentives to deviate.

More broadly, the results highlight the importance of treating energy communities as coordinated systems rather than collections of independent assets. Batteries, photovoltaic generation, flexible demand and local energy sharing interact with one another. Their combined value depends on timing, complementarity, group structure and market exposure. A practical energy-community design must therefore consider operation, allocation, formation and stability together. This integrated view is the main contribution of the thesis and provides a foundation for future work on scalable, incentive-compatible and flexibility-aware local energy systems.

The data and code used in this thesis are available in the GitHub repository.

Chapter A

Notation

Symbol	Description
Sets and indices	
$\mathcal{N}$	Set of users / players (participants)
i, j	User indices, with $i \neq j$ when relevant
S	Coalition, $S \subseteq \mathcal{N}$
t	Time index
$\mathcal{T}$	Set of time periods (discrete timesteps)
T	Final time index / number of periods in the horizon
Exogenous data and parameters	
$d_{i,t}$	Demand (load) of user i at time t
$g_{i,t}$	Local generation of user i at time t
π_t^{buy}	Grid import price at time t
π_t^{sell}	Grid export remuneration at time t
Δt	Duration of one time period
$\bar{E}_i$	Usable battery energy capacity of user i
η_i^{ch}	Charging efficiency of user i's battery
η_i^{dis}	Discharging efficiency of user i's battery
c_i^{deg}	Battery degradation-cost coefficient
α^{deg}	Battery-degradation scaling factor
F_i	Fixed cost term for user i
ε^{tr}	Small regularization coefficient for internal transfers
Decision variables (optimization model)	
$p_{i,t}^{\text{imp}}$	Grid import power of user i at time t
$p_{i,t}^{\text{exp}}$	Grid export power of user i at time t
$p_{i,t}^{\text{ch}}$	Battery charging power of user i at time t
$p_{i,t}^{\text{dis}}$	Battery discharging power of user i at time t
$E_{i,t}$	Battery state of charge (energy) of user i at time t

$q_{i,j,t}$	Internal transfer power from user i to user j at time t

Cost, value, and gain definitions

C_i^{base}	Baseline cost of user i (no battery optimization / no cooperation)
C_i^{solo}	Optimized cost of user i when scheduling independently
$C^{\text{coal}}(S)$	Optimized total cost of coalition S under coordination
$C^{\text{base}}(S)$	Baseline total cost of coalition S (typically additive)
$v(S)$	Value/saving of coalition S (as defined in the chapter)
$G^{\text{sched}}(\mathcal{N})$	Scheduling gain of $\mathcal{N}$ vs. baseline
$G^{\text{coord}}(\mathcal{N})$	Coordination gain of $\mathcal{N}$ vs. solo scheduling
$G^{\text{total}}(\mathcal{N})$	Total gain of $\mathcal{N}$ vs. baseline

Allocation rule and cooperative-game quantities

s_i	Stand-alone saving of player i vs. baseline
$g(S)$	Cooperation-surplus value of coalition S (beyond stand-alone benefits)
$G(\mathcal{N})$	Total cooperation surplus of the grand coalition
m_i	Leave-one-out marginal contribution of player i to $\mathcal{N}$
w_i	Nonnegative allocation weight (typically $w_i := m_i$)
b_i	Cooperation bonus allocated to player i
x_i	Final saving allocated to player i
α_i	Final cost share of player i after allocation

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